

Effective regions and kernels in continuous sparse regularisation, with application to sketched mixtures

Yohann De Castro*

Rémi Gribonval[†]

Nicolas Jouvin[‡]

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Abstract

This paper advances the general theory of continuous sparse regularisation on measures with the Beurling-LASSO (BLASSO). This TV-regularised convex program on the space of measures allows one to recover a sparse measure using a noisy observation from a measurement operator. While previous works have uncovered the central role played by this operator and its associated kernel in order to get estimation error bounds, the latter requires a technical local positive curvature (LPC) assumption to be verified on a case-by-case basis. In practice, this yields only a few LPC-kernels for which this condition is proved. In this paper, we prove that the “sinc-4” kernel, used for signal recovery and mixture problems, does satisfy the LPC assumption. Furthermore, we introduce the *kernel switch* analysis, which allows one to leverage a known LPC-kernel as a *pivot* kernel to prove error bounds. Together, these results provide easy-to-check conditions to get error bounds for a large family of translation-invariant model kernels. Besides, we also show that known BLASSO guarantees can be made adaptive to the noise level. This improves on known results where this error is fixed with some parameters depending on the model kernel. We illustrate the interest of our results in the case of mixture model estimation, using band-limiting smoothing and sketching techniques to reduce the computational burden of BLASSO.

Keywords: Continuous sparse regression; Beurling Lasso; Sketching; Mixture models

1 Introduction

Sparse regularisation, particularly methods based on ℓ^1 minimisation, has emerged as a cornerstone technique to solve high-dimensional linear inverse problems across numerous domains. These approaches have revolutionized signal processing, statistical learning, and computational imaging by enabling reconstruction from severely limited observations [Foucart and Rauhut, 2013, Candès et al., 2006]. Despite their remarkable success, traditional sparse methods rely on discretisation, introducing inherent limitations in both computational efficiency and theoretical precision. Consequently, there has been growing interest in continuous counterparts to ℓ^1 minimisation that operate in a grid-free manner. These “off-the-grid” approaches not only mitigate discretisation errors but also potentially offer more efficient solvers, sharper theoretical guarantees, and increased robustness to sampling limitations [see e.g. Candès and Fernandez-Granda, 2014, Tang et al., 2013, Duval and Peyré, 2015, 2017]. Such continuous formulations have proved particularly valuable when the underlying target structure inherently exists in a

*Institut Camille Jordan – IUF – Centrale Lyon, 36 Av. Guy de Collongue, 69130 Écully, France. Corresponding author: yohann.de-castro@ec-lyon.fr.

[†]Inria, CNRS, ENS de Lyon, Université Claude Bernard Lyon 1, LIP, UMR 5668, Lyon cedex 07, France.

[‡]INRAE, AgroParisTech, Université Paris-Saclay, MIA-Paris, 22 place de l’agronomie, 91120 Palaiseau, France.

continuous domain, such as in the case of mixtures of distributions or when the parameters of interest are not naturally discretised.

A promising extension of ℓ^1 minimisation to continuous domains involves encoding both positions and weights as a discrete Radon measure, replacing the ℓ^1 norm with the total variation norm. This approach treats a measure as "sparse" when it is a sum of few Dirac masses. The resulting infinite-dimensional optimisation problem is cast over the space of measures and known as Beurling-LASSO (BLASSO). The latter has emerged as a powerful framework across multiple disciplines. In signal processing, BLASSO has demonstrated effectiveness in sparse deconvolution and super-resolution [Candès and Fernandez-Granda, 2014, Duval and Peyré, 2015, Boyer et al., 2017]. Its applications extend to statistics for continuous sparse regression and off-the-grid compressed sensing [De Castro and Gamboa, 2012, Poon et al., 2023], and to machine learning for mixture model estimation [De Castro et al., 2021]. Practical implementations have brought significant advances in diverse fields: single-molecule fluorescent microscopy [Boyd et al., 2017], neuronal spike sorting [Ekanadham et al., 2014], statistical mixture modelling [Gribonval et al., 2021b, De Castro et al., 2021], and training of shallow neural networks [Bach, 2017]. In this paper, we motivate our analysis and illustrate our results in the case of mixture model estimation [De Castro et al., 2021], although our theoretical contributions apply broadly to all BLASSO applications.

The BLASSO objective is a TV-regularised data-fitting term, the latter inducing a *model* kernel which is tied to the measurement process—or model—at hand, *e.g.* compressed sensing, deconvolution, mixture model estimation, etc. One of the main practical limitations of BLASSO is the computational cost of large kernel matrices, see for instance De Castro et al. [2023, Section 1.3]. This well-known issue in kernel methods has motivated the use of sketching techniques [Rahimi and Recht, 2007, Gribonval et al., 2020], whose principle is to compress a dataset into a single vector, the *sketch*, by computing averages (over the samples of the dataset) of m non-linear random features. This compression is designed to approximate the kernel computations while retaining statistical guarantees for the underlying learning task. For continuous sparse regression, BLASSO analyses for *sketched* model kernels defined via random features have been investigated in Poon et al. [2019, 2023], leading to the development of a corresponding *sketched* BLASSO formulation. Related recent studies [Gribonval et al., 2021b, Belhadji and Gribonval, 2024] have proven that the model sketching operator, mapping from a dataset to its sketch, acts as a quasi-isometry when restricted to the set of sparse measures whose support points are sufficiently separated.

1.1 Main contributions

We begin by presenting a schematic view of BLASSO theoretical analysis presented in this paper. For the sake of presentation, most of the objects will not be properly defined as we wish to focus on the articulation between the state-of-the-art and our contributions.

The BLASSO is a convex optimisation program of the space of measures, denoted by $\mathcal{P}_K$ hereafter, which depends on a reproducing kernel K . The goal of BLASSO theoretical analysis is to obtain error bounds on the recovery of target sparse measures μ^0 belonging to a model set $\mathcal{M}_{\Delta,K}$. This model set involves a minimal separation $\Delta > 0$ between atoms of μ^0 , measured with respect to the Fisher-Rao distance $\mathfrak{d}_K$ of the kernel K . The crux of every theoretical work is to prove the existence of so-called *non-degenerate dual certificates* for the BLASSO program $\mathcal{P}_K$ and the model set $\mathcal{M}_{\Delta,K}$. These objects guarantee that the support of the BLASSO estimator “localizes” around the support of the target sparse measure μ^0 . This localisation error is stated in terms of *near regions* with a radius $r > 0$ around the true atoms, the lower the better the recovery. In their comprehensive treatment, Poon et al. [2023] formalized the local positive curvature assumption, $\text{LPC}(K, \Delta)$, a technical condition on kernel K to ensure the existence of such certificates. Their LPC analysis proves the existence of certificates for the program $\mathcal{P}_K$, as well as the *sketched* program $\mathcal{P}_{\tilde{K}}$ when using sketching techniques to approximate the kernel

$K \approx \tilde{K}$ with random features [Rahimi and Recht, 2007, Gribonval et al., 2020]. Notably, in the sketching case, the existence of certificates holds with high-probability with respect to the sketch size, however the model set $\mathcal{M}_{\Delta, K}$ still depends on the kernel K and not on its sketched version $\tilde{K}$.

Although appealing, this analysis requires proving the $\text{LPC}(K, \Delta)$ and computing the Fisher-Rao distance $\mathfrak{d}_K$ which can be cumbersome and has to be done on a case-by-case basis. To date, only a limited set of kernels have been rigorously proved to enjoy the LPC property. Furthermore, the radius of the localisation result is fixed and cannot adapt to the noise level.

We address these issues by introducing the notion of *effective* near regions and the *kernel switch* technique. Additionally, we prove the LPC for the sinc-4 kernel, thereby expanding the list of known LPC-kernels. We also demonstrate the interest of these results for mixture model estimation.

Effective near-regions and near-minimizers We show that the radius r of the near region is *adaptive to the noise level*, denoted by γ , and that BLASSO can localize *effective* near-regions up to $r \sim \sqrt{\gamma\sqrt{s_0}}$, where s_0 is the sparsity level of the target measure μ^0 . This is particularly relevant in statistical applications where the noise level decreases with sample size, denoted by n . As an example, in the mixture problem, the noise $\gamma = \mathcal{O}(n^{-1/2})$, and one gets effective near regions of size essentially $\mathcal{O}(n^{-1/4})$, narrowing around the true parameters as n increases. Additionally, we also uncover an interesting feature of our convex programming problem. Namely, we establish that near minimizers of the BLASSO problem—defined as solutions with an objective value no greater than that of the target—share the same guarantees as the exact minimizer itself. In practice, this means one does not need to solve the BLASSO exactly, any sufficiently *near solution* will do.

Kernel switch This paper introduces a significant advancement by demonstrating that an existing kernel K_0 satisfying $\text{LPC}(K_0, \Delta)$ —termed a *pivot kernel*—can be used to derive guarantees for BLASSO problems of interest $\mathcal{P}_K$ involving a *different* model kernel K , including the case of a sketched model kernel $\tilde{K}$ defined via sketching. This *kernel switch* principle, illustrated in Figure 1, is possible provided that the RKHS of K_0 is continuously embedded into the RKHS of K , which is ensured by a finite embedding constant: $C_{\text{switch}}(K, K_0) < +\infty$. Under this condition, we prove the existence of certificates both for the program $\mathcal{P}_K$ and the sketched program $\mathcal{P}_{\tilde{K}}$, with the model set $\mathcal{M}_{\Delta, K_0}$ defined via the pivot kernel K_0 and its $\text{LPC}(K_0, \Delta)$ condition. Our result yields the same localisation error bounds as in the direct approach of proving LPC, only factoring in the embedding constant C_{switch} . This substantially broadens the scope of continuous sparse regression by allowing practitioners to leverage the established theoretical guarantees of well-characterized pivot kernels, while employing model kernels that might be more computationally advantageous or better suited to specific problem structures.

The “sinc-4” pivot We rigorously show for the first time that a specific kernel K_0 , the sinc-4 defined as the fourth power of the sinus cardinal kernel [De Castro et al., 2021], satisfies the LPC assumption and is amenable to sketching with random features. This is particularly helpful as: *a) its RKHS is included in the well-known space of band-limited functions*; and *b) its Fisher-Rao distance is a simple rescaled Euclidean distance*, leading to a simple model set $\mathcal{M}_{\Delta, \text{sinc-4}}$ defined in terms of Euclidean separation between atoms. Coupled with our kernel switch analysis, this result implies that *any BLASSO problem $\mathcal{P}_K$ with K a translation-invariant kernel whose RKHS contains band-limited functions verifies the theoretical guarantees* resulting from our analysis. This also extends to sketched translation-invariant kernels under mild conditions on the random features.

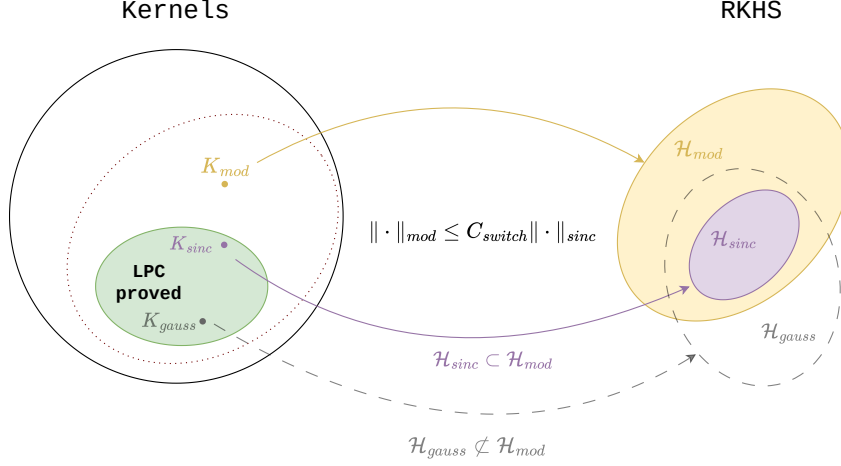


Figure 1: Illustration of the kernel switch principle. Left: A small set of pivot kernels known to satisfy the local positive curvature assumption (LPC) enables theoretical guarantees for a much larger class of model kernels. Right: A model kernel K_{mod} is admissible for our statistical guarantees if there exists a pivot LPC kernel (here K_{sinc}) whose RKHS is continuously embedded into the model RKHS. The error bounds remain valid with an additional scaling factor $C_{\text{switch}}(K_{\text{mod}}, K_{\text{sinc}})$. In this example, using the sinc kernel as pivot is valid because its RKHS is included in the model RKHS, while the Gaussian kernel cannot serve as a pivot since its RKHS contains functions with faster-decaying Fourier transforms than the model allows.

Contributions for Mixture Modelling Our contribution on sketching sparse regularisation of translation-invariant mixtures is detailed in Section 3 with a special focus on the sinc-4 kernel. This leads to S2Mix, a more practical algorithm for the *sketched* Supermix problem of De Castro et al. [2021]. The consequences of our results are as follows: S2Mix is a computationally efficient algorithm to estimate the parameters of a mixture model, even for large datasets. For a mixture with s_0 components, it achieves a sketch size complexity of $\mathcal{O}(s_0 \log^2(s_0))$, which can be significantly lower than the original sample complexity n . Furthermore, S2Mix covers a wide range of mixture models, including non-Gaussian template distributions, since it only requires that its characteristic function does not vanish at low frequencies, a mild condition that is easy to verify. This makes it a versatile tool for various applications in machine learning and statistics. The algorithm is robust to noise and model misspecification, as it can handle a wide range of noise levels and model assumptions. This robustness is particularly important in real-world applications where data may be noisy or incomplete. Finally, S2Mix provides theoretical guarantees for parameter recovery, ensuring that the estimated parameters are close to the true model parameters with high probability.

It is worth noting that our theoretical guarantees are stated under a *well-separated* assumption, requiring the minimal distance between true parameters to be above a certain threshold (the Rayleigh limit). While continuous sparse regularisation can sometimes succeed below this limit under specific conditions, the well-separated regime is the standard setting for robust recovery bounds, as resolution limits inherently restrict reconstruction precision when sources are arbitrarily close [Liu and Zhang, 2021, Batenkov et al., 2021, Diab and Batenkov, 2024].

1.2 Related works

In their pioneering work, Candès and Fernandez-Granda [2014] studied the super-resolution deconvolution problem on the torus. Their proof technique can be reread under the viewpoint developed in this paper as using a kernel switch with the Dirichlet kernel as model kernel and the Jackson kernel (Dirichlet kernel raised to the power of 4) as pivot kernel. Building directly

on this methodology, De Castro et al. [2021] utilized a similar proof technique in their detailed analysis of mixture model estimation with BLASSO, relying on the sinc-4 kernel as pivot. Notably, both papers implicitly leverage a kernel switch analysis without formally characterizing the general principles of this phenomenon. Furthermore, De Castro et al. [2021] did not provide a proof that the sinc-4 kernel satisfies the LPC assumption, making it impossible to derive sketching statistical error bounds using the framework later developed by Poon et al. [2023].

For Gaussian mixture models with fixed covariance, Poon et al. [2023, Section 2.3 and Appendix D.5.1] showed that the Gaussian kernel with the same covariance satisfies the LPC assumption. Their BLASSO analysis, however, is essentially restricted to Gaussian mixtures where the empirical distribution is convolved with a prescribed Gaussian (referred to as smoothing kernel in this paper, see Section 3.1). In contrast, our kernel switch analysis (under the mild assumptions in Section 3) covers a much broader class of mixture densities and smoothing kernels. Recently, building on our effective near-region analysis, Giard et al. [2025] further established BLASSO error bounds for Gaussian mixtures with *unknown* diagonal covariances by leveraging the Fisher-Rao geometry of the model kernel via a novel semi-distance tailored to the LPC condition.

Although the near-minimizer property is implicit in the proofs of previous works on Bregman divergence [Candès and Fernandez-Granda, 2014, Duval and Peyré, 2015, 2017, Azais et al., 2015, Poon et al., 2023], it has not been explicitly stated. A similar concept exists in the Compressed Sensing (CS) literature, where guarantees for exact minimizers of LASSO-related problems are often proven via more general results that also apply to near minimizers (see *e.g.*, Foucart and Rauhut [2013, Theorems 4.18 & 4.19, 4.21 & 4.23]). The near-minimizer property is of great practical importance, as it justifies the use of approximate solvers. This is especially relevant for large-scale or high-dimensional problems where finding the exact minimizer is computationally infeasible. Recent algorithms to solve the BLASSO problem, such as Sliding Frank-Wolfe [Denoyelle et al., 2019] and stochastic versions of Conic Particle Gradient Descent (CPGD) [Chizat, 2019, De Castro et al., 2023], are designed precisely to find such near minimizers with convergence guarantees. However, while these CPGD-based methods can find a near minimizer in a finite number of steps, the required number of steps scales exponentially with the dimension d , currently limiting their application to moderate-dimensional regimes where d is less than approximately 100.

We illustrate our contribution on mixture models. The latter represent a fundamental framework in modern statistics, with diverse applications spanning machine learning, biology, genetics, and astronomy [McLachlan and Peel, 2004, Fruhwirth-Schnatter et al., 2019]. Traditional inference for these models typically relies on the non-convex maximum-likelihood paradigm solved via the Expectation-Maximisation (EM) algorithm [McLachlan and Krishnan, 2007]. A critical challenge in mixture modelling—determining the optimal number of components s_0 —is conventionally addressed post-inference as a model selection problem [Celeux et al., 2019]. To handle large-scale datasets, the computational statistics community has developed sophisticated methodological tools including variational inference [Blei et al., 2017] and amortized inference techniques [Kingma and Welling, 2014, Tomczak, 2024], though theoretical guarantees for parameter recovery remain relatively limited. The sparse continuous regression perspective on mixture modelling, recently introduced by De Castro et al. [2021], Poon et al. [2023], offers compelling theoretical guarantees for parameter recovery while demonstrating robustness to noise and model misspecification. This approach complements practical algorithms developed to solve the BLASSO optimisation problem, such as sliding Frank-Wolfe [Denoyelle et al., 2019] and particle-based methods [Chizat, 2019, De Castro et al., 2023], providing both theoretical foundations and computational efficiency.

Outline

The remainder of the paper is organized as follows:

- Section 2 introduces the BLASSO along with state-of-the-art theoretical guarantees that can be obtained. The notions of model kernel, model set and sketching are properly defined; we also review kernels known to satisfy the LPC assumption.
- Section 3 introduces the sketching BLASSO problem for mixture models. We first present a resolution-based approach to mixture modelling and show how it can be formulated as a continuous sparse regularisation problem. We then develop a sketched version of the framework, called S2Mix, which achieves significant computational efficiency through random Fourier features. We derive statistical guarantees for this approach, demonstrating how the sinc-4 kernel serves as an effective pivot kernel that enables parameter estimation with near-optimal sample complexity.
- Section 4 presents our three main theoretical contributions. First, our effective near regions analysis in Theorem 1 demonstrates that the size of near regions can adapt to the noise level, yielding localisation guarantees of order $\mathcal{O}(\sqrt{\gamma\sqrt{s_0}})$ that improve as noise decreases. Second, our kernel switch technique in Theorem 1 establishes that statistical error bounds hold when using a different kernel for analysis than for measurement, provided the RKHS embedding condition (Assumption 2) is satisfied. Theorem 2 extends the kernel switch to the case of sketched kernels under mild assumption on their random features. Third, we prove in both theorems that near-optimal solutions—*i.e.*, measures with objective value not exceeding that of the target measure—enjoy the same statistical guarantees as the exact minimizer. This section also includes a rigorous analysis proving that the sinc-4 kernel satisfies the LPC assumption, and can therefore be used as a pivot in a kernel switch analysis.

Technical proofs, additional mathematical background, and supporting material are provided in the appendix. A comprehensive list of notation is given in Section D.

2 State-of-the-art on continuous sparse regularisation with BLASSO

Let $\mathcal{X} \subset \mathbb{R}^d$ be a compact set, referred to as the *parameter space*, and consider $(\mathcal{M}(\mathcal{X}), \|\cdot\|_{TV})$ the space of Radon measures, defined as the topological dual space of $(\mathcal{C}(\mathcal{X}), \|\cdot\|_\infty)$, the continuous functions endowed with the infinity norm. Let $\mathcal{F}$ be a separable Hilbert space and let $F : \mathcal{M}(\mathcal{X}) \rightarrow \mathcal{F}$ be a linear map, referred to as the forward measurement operator. We define the BLASSO problem as

$$\hat{\mu} \in \arg \min_{\mu \in \mathcal{M}(\mathcal{X})} J_\kappa(\mu) \quad \text{where} \quad J_\kappa(\mu) := \frac{1}{2} \|\mathbf{y} - F\mu\|_{\mathcal{F}}^2 + \kappa \|\mu\|_{TV}, \quad (2.1-J_\kappa)$$

with $\kappa > 0$ a regularisation parameter and $\mathbf{y} \in \mathcal{F}$ some observation. One can prove that a solution $\hat{\mu} \in \mathcal{M}(\mathcal{X})$ to BLASSO (2.1- J_κ) exists under the forthcoming hypothesis (2.3- $\mathbf{H}_{\text{mod}}$), see for instance [De Castro et al., 2023, Theorem 1.1].

The criterion to be minimized decomposes as the sum of two terms: a *data-fitting* term, comparing some observation $\mathbf{y}$ with a candidate $F\mu$ in the separable Hilbert space $\mathcal{F}$; and a *TV-regularisation* term, which acts as a sparsity inducing penalty analogous to ℓ^1 -regularisation in finite dimension [Foucart and Rauhut, 2013, Boyer et al., 2019, Theorem 1.1].

2.1 Model kernel, class of sparse models and inverse problem under consideration

The continuous regression problem under consideration offers a powerful generalisation of classical sparse regression, encoding both weights and positions within a Radon measure framework. Central to our analysis is the *model kernel* K_{mod} , which characterizes the interaction between

point masses in the parameter space $\mathcal{X}$ through the measurement operator F . Although it is traditionally referred to simply as “the kernel” in the BLASSO literature [Poon et al., 2023], we choose to deliberately distinguish it from what we will call the *pivot kernel* later in our analysis.

The model kernel K_{mod} is formally defined as the $\mathcal{F}$ -inner product of the operator F applied to pairs of Dirac masses at different locations in the parameter space $\mathcal{X}$. Its continuity is a mild technical condition which ensures the existence of solutions to the BLASSO optimisation problem [De Castro et al., 2023]. This distinction between *model kernel* and *pivot kernel* represents one of our key contributions, allowing us to extend theoretical guarantees to a broader class of inverse problems than previously possible.

Definition 1 (Model kernel). *Define the model kernel by*

$$K_{\text{mod}}(\mathbf{s}, \mathbf{t}) := \langle F\delta_{\mathbf{s}}, F\delta_{\mathbf{t}} \rangle_{\mathcal{F}} \quad \text{for all } \mathbf{s}, \mathbf{t} \in \mathcal{X}, \quad (2.2)$$

where $\delta_{\mathbf{u}}$ is the Dirac mass at point $\mathbf{u} \in \mathcal{X}$. We assume that

$$K_{\text{mod}}(\cdot, \cdot) \text{ is continuous,} \quad (2.3\text{--}\mathbf{H}_{\text{mod}})$$

where continuity is understood with respect to the pair of variables. It is a positive definite kernel and we denote as $\mathcal{H}_{\text{mod}}$ its reproducing kernel Hilbert space (RKHS), which is continuously included in $\mathcal{F}$ as discussed in Section 4.

In the following, the class of models $\mathcal{M}_{s,\Delta,\mathfrak{d}}$ is defined as the set of measures that can be written as a sum of s Dirac masses with a *minimal separation* of Δ between two spikes for some distance $\mathfrak{d}$; the noise term Γ is defined as the difference between the observation $\mathbf{y}$ and the operator F applied to the *target* measure μ^0 ; and the *noise level* γ is defined as the norm of the noise term.

Definition 2 (Sparse models). *Let $\mathfrak{d}(\cdot, \cdot)$ be a distance defined on $\mathcal{X}$. Let Δ be positive and let s be greater than 1. We define the class of s -sparse measures with minimal separation Δ as*

$$\mathcal{M}_{s,\Delta,\mathfrak{d}} := \left\{ \mu : \mu = \sum_{k=1}^s a_k \delta_{\mathbf{x}_k} \text{ and } \min_{k,\ell} \mathfrak{d}(\mathbf{x}_k, \mathbf{x}_\ell) \geq \Delta \right\}, \quad (2.4)$$

where Δ is referred to as the *minimal separation*, and s as the *sparsity*. Note that throughout this work, we consider finite signed Radon measures, so the amplitudes a_k are taken in $\mathbb{R}$.

Remark 2.1. *The class of sparse models is usually defined with respect to the so-called Fisher-Rao distance associated to some positive type kernel $K(\cdot, \cdot) : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$. While previous works choose this kernel to be the model kernel [Poon et al., 2019, 2023], this paper shows error bounds choosing a possibly different kernel.*

The inverse problem under consideration is defined as follows. We consider a target measure $\mu^0 \in \mathcal{M}_{s_0,\Delta_0,\mathfrak{d}}$, for some sparsity s_0 , distance $\mathfrak{d}$ and minimal separation Δ_0 . Given a noisy observation $\mathbf{y}$ of μ^0 through the measurement operator F , we define the noise term and noise level respectively as

$$\Gamma := \mathbf{y} - F\mu^0, \quad (2.5a)$$

$$\text{resp. } \gamma_0 := \|\Gamma\|_{\mathcal{F}}. \quad (2.5b)$$

Given a *noisy* observation $\mathbf{y}$, the goal is to recover the true locations $\mathbf{x}_k^0$ and weights a_k^0 using a solution $\hat{\mu}$ of BLASSO (2.1– J_κ). The noise term Γ in (2.5) accounts for model misspecification, outliers, and sampling or measurement noise [De Castro et al., 2021, Poon et al., 2023]. Throughout this paper, we call *noise level* some upper bound γ on the noise γ_0 , namely $\gamma \geq \gamma_0$. Error bounds are stated in terms of the noise level γ and hold for any κ ; setting $\kappa \simeq \gamma/\sqrt{s_0}$ yields sharper constants, as presented in this paper. In statistical settings, standard concentration gives $\gamma = \mathcal{O}_{\mathbb{P}}(1/\sqrt{n})$ where n is the sample size (we will uncover such a result for mixture models later in this paper), while in deterministic inverse problems γ is often specified by the measurement device.

2.2 Estimation error bounds via dual certificates

The analysis of BLASSO estimation error bounds (2.7) below hinges on the concept of the so-called *far* and *near* regions, which define spatial partitioning around the support points of the target measure μ^0 . These regions serve a fundamental role in characterizing the behaviour of any solution $\hat{\mu}$ to (2.1– J_κ). Specifically, the BLASSO exhibits strong reconstruction properties in near regions—where $\hat{\mu}$ closely approximates μ^0 —while maintaining controlled behaviour in far regions. This spatial characterisation provides precise quantitative guarantees on the support recovery capabilities of the estimator, referred to as the localisation property of BLASSO. We formally define these regions as follows.

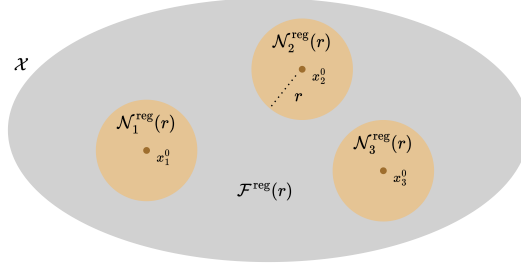


Figure 2: Two-dimensional illustration of near (light maroon) and far (light gray) regions with fixed radius r and a distance $\mathfrak{d}(\mathbf{s}, \mathbf{t}) \propto \|\mathbf{s} - \mathbf{t}\|$.

Definition 3 (Far and Near regions). *Let $\mu^0 \in \mathcal{M}_{s_0, \Delta_0, \mathfrak{d}}$, let $r > 0$, we define the near region of $\mathbf{x}_k^0$ as the closed ball centered at $\mathbf{x}_k^0$ for the distance $\mathfrak{d}$*

$$\mathcal{N}_k^{\text{reg}}(r) := \left\{ \mathbf{x} \in \mathcal{X}, \quad \mathfrak{d}(\mathbf{x}, \mathbf{x}_k^0) \leq r \right\}, \quad (2.6a)$$

and the far region as the complement

$$\mathcal{F}^{\text{reg}}(r) := \mathcal{X} \setminus \mathcal{N}^{\text{reg}}(r), \quad \text{with: } \mathcal{N}^{\text{reg}}(r) = \bigcup_{k=1}^{s_0} \mathcal{N}_k^{\text{reg}}(r). \quad (2.6b)$$

Existing bounds on estimation errors proved in the literature [Azais et al., 2015, Candès and Fernandez-Granda, 2014, Poon et al., 2023] depend on some fixed $r > 0$, that may depend on Δ_0 , and are expressed as follows:

- *Control of the far region:*

$$|\hat{\mu}|(\mathcal{F}^{\text{reg}}(r)) \lesssim_d \gamma \sqrt{s_0}, \quad (2.7a)$$

- *Control of all the near regions:*

$$|\hat{\mu}(\mathcal{N}_k^{\text{reg}}(r)) - a_k^0| \lesssim_d \gamma \sqrt{s_0}, \quad (2.7b)$$

- *Detection level:* For all Borelian $A \subset \mathcal{X}$ such that $|\hat{\mu}|(A) \gtrsim_d \gamma \sqrt{s_0}$, there exists $\mathbf{x}_k^0$ such that

$$\mathfrak{d}(A, \mathbf{x}_k^0) := \min_{t \in A} \mathfrak{d}(t, \mathbf{x}_k^0) \lesssim_d r, \quad (2.7c)$$

where $\lesssim_d$ denotes the inequality up to a multiplicative constant that may depend on the dimension d , $\hat{\mu}$ is a solution to (2.1– J_κ) with regularisation parameter $\kappa \simeq \gamma/\sqrt{s_0}$ and $|\hat{\mu}|$ denotes the absolute part of $\hat{\mu}$.

Remark 2.2 (On regularisation parameter calibration). *The regularisation parameter κ is crucial for the performance of BLASSO. In practice, it is often calibrated using cross-validation techniques or other model selection criteria. The choice of κ can significantly impact the balance between bias and variance in the estimation process. In our analysis, we assume that κ is chosen to be proportional to the noise level γ and the sparsity s_0 , specifically $\kappa \simeq \gamma/\sqrt{s_0}$. This choice ensures that the regularisation term effectively controls the trade-off between fitting the data and promoting sparsity in the estimated measure.*

When s_0 is not known, other choices of the regularisation parameter can be considered, such as $\kappa \simeq \gamma$ which is a more realistic choice in practice, yet less optimal in terms of the theoretical guarantees. In this case, the error bounds (2.7) scale as s_0 instead of $\sqrt{s_0}$, which is a less favorable scaling, yet still provides useful information about the localisation of the solution, see Remark 4.2 below.

This error analysis of BLASSO (2.1– J_κ) relies on controlling both the noise level γ and the Bregman divergence of the total variation norm

$$\mathcal{D}_{\eta^0}(\mu || \mu^0) := \|\mu\|_{\text{TV}} - \|\mu^0\|_{\text{TV}} - \langle \eta^0, \mu - \mu^0 \rangle_{\mathcal{C}(\mathcal{X}) \times \mathcal{M}(\mathcal{X})} \quad (2.8)$$

between the underlying $\mu^0 \in \mathcal{M}_{s_0, \Delta_0, \mathfrak{d}}$ and a solution $\mu = \hat{\mu}$ [De Castro and Gamboa, 2012, Azais et al., 2015] or, as we will see, any *approximate* solution to (2.1– J_κ).

This error analysis requires that

- η^0 is an element of the RKHS $\mathcal{H}_{\text{mod}}$,
- η^0 is a sub-gradient of the total variation norm at point μ^0 with specific properties on the near regions $\mathcal{N}_k^{\text{reg}}(r)$ and far region $\mathcal{F}^{\text{reg}}(r)$,

Such η^0 is referred to as a *non-degenerate dual certificate of radius r* [De Castro and Gamboa, 2012, Candès and Fernandez-Granda, 2014, Duval and Peyré, 2015]. When such a certificate exists, the control on Bregman divergence directly translates into the estimation error bounds of Equations (2.7), yielding precise guarantees on both support recovery and parameter estimation accuracy [Duval and Peyré, 2017].

2.3 Construction of dual certificates under local positive curvature assumption (LPC)

Crucially, a sufficient condition for BLASSO error analysis to hold is that the model kernel verifies a so-called *local positive curvature* assumption [LPC, Poon et al., 2023, Assumption 1], given in Section 4.1. The LPC ensures the existence of a non-degenerate dual certificate η^0 of radius r for any target measure μ^0 in the sparse model class $\mathcal{M}_{s_0, \Delta_0, \mathfrak{d}_{\text{FR}}}$ where Δ_0 is taken sufficiently large with respect to r . There, the distance $\mathfrak{d}_{\text{FR}}$ is the so-called *Fisher-Rao distance* induced by the model kernel, which we will detail later on in Section 4.1.

To the best of our knowledge, the literature [Azais et al., 2015, Candès and Fernandez-Granda, 2014, Poon et al., 2019, 2023] has proved that the estimation error bounds of Equations (2.7) hold, but only for *fixed* near regions of size $r = r_0$ *irrespective of the noise level*. However, in view of Equations (2.7), the smaller the radius r , the closer the BLASSO localizes around the true support, hence the sharper the error bounds. One of our key contributions is showing that *these bounds extend to near regions whose size can adapt to the noise level γ , decreasing as $\mathcal{O}(\sqrt{\gamma\sqrt{s_0}})$* , which we call *effective* near regions. This is a substantial improvement for small noise levels since the LPC is a property *intrinsic* to the model kernel, with parameters independent of μ^0 and $\mathbf{y}$, and thus of the noise level γ .

2.4 Previous studies on kernels for BLASSO

Fortunately, the existence of dual certificates has already been proved for some kernels. Some of these have also been shown to verify the LPC assumption. Pioneering work [Candès and Fernandez-Granda, 2014] has proved that the non-degenerate dual certificate exists for the Jackson kernel:

- **Jackson kernel** [Candès and Fernandez-Granda, 2014] on $\mathcal{X} = [0, 1)^d$ identified to the d -dimensional torus $(\mathbb{R} \setminus \mathbb{Z})^d$. The kernel is the fourth power of the Dirichlet kernel, i.e.

$$K(\mathbf{s}, \mathbf{t}) = \prod_{i=1}^d \kappa(s_i - t_i), \quad \text{with } \kappa(t) = \left(\frac{\sin((\frac{N}{2} + 1)\pi t)}{(\frac{N}{2} + 1) \sin(\pi t)} \right)^4 \text{ and } N \geq 1. \quad (2.9a)$$

This kernel satisfies the LPC [Poon et al., 2023, Appendix C].

Using a similar technique as in Candès and Fernandez-Granda [2014], a low-pass spherical harmonic kernel has been investigated in Bendory et al. [2015] for *spike detection* on the sphere.

- **Low-pass spherical harmonic** [Bendory et al., 2015] on the 2-sphere $\mathcal{X} = \mathbb{S}^2 \subset \mathbb{R}^3$, given by

$$K(\mathbf{s}, \mathbf{t}) = C(N) \sum_{k=0}^N \zeta(k/N) P_{k,3}(\langle \mathbf{s}, \mathbf{t} \rangle),$$

where $N \geq 1$ is some degree, $C(N)$ is a normalizing constant, $P_{k,3}$ is the univariate ultraspherical Gegenbauer polynomial of order 3 and degree k , and $\zeta(t)$ is a smooth non-negative univariate function equal to 1 for t between 0 and $1/2$, 0 for t greater than 1 and less than 1 otherwise.

The LPC has not been proved for this kernel although the existence of a dual certificate has been proven via alternative techniques.

Recently, two other kernels have been proved to satisfy the LPC, namely

- **Gaussian kernel** [Poon et al., 2023, Appendix D] given by

$$K(\mathbf{s}, \mathbf{t}) = \Theta_{\Omega}(\mathbf{s} - \mathbf{t}) \quad \text{where} \quad \Theta_{\Omega} := \exp\left(-\frac{1}{2} \|\cdot\|_{\Omega}^2\right), \quad (2.9b)$$

and $\|\mathbf{t}\|_{\Omega}^2 = \langle \mathbf{t}, \Omega^{-1} \mathbf{t} \rangle$ for some positive definite matrix Ω ;

- **Laplace transform kernel** [Poon et al., 2023, Appendix E]. This kernel is defined on $\mathcal{X} = \mathbb{R}_+^d$ and arises when studying continuous sampling of the Laplace transform of a positive signal, defined as $\mathcal{L}[\mu](\omega) = \int_{\mathcal{X}} e^{-\omega^\top \mathbf{t}} d\mu(\mathbf{t})$. It was studied in Denoyelle et al. [2019], Poon et al. [2019] and is not to be confused with the *Laplace kernel*. In particular, it is not translation-invariant and is given by

$$K_{\alpha}(\mathbf{s}, \mathbf{t}) = \prod_{i=1}^d \kappa(s_i + \alpha_i, t_i + \alpha_i), \quad \text{with } \kappa(a, b) = \frac{2\sqrt{ab}}{a+b} \text{ and } \alpha_i > 0. \quad (2.9c)$$

Importantly, these kernels (except the low-pass spherical harmonic) will serve as “pivot kernels” in our kernel switch analysis—a concept central to our contribution. As we will show, *a pivot kernel that satisfies the LPC enables the construction of non-degenerate dual certificates with other model kernels.*

Kernel name	Domain $\mathcal{X}$	$K(\mathbf{s}, \mathbf{t})$	Assumption 1 (LPC)	LPC-parameters
Jackson	$(\mathbb{R} \setminus \mathbb{Z})^d$	(2.9a)	Poon et al. [2023, Appendix C]	$r_0 = \mathcal{O}(1), \Delta_0 = \mathcal{O}(d^{\frac{1}{2}} s_0^{\frac{1}{4}})$
Gaussian	$\mathbb{R}^d$	(2.9b)	Poon et al. [2023, Appendix D]	$r_0 = \mathcal{O}(1), \Delta_0 = \mathcal{O}(\log(s_0))$
Laplace	$\mathbb{R}^d$	(2.9c)	Poon et al. [2023, Appendix E]	$r_0 = \mathcal{O}(1), \Delta_0 = \mathcal{O}(d + \log(d^{\frac{3}{2}} s_0))$
<i>Sinc-4</i>	$\mathbb{R}^d$	(2.9d)	<i>This work</i> - Theorem 3	$r_0 = \mathcal{O}(\frac{1}{d}), \Delta_0 = \mathcal{O}(d^{\frac{7}{4}} s_0^{\frac{1}{4}})$

Table 1: Table of known kernels verifying the LPC assumption, with associated references for proofs. The parameters s_0 and r_0 are the sparsity and radius of the near regions, respectively. The parameter Δ_0 is the minimal separation between two spikes in the target measure.

2.5 The sinc-4 kernel

Another important kernel introduced by De Castro et al. [2021, Section 4.1] to derive recovery guarantees for mixture models is the fourth power of the sinus cardinal kernel, coined sinc-4.

- **Sinc-4 kernel** where the sinc^4 function is defined on $\mathbb{R}^d$ as the product over coordinates of the fourth power of the univariate sinus cardinal $\sin(u)/u$, and

$$K(\mathbf{s}, \mathbf{t}) = \Psi_\tau(\mathbf{t} - \mathbf{s}) \quad \text{where} \quad \Psi_\tau(\mathbf{u}) := \text{sinc}^4\left(\frac{\mathbf{u}}{4\tau}\right), \quad \tau > 0. \quad (2.9d)$$

This kernel is particularly interesting as pivot, since it is bandlimited so that the kernel switch analysis is valid with any translation invariant kernel with large enough spectral support. However, the analysis of De Castro et al. [2021, Section 4.1] does not directly prove the LPC for this kernel. One of the important contributions of our paper is Theorem 3 which proves that this kernel indeed satisfies the LPC, yielding BLASSO guarantees for a wide range of translation-invariant kernels.

2.6 Continuous sparse regression and sketching

A key motivation for the introduction of pivot kernels is to address model kernels arising from sketched problems, which we now detail.

The general goal is to estimate a sparse target measure $\mu^0 \in \mathcal{M}_{s, \Delta, \mathfrak{D}}$ from a sketch $\mathbf{y}_{\text{sketch}} \in \mathbb{C}^m$, by solving the corresponding BLASSO program (2.1– J_κ), with the sketched forward operator F_{sketch} , whose definition involves the sketching law Λ . Thus the corresponding setting is as follows:

- *Hilbert space* $\mathcal{F}_{\text{sketch}}$: consider the standard dot product on $\mathbb{C}^m$, $\mathcal{F}_{\text{sketch}} = \mathbb{C}^m$,
- *Forward measurement* $F_{\text{sketch}} : \mathcal{M}(\mathcal{X}) \rightarrow \mathbb{C}^m$ given by

$$(F_{\text{sketch}}\mu)_i = \frac{1}{\sqrt{m}} \int_{\mathcal{X}} \varphi_{\omega_i}(\mathbf{t}) d\mu(\mathbf{t}), \quad i = 1, \dots, m, \quad (2.10a)$$

where $\varphi_{\omega} : \mathcal{X} \rightarrow \mathbb{C}$ is a complex-valued continuous function, referred to as the *sketching function*.

- *Sketching law* Λ : the sequence of *i.i.d.* random vectors $(\omega_i)_i$ is drawn with respect to Λ ,

$$\omega_i \underset{\text{i.i.d.}}{\sim} \Lambda, \quad i = 1, \dots, m. \quad (2.10b)$$

Thus, in line with Definition 1, we define the *sketched* model kernel by

$$K_{\text{sketch},\text{mod}}(\mathbf{s}, \mathbf{t}) := \left\langle F_{\text{sketch}}\delta_{\mathbf{s}}, F_{\text{sketch}}\delta_{\mathbf{t}} \right\rangle_{\mathbb{C}^m} = \frac{1}{m} \sum_{i=1}^m \varphi_{\omega_i}(\mathbf{s}) \overline{\varphi_{\omega_i}(\mathbf{t})}, \quad (2.11a)$$

where $\overline{\varphi_{\omega}}$ denotes complex conjugation. This kernel depends on the draw of the sketch, and taking its expectation naturally leads to the *population* model kernel

$$K_{\text{mod}}(\mathbf{s}, \mathbf{t}) := \mathbb{E}_{\omega \sim \Lambda} [\varphi_{\omega}(\mathbf{s}) \overline{\varphi_{\omega}(\mathbf{t})}]. \quad (2.11b)$$

Note that every draw of the sketched model kernel meets Definition 1 and satisfies the continuity assumption (2.3– $\mathbf{H}_{\text{mod}}$) with the construction (2.10). Furthermore, its limit as m tends to infinity is almost surely the population model kernel, for fixed $\mathbf{s}$ and $\mathbf{t}$.

An interesting feature of the sketched problem is to reduce the computational cost induced by the model kernel by considering an empirical version of rank m . Moreover, Poon et al. [2023] showed that the error bound analysis of the BLASSO also holds for its sketched counterpart, provided that K_{mod} satisfies the LPC assumption, under additional probabilistic assumptions on the tails of the feature function φ_{ω} and its derivatives, which are detailed later in Assumption 3.

However, proving the LPC assumption has to be done on a case-by-case basis, possibly leading to tedious calculations. Instead, we propose in this paper to *switch* to another kernel (known to satisfy the LPC) using an alternative (continuous) sketching function $\psi_{\omega} : \mathcal{X} \rightarrow \mathbb{C}$. We define the associated kernel by

$$K(\mathbf{s}, \mathbf{t}) := \mathbb{E}_{\omega \sim \Lambda} [\psi_{\omega}(\mathbf{s}) \overline{\psi_{\omega}(\mathbf{t})}], \quad (2.12-\mathbf{H}_{\psi_{\omega}, \Lambda})$$

where we assume that $K(\cdot, \cdot)$ is real valued and the expectation is taken with respect to the same distribution Λ . Given a model kernel K_{mod} , we characterize a broader class of kernels K that can be used as *pivot* in the BLASSO analysis, provided that $\mathcal{H}_{\text{mod}}$ is continuously included in their RKHS (see Figure 1). Our result offers an alternative route to obtain the controls of Equations (2.7) when considering a specific BLASSO problem.

3 Sketching mixture models via continuous sparse regression

This section focuses on the application of our general theoretical framework to the problem of mixture model estimation. We present specific consequences of the broader results that will be detailed in Section 4. The key novelties for mixture modelling include the ability of our proposed algorithm, S2Mix, to handle a wide variety of template densities beyond the commonly assumed Gaussian. Furthermore, we establish new localisation error bounds for parameter recovery, demonstrating improved scaling as the sample size increases. These advancements significantly enhance the practical utility and theoretical understanding of continuous sparse regularisation for mixture models.

3.1 A resolution-based approach to mixture modelling

We demonstrate our theoretical contributions through their application to the important problem of mixture model estimation. Mixture models represent fundamental statistical tools used across machine learning, signal processing, and computational biology to model heterogeneous populations. Before introducing the technical details, let us sketch the key ideas: we reformulate mixture estimation as a continuous sparse regression problem where the unknown mixture components—both their weights and locations—are encoded as a sparse discrete measure. This measure can then be recovered using BLASSO, even when working with compressed data through sketching techniques.

In this section, we first present a resolution-based approach to mixture modelling via continuous sparse regression. We then develop an efficient sketching algorithm, coined S2Mix, that enables parameter estimation from large datasets with near-optimal sample complexity. Crucially, we establish novel statistical guarantees for this approach by leveraging the sinc-4 kernel Ψ_τ as a pivot kernel within our theoretical framework. This demonstrates how our effective near regions analysis and kernel switch technique deliver practical improvements for an important class of statistical models.

We are interested in estimating the mixture positive weights $(a_k^0)_k$ and parameters $(\mathbf{x}_k^0)_k$ from an n -sample $\mathbf{Z} := \{\mathbf{z}_1, \dots, \mathbf{z}_n\}$ of a translation invariant mixture model with s_0 -components:

$$f^0 := \sum_{k=1}^{s_0} a_k^0 \phi(\cdot - \mathbf{x}_k^0) \quad \text{with} \quad \sum_k a_k^0 = 1 \text{ and } \mathbf{x}_k^0 \in \mathcal{X}, \quad (3.1)$$

where $\mathcal{X}$ is a compact subset of $\mathbb{R}^d$.

Here, ϕ is a probability density function on $\mathbb{R}^d$ called the *template distribution*. Popular instances include mixtures of Gaussians with fixed and known covariance, or mixtures of Laplace distributions. More generally, our proposed framework will also encompass mixtures of any distribution with known characteristic function, such as mixtures of stable distributions for instance.

From a non-parametric statistics perspective, estimating a mixture model amounts to estimating the density f^0 , which can be done via kernel density estimation (*i.e.*, smoothing the empirical sample distribution), namely

$$L_\tau \hat{f}_n := \lambda_\tau \star \hat{f}_n \quad \text{where} \quad \hat{f}_n := \frac{1}{n} \sum_{j=1}^n \delta_{\mathbf{z}_j}.$$

The *smoothing* function $\lambda_\tau(\cdot) = \tau^{-d} \lambda(\cdot/\tau)$ is defined by the smoothing kernel λ and its *width* parameter τ . In this work, we focus on the sinus cardinal smoothing kernel which is a standard choice in the literature [De Castro et al., 2021], given by

$$\lambda(\mathbf{u}) := \text{sinc}(\mathbf{u}) = \prod_{i=1}^d \frac{\sin(u_i)}{u_i} \quad \text{so that} \quad \lambda_\tau(\mathbf{x}) = \tau^{-d} \text{sinc}\left(\frac{\mathbf{x}}{\tau}\right). \quad (3.2)$$

This function acts as a low-pass filter in the Fourier domain; other filters are possible, but we will consider the sinus cardinal function in this section. The crux of non-parametric methods is then to optimise the mean quadratic loss of the estimator with respect to the width parameter τ_n , usually depending on n , the latter translating into a bias-variance trade-off.

In this work, we follow a different path considering that the width parameter τ will be fixed by the practitioner and will not depend on the sample size n but rather on the *minimal separation* between parameters $(\mathbf{x}_k^0)_k$. The difficulty of the problem is governed by the ratio of this minimal separation to τ , which situates the recovery problem with respect to the *Rayleigh limit* (which is proportional to τ). Precisely, we assume that the width $\tau > 0$ is such that

$$\tau \leq \tau_{\max}(\mu^0) := \frac{\min_{1 \leq k \neq \ell \leq s_0} \|\mathbf{x}_k^0 - \mathbf{x}_\ell^0\|_2}{147.77 s_0^{1/4} d^{7/4}}. \quad (3.3\text{--}\mathbf{H}_\tau)$$

As we will see, this hypothesis ensures that the width is small enough to resolve the mixture components. We stress that τ is fixed a priori by the practitioner as a prior scale matching the smallest separation they hope to resolve, so that (3.3-- $\mathbf{H}_\tau$) should be read as a self-consistency condition on this choice rather than an a posteriori prescription; see Remark 3.1 for a discussion of the joint constraint with C_{switch} and the resulting sample-complexity reading.

Now, consider a putative parameter θ given by

$$\theta := \{s; a_1, \dots, a_s; x_1, \dots, x_s\} \quad \text{and} \quad \mu_\theta := \sum_{k=1}^s a_k \delta_{x_k},$$

with $s \geq 1$ the number of mixing components and μ_θ is the lifting of the putative parameter on the space of measures, where δ_t denotes the Dirac mass at location $t \in \mathcal{X}$.

The mixture model density defined in Equation (3.1) can be seen as an image of a discrete measure through the convolution operator:

$$\Phi : \mu \in \mathcal{M}(\mathcal{X}) \mapsto \Phi\mu := \phi \star \mu = \int \phi(\cdot - \mathbf{x}) d\mu(\mathbf{x}) = \int \phi_{\mathbf{x}} d\mu, \quad (3.4)$$

where $\phi_{\mathbf{x}} := \phi(\cdot - \mathbf{x})$. This mapping is reminiscent of the so-called Kernel Mean Embedding of μ in Machine Learning, and they exactly correspond when the template ϕ is of positive type [Muandet et al., 2017].

The mixture problem seeks to fit a parameter θ from the observation of $\hat{f}_n$, namely

$$\theta \xrightarrow{\text{Non-linear}} \left[f_\theta := \Phi\mu_\theta = \sum_{k=1}^s a_k \phi(\cdot - t_k) \right] \xrightarrow{\text{Sampling}} \hat{f}_n.$$

where the step $\theta \rightarrow f_\theta$ is a non-linear mapping of the parameters to the mixture density. Directly solving the corresponding inverse problem is hard, both because of the non-linearity as well as the fact that those two objects do not belong to the same functional space. On the other hand, lifting the problem on the space of measures yields the following linear problem

$$\mu_\theta \xrightarrow{\text{Linear}} [f_\theta = \Phi\mu_\theta] \xrightarrow{\text{Embedding}} [L_\tau f_\theta = (L_\tau \circ \Phi)\mu_\theta] \approx L_\tau \hat{f}_n \xleftarrow{\text{Embedding}} \hat{f}_n, \quad (3.5)$$

where we have embedded the candidate $L_\tau f_\theta$ and the observation $L_\tau \hat{f}_n$ into the same Hilbert space $\mathcal{F}_\tau$, the RKHS of λ_τ .

3.2 Continuous sparse regularisation for mixture estimation – Supermix

Recent advances have established a powerful connection between continuous sparse regularisation techniques developed for super-resolution problems and the statistical challenge of mixture estimation [De Castro et al., 2021, Poon et al., 2023]. Indeed, as we have just seen, mixture modelling can be reformulated as the problem of recovering an unknown sparse discrete measure μ^0 from observations obtained through a linear measurement operator $F = L_\tau \circ \Phi$ mapping from the space of finite Radon measures $\mathcal{M}(\mathcal{X})$ to a Hilbert space $\mathcal{F}$, a possible approach to address it is to seek a (near) optimum of the BLASSO problem (2.1– J_κ).

This formulation offers the significant advantage that the number of components s_0 need not be specified in advance. Specifically, this approach involves

- *Hilbert space:* $\mathcal{F} := \mathcal{F}_\tau$, the RKHS associated to the smoothing kernel $\lambda_\tau(t - s)$ defined in (3.2). As a translation invariant reproducing kernel, it admits an explicit characterisation in the Fourier domain (see Section C.3).
- *Forward measurement operator* $F := L_\tau \circ \Phi : \mathcal{M}(\mathcal{X}) \rightarrow \mathcal{F}_\tau$.
- *Observation* $\mathbf{y} := L_\tau \hat{f}_n$ which is of the form

$$\mathbf{y} = F\mu^0 + \Gamma_n \quad \text{with} \quad \Gamma_n := L_\tau \hat{f}_n - \mathbb{E}[L_\tau \hat{f}_n],$$

the latter being a centered noise term due to n -sampling with $\mathbb{E}[L_\tau \hat{f}_n] = L_\tau f^0 = F\mu^0$.

Hence, one can use a sparse regression estimation procedure to fit the mixture model, which was coined *Supermix* by De Castro et al. [2021]. For the specific problem at hand, the model kernel defined in (2.2) is translation invariant and can be described in the Fourier domain (see Section C.3). Indeed, one has

$$\begin{aligned} K_{\text{mod}}(\mathbf{s}, \mathbf{t}) &:= \langle F\delta_{\mathbf{s}}, F\delta_{\mathbf{t}} \rangle_{\mathcal{F}} = \langle \lambda_{\tau} \star \phi_{\mathbf{s}}, \lambda_{\tau} \star \phi_{\mathbf{t}} \rangle_{\mathcal{F}_{\tau}}, \\ &= \frac{1}{(2\pi)^d} \int \mathfrak{F}[\lambda_{\tau}](\boldsymbol{\omega}) \cdot |\mathfrak{F}[\phi]|^2(\boldsymbol{\omega}) \cdot e^{+i\boldsymbol{\omega}^{\top}(\mathbf{t}-\mathbf{s})} d\boldsymbol{\omega}, \\ &= \mathfrak{F}^{-1}[\mathfrak{F}[\lambda_{\tau}] \cdot |\mathfrak{F}[\phi]|^2](\mathbf{t} - \mathbf{s}), \\ &= (\lambda_{\tau} \star \phi \star \check{\phi})(\mathbf{t} - \mathbf{s}), \end{aligned}$$

where $\check{\phi}(\mathbf{x}) = \phi(-\mathbf{x})$ and we used the fact that $\langle \lambda_{\tau} \star f, \lambda_{\tau} \star g \rangle_{\mathcal{F}_{\tau}} = (2\pi)^{-d} \langle \mathfrak{F}[f], \mathfrak{F}[\lambda_{\tau}] \overline{\mathfrak{F}[g]} \rangle_{L^2}$ for any translation-invariant kernel (see Section C.3). In the specific case of the sinc smoothing kernel considered here, $\mathfrak{F}[\lambda_{\tau}]$ is the indicator function of the hypercube $[-1/\tau, 1/\tau]^d$, which clarifies why it appears unsquared in the integrand above. At first glance, the natural road to derive estimation error bounds is trying to prove LPC (Assumption 1) for this translation invariant kernel. However, this would require controlling non-trivial quantities involving the model kernel and its associated Fisher-Rao metric. Alternatively, De Castro et al. [2021] proposed to use the sinc-4 kernel $K_{\text{pivot}}(\mathbf{s}, \mathbf{t}) = \Psi_{\tau}(\mathbf{t} - \mathbf{s})$ defined in Equation (2.9d) as a proxy to build non-degenerate certificates, implicitly relying on a *kernel switch* analysis. The latter holds under the assumption that the RKHS $\mathcal{H}_{\text{pivot}}$, defined by K_{pivot} , is continuously included in $\mathcal{H}_{\text{mod}}$, defined by K_{mod} . In the special case where both K_{pivot} and K_{mod} are translation invariant, this translates into an assumption on the boundedness of the ratio of their Fourier transforms (see e.g. Zhang and Zhao [2013, Corollary 3.2] and Proposition 4.2):

$$C_{\text{switch}} = C_{\text{switch}}(\tau, \phi) = \sup_{\boldsymbol{\omega} \in [-1/\tau, 1/\tau]^d} \sqrt{\frac{\mathfrak{F}[\Psi_{\tau}]}{|\mathfrak{F}[\lambda_{\tau}]| |\mathfrak{F}[\phi]|^2}}(\boldsymbol{\omega}) < +\infty, \quad (3.6\text{--}\mathbf{H}_{\phi})$$

with the convention $\frac{0}{0} = 0$. This is a special instance of the general results discussed in Section 4. The scaling of this constant C_{switch} with respect to the width parameter τ depends on the smoothness of $\mathfrak{F}[\phi]$ as discussed in De Castro et al. [2021, Section 5.2]. In particular, the case of *supersmooth* densities defined hereafter leads to a scaling in

$$\exists p \in [1, +\infty], \alpha, \beta > 0, \quad \mathfrak{F}[\phi](\boldsymbol{\omega}) \propto e^{-\alpha \|\boldsymbol{\omega}\|_p^{\beta}} \implies C_{\text{switch}} = \mathcal{O}_d\left(\tau^{d/2} e^{\alpha \left(\frac{d^{1/p}}{\tau}\right)^{\beta}}\right),$$

where the constant in $\mathcal{O}_d$ may depend exponentially in the dimension d , but it does not depend on τ . The parameters $p \in [1, \infty]$, $\alpha > 0$, and $\beta > 0$ characterize the smoothness of the density and are obtained directly by writing its characteristic function $\mathfrak{F}[\phi]$, which makes the constant C_{switch} completely tractable. This case encompasses e.g. Gaussian, multivariate Cauchy, product of univariate Cauchy, or more generally of centered stable distributions with known scale parameter and zero skewness. As an example, when ϕ is a standard Gaussian density, one has $p = 2, \alpha = 1/2, \beta = 2$, and the scaling is simply given by $C_{\text{switch}} = \mathcal{O}_d(\tau^{d/2} e^{d/2\tau^2})$.

Remark 3.1 (Choice of τ and sample-complexity). *Condition (3.3--H $_{\tau}$) and the supersmooth scaling of C_{switch} impose two seemingly competing constraints on the width parameter τ : the former pushes τ below the minimal separation to make the components resolvable, while the latter penalises values of τ that are too small for supersmooth templates. These two constraints should be read jointly, with τ treated as a prior resolution scale rather than a data-driven knob.*

◦ In contrast with classical kernel density estimation, τ is not optimised through an n -dependent bias-variance trade-off. Instead, it is fixed by the practitioner as an a priori lower

bound on the minimal separation one hopes to resolve: τ encodes the typical scale below which two mixture components cannot (and should not) be disentangled by the procedure.

◦ In this reading, (3.3–H $_{\tau}$) is not a prescription to be optimised over a posteriori, but a self-consistency check on the user’s prior guess: if the chosen τ is indeed compatible with the true minimal separation Δ_0 , the LPC-based certificate construction goes through and the statistical guarantees apply. If τ is chosen too aggressively (larger than allowed by (3.3–H $_{\tau}$)), nearby spikes are merged at the chosen resolution and the theorem is silent.

◦ The quantity actually appearing in the error bounds (see Proposition 3.1 and Proposition 3.2) is the product $C_{\text{switch}} \cdot \tau^{-d/2}$. For a Gaussian template, this product simplifies to $\mathcal{O}_d(e^{d/(2\tau^2)})$. Combined with the noise scaling $\gamma_n \propto \tau^{-d/2}/\sqrt{n}$ of Lemma 3.1 and the effective near-region radius $r_n = \delta_n n^{-1/4}$, the bounds are informative only when

$$n \gtrsim s_0 e^{d/\tau^2} \quad \Leftrightarrow \quad \frac{1}{\tau} \lesssim \sqrt{\frac{\log(n/s_0)}{d}},$$

i.e., the sample size needed to resolve at scale τ grows exponentially. This reflects the fundamental difficulty of deconvolving a supersmooth template at scales much smaller than its natural width.

◦ Combining both points yields a transparent recipe: the practitioner picks τ of the order of the smallest separation they hope to resolve, having in mind that the sample size must satisfy $n \gtrsim s_0 e^{d/\tau^2}$ for the bounds to be non-vacuous. This also clarifies that our framework targets the well-separated regime: the high-resolution regime $\tau \ll 1$ is not forbidden by (3.3–H $_{\tau}$), but it becomes sample-complexity prohibitive under our analysis, which is why different techniques are required there.

◦ The exponential factor is specific to supersmooth templates. For templates with only algebraic decay of $\mathfrak{F}[\phi]$ (e.g. Laplace), C_{switch} scales polynomially in $1/\tau$, and the constraint above is compatible without an exponential sample-complexity cost. The supersmooth-specific trade-off discussed here is thus a feature of templates whose characteristic function decays faster than any polynomial.

In addition, the scaling of the noise level $\gamma_n \geq \|\Gamma_n\|_{\mathcal{F}_{\tau}}$ defined in Equations (2.5) can be controlled with high probability over the draw of the sample $\mathbf{Z}$ given by the following Lemma.

Lemma 3.1 (Control of the noise level γ_n). *Let $\alpha > 0$. Then with probability at least $1 - \alpha$, it holds that*

$$\|\Gamma_n\|_{\mathcal{F}_{\tau}} \leq \gamma_n := C_{\alpha} \frac{\tau^{-\frac{d}{2}}}{\sqrt{n}}, \quad (3.7)$$

where $C_{\alpha} := 2\sqrt{1 + C_1 \log(C_2/\alpha)}$ is a constant only depending on α , while C_1 and C_2 are universal constants.

Proof. The result is proved in De Castro et al. [2021, Lemma 16] using concentration of U-process and Arcones and Giné [1993, Proposition 2.3]. A closely related proof is given for the sketched case in Lemma 3.2, see also Remark 3.7. $\square$

Thus, we can tune the regularisation κ of the BLASSO problem (2.1–J $_{\kappa}$), which is crucial for the statistical recovery result. This leads to the following proposition, which is an instance of the upcoming Theorem 1 for the Supermix problem, using the sinc-4 kernel Ψ_{τ} as pivot. It can also be seen as a refined version of De Castro et al. [2021, Theorem 10] as commented below.

Proposition 3.1 (Statistical guarantees for the Supermix problem). *Let $\mathcal{X} \subset \mathbb{R}^d$ be a compact set, and the target $\mu^0 \in \mathcal{M}(\mathcal{X})$ be a sparse measure of s_0 spikes $\{\mathbf{x}_1^0, \dots, \mathbf{x}_{s_0}^0\}$, with minimal separation $\Delta = \min_{k \neq l} \|\mathbf{x}_k^0 - \mathbf{x}_l^0\|$. Fix the width parameter $\tau \leq \tau_{\max}(\mu^0)$ as in Equation (3.3–H $_\tau$). Take any template distribution ϕ such that it achieves a finite kernel switch constant $C_{\text{switch}}(\tau, \phi) < +\infty$ in Equation (3.6–H $_\phi$). Consider the forward measurement operator $F : \mu \in \mathcal{M}(\mathcal{X}) \rightarrow \lambda_\tau \star \phi \star \mu \in \mathcal{F}_\tau$, where λ_τ is the smoothing kernel defined in Equation (3.2). Let $\mathbf{Z} = \{\mathbf{z}_1, \dots, \mathbf{z}_n\}$ be an i.i.d. sample from $\phi \star \mu^0$, and $\hat{f}_n$ its associated empirical distribution. Define the following BLASSO problem, named Supermix, with observation $\mathbf{y} = \lambda_\tau \star \hat{f}_n$*

$$\min_{\mu \in \mathcal{M}(\mathcal{X})} \left\{ J_\kappa(\mu) = \frac{1}{2} \|\mathbf{y} - F\mu\|_{\mathcal{F}_\tau}^2 + \kappa \|\mu\|_{\text{TV}} \right\}. \quad (3.8\text{--Supermix})$$

Take any diverging sequence δ_n and set the near and far regions' effective radius $r_n = \delta_n n^{-1/4}$. Fix the regularisation $\kappa = \frac{1}{\sqrt{2}C_{\text{switch}}} \frac{C_\alpha \tau^{-\frac{d}{2}}}{\sqrt{n}} \frac{1}{\sqrt{s_0}}$. Then, for any near optimal $\mu \in \mathcal{M}(\mathcal{X})$ such that $J_\kappa(\mu) \leq J_\kappa(\mu^0)$, $n \geq c_d \delta_n^4$ where c_d depends polynomially on d , and $\alpha > 0$, we have with probability $1 - \alpha$ over the drawing of $\mathbf{Z}$:

- Control of the far region:

$$|\mu|(\mathcal{F}^{\text{reg}}(r)) \leq \frac{256\sqrt{2} C_\alpha C_{\text{switch}} \tau^{-\frac{d}{2}}}{23 \delta_n^2} \sqrt{s_0}, \quad (3.9a)$$

- Control of all the near regions: for all $k \in [s_0]$,

$$\left| \mu(\mathcal{N}_k^{\text{reg}}(r)) - a_k^0 \right| \leq \frac{1536\sqrt{2} C_\alpha C_{\text{switch}} \tau^{-\frac{d}{2}}}{23 \delta_n^2} \sqrt{s_0} \quad (3.9b)$$

- Detection level: for all Borelian $A \subset \mathcal{X}$ such that $|\mu|(A) > \frac{256\sqrt{2} C_\alpha C_{\text{switch}} \tau^{-\frac{d}{2}}}{23 \delta_n^2} \sqrt{s_0}$,

$$\exists k \text{ s.t. } \min_{\mathbf{t} \in A} \frac{1}{2\sqrt{3}\tau} \|\mathbf{t} - \mathbf{x}_k^0\|_2 \leq r_n. \quad (3.9c)$$

The proof is given in Section A.7, and is an application of the main theorem of Section 4, Theorem 1. This can be adjusted with regularisation κ independent of s_0 , with the price of slightly degraded error bounds as discussed in Section 4 after Theorem 1. Moreover, this result is a digest and improvement of De Castro et al. [2021, Theorem 10] as made clear by the following remarks.

Remark 3.2 (Notational comparison with SuperMix). *When comparing with De Castro et al. [2021], the following notational equivalences are relevant:*

- The expression $1/(4\tau)$, related to inverse width (or spatial resolution) in this paper, corresponds to their notation m .
- Our noise level, $\gamma_n \geq \|\Gamma_n\|$, corresponds to their notation ρ_n .
- Our constant $C_{\text{switch}}(\tau, \phi)$ from Equation (3.6–H $_\phi$) corresponds to their notation $C_m(\varphi, \lambda)$.
- Their radius is denoted as ϵ and does not depend on n (see Remark 3.4 below).

Remark 3.3 (Model set with sinc-4 pivot). *The sinc-4 pivot kernel Ψ_τ induces the so-called Fisher-Rao distance, properly defined in Equation (4.1), which is here just a rescaled version of the Euclidean distance by a factor $2\sqrt{3}\tau$. Consequently, for a measure to belong to the model class $\mathcal{M}_{s_0, \Delta_0, \mathfrak{d}_g}$, its support points $\{\mathbf{x}_k^0\}$ must satisfy $\|\mathbf{x}_k^0 - \mathbf{x}_l^0\|_2 \geq 2\sqrt{3}\tau\Delta_0$. The parameter Δ_0 is the minimal separation inherent to the Local Positive Curvature (LPC) property of Ψ_τ , which is established for this kernel in Theorem 3. Assumption (3.3-H $_\tau$) ensures that the chosen width τ aligns the actual separations in the target measure μ^0 with this LPC-derived requirement for Ψ_τ . Since the width τ is chosen by the practitioner, this assumption means that they must choose τ small enough to resolve the mixture components, which is a standard assumption in the literature [De Castro et al., 2021].*

Remark 3.4 (r -parameterised form and tradeoffs for the near regions). *One of the novelties of Theorems 1 and 2 is that the radius $r > 0$ of the near and far regions is kept as a free parameter of the analysis. Propositions 3.1 and 3.2 specialise this radius to the noise-adaptive effective radius given by $r_n = \delta_n n^{-1/4}$, where δ_n is a user-chosen diverging sequence.*

◦ *The fully r -parameterised form underlying these specialisations remains accessible: under the assumptions of Proposition 3.1, and with the population noise level bounded by $\gamma_n \leq C_\alpha \tau^{-d/2}/\sqrt{n}$ as in Lemma 3.1, for any $r > 0$ satisfying $r < \min(r_0, \sqrt{\bar{\varepsilon}_0/\bar{\varepsilon}_2})$, the following bounds hold with probability at least $1 - \alpha$:*

$$|\mu|(\mathcal{F}^{\text{reg}}(r)) \leq \frac{256\sqrt{2}}{23} C_\alpha C_{\text{switch}} \tau^{-d/2} r^{-2} n^{-1/2} \sqrt{s_0}, \quad (3.10a)$$

$$\max_k |\mu(\mathcal{N}_k^{\text{reg}}(r)) - a_k^0| \leq \frac{1536\sqrt{2}}{23} C_\alpha C_{\text{switch}} \tau^{-d/2} r^{-2} n^{-1/2} \sqrt{s_0} + \hat{c}_\kappa C_\alpha \tau^{-d/2} n^{-1/2}. \quad (3.10b)$$

The specialisation $r = r_n = \delta_n n^{-1/4}$, together with the constraint $n \geq c_d \delta_n^4$, absorbs the r -independent bias term $\hat{c}_\kappa C_\alpha \tau^{-d/2} n^{-1/2}$ into the leading $r^{-2} n^{-1/2}$ term and recovers the bounds of Proposition 3.1. An analogous r -parameterised statement for the sketched case follows directly from Theorem 2 and underlies Proposition 3.2.

◦ *As a consequence of this specialisation, the radius of the effective near regions $r_n = \delta_n n^{-1/4}$ now depends on the number of samples, and any sequence δ_n diverging to infinity is acceptable provided that $r_n \rightarrow 0$. The sequence δ_n controls a trade-off between how fast effective near regions narrow around the true support of μ^0 at a rate $r_n = \delta_n n^{-1/4}$, and how slow the estimator $\hat{\mu}$ detects the effective near and far regions at a rate $v_n = \delta_n^{-2}$. One thus obtains different scalings of the near regions depending on the choice of the diverging sequence δ_n :*

- *Polynomial:* $\delta_n = n^\alpha$ where $0 < \alpha < 1/4$. The effective near regions scales in $n^{-1/4+\alpha}$ and $v_n = n^{-2\alpha}$.
- *Logarithmic:* $\delta_n = \sqrt{\log n}$. In this case the effective near regions narrows at the rate $(\log(n)/n)^{1/4}$ while $v_n = \log(n)$.
- *Log-Polynomial:* $\delta_n = n^\alpha \log^\beta n$ where $0 < \alpha < 1/4$ and $\beta \geq 0$. The effective near regions scales in $n^{-1/4+\alpha} \log^\beta n$ and $v_n = n^{-2\alpha} \log^{-2\beta} n$.

Remark 3.5 (Scaling of the separation condition and comparison with the literature). *The separation condition (3.3-H $_\tau$) involves a $s_0^{1/4}$ factor and a rather large numerical constant (here $147.77 = 2\sqrt{3} \cdot 42.66$ in dimension $d = 1$). A few remarks are in order regarding the scaling with s_0 and the comparison with existing results on a related model given by Fourier coefficients and the one dimensional torus.*

◦ *The $s_0^{1/4}$ factor can be traced back to the proof of Theorem 1, where the sharp control of the RKHS norm of the dual certificate scales as $\|c^0\|_{\mathcal{F}_\tau} \lesssim \sqrt{s_0}$. In the large-separation regime, this estimate is tight in the sense that each spike is isolated and the dual certificate behaves locally like*

a sinc-type bump whose total RKHS norm scales as $\sqrt{s_0}$. This precisely yields the $\sqrt{s_0}$ -factor appearing in our error bounds (3.9b) for mixtures and (2.7) in the general case.

◦ In the specific context of line spectral estimation from compressed Fourier measurements, the error bounds (2.7) match the rates of Tang et al. [2015], and our refined analysis (see Theorems 1 and 2 below) further sharpens them by a $\sqrt{s_0}$ factor in the relevant regime.

◦ As a comparison with the super-resolution literature, on the one-dimensional torus with Fourier coefficients, Candès and Fernandez-Granda [2014] proved that exact recovery from TV regularisation holds under the condition $1/(\pi\tau) \cdot \Delta \geq 8\pi$, with no dependence on s_0 . In our setting (where the signal is not periodic and we have a Fourier transform rather than Fourier coefficients), our condition reads $\Delta \geq 147.77\tau s_0^{1/4}$. This requirement is both more stringent and more general: it holds in arbitrary dimension d and for any kernel satisfying the LPC condition (Assumption 1), whereas Candès and Fernandez-Granda [2014] specifically exploit the analytic structure of the Jackson kernel (2.9a) on the torus and a coefficient-based Fourier measurement model. The large constant of 147.77 is, to the best of our understanding, a by-product of our analysis; a refined argument tailored to the sinc-4 kernel could plausibly yield a much lower bound. We leave this refinement for future work.

3.3 Statistical error bounds for *sketched* mixture models

While the preceding results are interesting from a theoretical perspective, a major practical difficulty arises when trying to solve the Supermix as the objective function J_κ , or its gradient, involves d -dimensional integrals. In this work, we focus on a *sketched* version of the Supermix BLASSO problem leading to computational gains.

Random Fourier features The main idea revolves around an approximation of the translation invariant model kernel K_{mod} using weighted random Fourier features [Rahimi and Recht, 2007, Gribonval et al., 2021a]. The latter stems from the representation of a translation invariant K_{mod} as an expectation over a free *sketch* distribution, which is then approximated by m Monte-Carlo samples. Akin to importance sampling in the Fourier domain, this *sketching distribution* Λ may be chosen different from (a properly normalized version of) the spectral measure $U_\tau := \mathfrak{F}[\lambda_\tau]/(2\pi)^d$ of the smoothing kernel. Formally, let Λ be *any p.d.f. with support containing the hypercube* $\|\omega\|_\infty \leq 1/\tau$, which is the support of U_τ . Then, the model kernel is amenable to sketching as it can be written in the form of Equation (2.11b) as follows:

$$\begin{aligned} K_{\text{mod}}(\mathbf{t}, \mathbf{s}) &= \int \overbrace{\frac{\mathfrak{F}[\lambda_\tau]}{(2\pi)^d}(\omega)}^{U_\tau} |\mathfrak{F}[\phi]|^2(\omega) e^{+i\omega^\top(\mathbf{t}-\mathbf{s})} d\omega, \\ &= \int \Lambda(\omega) \frac{U_\tau}{\Lambda}(\omega) |\mathfrak{F}[\phi]|^2(\omega) e^{+i\omega^\top(\mathbf{t}-\mathbf{s})} d\omega, \\ &= \mathbb{E}_{\omega \sim \Lambda} \left[\varphi_\omega(\mathbf{s}) \overline{\varphi_\omega(\mathbf{t})} \right], \end{aligned} \tag{3.11a}$$

$$\text{where } \varphi_\omega(\mathbf{t}) := W(\omega) \mathfrak{F}[\phi](\omega) e^{-i\omega^\top \mathbf{t}}, \quad \text{with: } W(\omega) = \sqrt{\frac{U_\tau}{\Lambda}(\omega)}. \tag{3.11b}$$

Then, this leads to a Monte-Carlo approximation of the kernel, defining the *sketched* model kernel as in (2.11a)

$$K_{\text{sketch,mod}}(\mathbf{s}, \mathbf{t}) := \frac{1}{m} \sum_{i=1}^m \varphi_{\omega_i}(\mathbf{s}) \overline{\varphi_{\omega_i}(\mathbf{t})}, \tag{3.11c}$$

where the sequence of *i.i.d.* random vectors $(\omega_i)_i$ is drawn with respect to Λ .

Sketching the supermix problem Recalling the definition of Equations (2.10), we are in presence of a sketched version of the Supermix BLASSO problem, coined S2Mix hereafter, with the following characteristics

- *Finite dimensional Hilbert space*: $\mathcal{F}_{\text{sketch}} = \mathbb{C}^m$.
- *Sketching law*: the sequence of *i.i.d.* random vectors $(\omega_i)_i$ is drawn with respect to Λ ,

$$\omega_i \underset{\text{i.i.d.}}{\sim} \Lambda, \quad i = 1, \dots, m. \quad (3.12a)$$

- *Sketched¹ forward operator* $F_{\text{sketch}} : \mathcal{M}(\mathcal{X}) \rightarrow \mathbb{C}^m$ defined as

$$F_{\text{sketch}}\mu = \frac{1}{\sqrt{m}} \left(\int \varphi_{\omega_i}(\mathbf{t}) \, \mathrm{d}\mu(\mathbf{t}) \right)_{i=1}^m. \quad (3.12b)$$

This forward sketched operator yields the sketched model kernel given by Equation (3.11c), which can in turn be viewed as a Monte-Carlo approximation of the model kernel given by Equation (3.11a).

- *Sketch vector*: $\mathbf{y}_{\text{sketch}}$, a summary of the dataset easily computed as a weighted version of the empirical distribution's Fourier transform taken at the random frequencies ω_i

$$\mathbf{y}_{\text{sketch}} := \frac{1}{\sqrt{m}} \left(W(\omega_i) \frac{1}{n} \sum_{j=1}^n e^{-i\omega_i^\top \mathbf{z}_j} \right)_{i=1}^m. \quad (3.12c)$$

The latter is of the form:

$$\mathbf{y}_{\text{sketch}} = F_{\text{sketch}}\mu^0 + \Gamma_{\text{sketch}} \quad \text{with: } \Gamma_{\text{sketch}} = \mathbf{y}_{\text{sketch}} - F_{\text{sketch}}\mu^0.$$

The noise term is centered as $\mathbb{E}_{\mathbf{Z}}[\mathbf{y}_{\text{sketch}}] = \frac{1}{\sqrt{m}} \left(W(\omega_i) \mathbb{E}_{\mathbf{z} \sim \phi \star \mu^0} [e^{-i\omega_i^\top \mathbf{z}}] \right)_{i=1}^m = F_{\text{sketch}}\mu^0$.

Hence the data sketch is an empirical version of F_{sketch} at the target point μ^0 .

Remark 3.6 (Sketching and compression). *Crucially, computing $\mathbf{y}_{\text{sketch}}$ acts as a compression of the original $n \times d$ dataset $\mathbf{Z}$ into an m -dimensional sketch in $\mathbb{C}^m$, and is done only once prior to learning. This is the essence of sketching and compressive learning methods [Gribonval et al., 2021b], seeking statistical guarantees with respect to the sketch size m .*

As in the population case, we can use concentration arguments to control the sketched noise level γ_{sketch} .

Lemma 3.2. *Let $\alpha \in (0, 1)$, $\mathbf{Z} = \{\mathbf{z}_j\}_{j=1}^n$ be an *i.i.d.* n -sample from f^0 and $\{\omega_i\}_{i=1}^m$ be an m -sample from Λ . Then, with probability at least $1 - \alpha$ over the joint draw of the sample and the sketch, it holds that*

$$\|\Gamma_{\text{sketch}}\|_{\mathbb{C}^m} \leq C_{\alpha, m} \frac{1}{\sqrt{n}},$$

with

$$C_{\alpha, m} = 2\sqrt{\left[\left(\frac{1}{\tau} \right)^d + \frac{1}{2\sqrt{m}} \left\| \frac{U_\tau}{\Lambda} \right\|_\infty \log \left(\frac{2}{\alpha} \right) \right] \left[1 + C_1 \log \left(\frac{2C_2}{\alpha} \right) \right]},$$

and $C_1, C_2 > 0$ the same universal constants as in Lemma 3.1.

¹Related to the sketching operator $\mathcal{A}$ [Gribonval et al., 2021b] which verifies $\mathbf{y}_{\text{sketch}} := \mathcal{A}\hat{f}_n \approx \mathcal{A}(\phi \star \mu^0) = F_{\text{sketch}}\mu^0$.

Remark 3.7. Note that, as m goes to infinity, we uncover that $C_{\alpha,m}$ tends to $C_{\alpha/2} \left(\frac{4}{\tau}\right)^{\frac{d}{2}}$ and we recover Lemma 3.1, which bounds the noise level in the population case.

The proof is given in Section A.8.1 and, again, we may tune the optimal regularisation κ . Applying the upcoming Theorem 2 to this specific sketched BLASSO, using the sinc-4 pivot kernel Ψ_τ , one can prove the following results for the S2Mix problem.

Proposition 3.2 (Statistical guarantees for S2Mix). *Let $\mathcal{X} \subset \mathbb{R}^d$ be a compact set, and the target $\mu^0 \in \mathcal{M}(\mathcal{X})$ be a sparse measure of s_0 spikes $\{\mathbf{x}_1^0, \dots, \mathbf{x}_{s_0}^0\}$, with minimal separation $\Delta = \min_{k \neq l} \|\mathbf{x}_k^0 - \mathbf{x}_l^0\|$. Fix the width parameter $\tau \leq \tau_{\max}(\mu^0)$ as in Equation (3.3–H $_\tau$). Take any template distribution ϕ such that it achieves a finite kernel switch constant $C_{\text{switch}}(\tau, \phi) < +\infty$ in Equation (3.6–H $_\phi$). Take any sketching distribution Λ with support containing $[-1/\tau, 1/\tau]^d$ and such that $\|U_\tau/\Lambda\|_\infty < +\infty$. Let $m \in \mathbb{N}^*$ be the sketch size and consider the following sketched BLASSO problem, denoted as sketched Supermix (S2Mix), with sketched forward measurement operator F_{sketch} and observation (data sketch) $\mathbf{y}_{\text{sketch}}$ defined in Equations (3.12b) and (3.12c) respectively:*

$$\min_{\mu} \left\{ J_{\text{sketch}, \kappa}(\mu) = \frac{1}{2} \|\mathbf{y}_{\text{sketch}} - F_{\text{sketch}}\mu\|_{\mathbb{C}^m}^2 + \kappa \|\mu\|_{\text{TV}} \right\}. \quad (3.13\text{--S2Mix})$$

Suppose the constant $C_\Lambda := \|f_\tau^{(4)}/\Lambda\|_\infty < +\infty$, where $f_\tau^{(4)} = \mathfrak{F}[\Psi_\tau]/(2\pi)^d$ is the normalized spectral measure of the sinc-4 kernel Ψ_τ (Section C.5). Let $C_d, C'_{\text{pivot}} > 0$ be constants that may depend polynomially on the dimension d . Fix the regularisation

$$\kappa = \frac{C_{\alpha,m}}{C'_{\text{pivot}} C_{\text{switch}} \sqrt{n}} \frac{1}{\sqrt{s_0}}.$$

Take any diverging sequence $\delta_n = o(n^{1/4})$ and set the near and far regions' effective radius $r_n = \delta_n n^{-1/4}$. Let $\alpha > 0$ and the sketch size m be such that

$$m \geq C_d C_\Lambda \max(d, \log(s_0)) s_0 \log \left(\frac{\max(1, |\mathcal{X}|) s_0}{\alpha} \right), \quad (3.14a)$$

Then, for any near optimal $\mu \in \mathcal{M}(\mathcal{X})$ such that $J_{\text{sketch}, \kappa}(\mu) \leq J_{\text{sketch}, \kappa}(\mu^0)$, and $n \geq c_d \delta_n^4$, where c_d depends polynomially on d , the following controls hold with probability $1 - \alpha$ over the joint draws of the sample $\mathbf{Z}$ and the sketches $(\omega_1, \dots, \omega_m)$.

- Control of the far region:

$$|\mu|(\mathcal{F}^{\text{reg}}(r)) \leq C'_{\text{pivot}} C_{\text{switch}} \frac{512}{69} C_{\alpha,m} \left(\frac{1}{\delta_n^2} \right) \sqrt{s_0}, \quad (3.14b)$$

- Control of all the near regions: for all $k \in [s_0]$,

$$|\mu(\mathcal{N}_k^{\text{reg}}(r)) - a_k^0| \leq C'_{\text{pivot}} C_{\text{switch}} \frac{1536}{69} C_{\alpha,m} \left(\frac{1}{\delta_n^2} \right) \sqrt{s_0}, \quad (3.14c)$$

- Detection level: for all Borelian $A \subset \mathcal{X}$ such that $|\mu|(A) > C'_{\text{pivot}} C_{\text{switch}} \frac{512}{69} C_{\alpha,m} \left(\frac{1}{\delta_n^2} \right) \sqrt{s_0}$, there exists $\mathbf{x}_k^0$ such that

$$\min_{t \in A} \mathfrak{d}_g(t, \mathbf{x}_k^0) \leq r. \quad (3.14d)$$

3.4 Extensions

An important consequence of the kernel switch is that one can leverage the previous results for any translation invariant mixture model beyond Gaussian, and any smoothing kernel λ_τ beyond the sinus cardinal, as long as the resulting K_{mod} verifies Equation (3.6–H $_\phi$). Since the latter is translation invariant, it suffices to show that its Fourier transform is not “too localized”. For the sinc-4 pivot, this means $\mathfrak{F}[K_{\text{mod}}]$ should contain low-frequencies and be bounded below so that C_{switch} is finite. One could also consider using another translation invariant pivot kernel; for the Gaussian pivot of Equation (2.9b) (not treated in this paper), this means $\mathfrak{F}[K_{\text{mod}}]$ has a slower decay than the Gaussian.

4 Effective near regions, kernel switch and near-optimal solutions

This section presents the core theoretical contributions of our work, establishing rigorous guarantees for the BLASSO estimator. We begin by detailing the key assumptions underpinning our analysis, notably the local positive curvature (LPC) property for pivot kernels and conditions for sketching. Our main results then demonstrate three significant advancements: first, we introduce the concept of *effective near regions*, showing that the localisation accuracy of BLASSO adapts to the noise level, improving as more data becomes available. Second, we formalize the *kernel switch* principle, which allows leveraging well-characterized pivot kernels to analyze BLASSO problems with different, potentially more complex, model kernels, provided a specific embedding condition holds. This greatly expands the range of problems amenable to theoretical guarantees. Third, we prove that these statistical error bounds apply not only to exact minimizers of the BLASSO objective but also to *near-optimal solutions*, a crucial finding for practical algorithm design where exact optimisation is often infeasible. Finally, we provide a dedicated analysis for the sinc-4 kernel, proving it satisfies the necessary LPC and sketching assumptions, thereby enabling its use as a robust pivot kernel, particularly for mixture model estimation as discussed in Section 3.

4.1 Assumptions

First, we introduce the local positive curvature assumption, which is a key property for BLASSO analysis. This assumption is crucial to establish the statistical guarantees of our results. We also define the Fisher-Rao distance, which is a Riemannian distance associated with a metric tensor denoted $\mathfrak{g}_x$ and a kernel $K(\cdot, \cdot)$.

Fisher-Rao distance for general kernels Let $K(\cdot, \cdot)$ be a real valued kernel of positive type. Define the metric tensor (identified as a psd matrix) $\mathfrak{g}_x := \nabla_1 \nabla_2 K(\mathbf{x}, \mathbf{x}) \in \mathbb{R}^{d \times d}$, where ∇_i is the gradient with respect to the i th variable. Define the Fisher-Rao distance by

$$\mathfrak{d}_{\mathfrak{g}}(\mathbf{s}, \mathbf{t}) := \inf_p \int_0^1 \sqrt{p'(u)^\top \mathfrak{g}_{p(u)} p'(u)} \, du, \quad (4.1)$$

where the infimum is taken over smooth paths $p : [0, 1] \rightarrow \mathcal{X}$ such that $p(0) = \mathbf{s}$ and $p(1) = \mathbf{t}$. The Fisher-Rao distance is thus the infimum of the length of paths connecting two points in the parameter space $\mathcal{X}$.

Remark 4.1 (Metric of translation invariant kernels). *The Fisher-Rao distance is a Riemannian distance associated with the metric tensor $\mathfrak{g}_x$ and the kernel $K(\cdot, \cdot)$. For translation invariant kernels defined by*

$$K(\mathbf{s}, \mathbf{t}) = \rho(\mathbf{s} - \mathbf{t}), \quad (4.2a)$$

the metric tensor is constant and the Fisher-Rao distance is the Mahalanobis distance given by a positive definite matrix $\mathbf{g}$ which does not depend on the points $\mathbf{x}$, it holds

$$\mathbf{g} = -\nabla^2 \rho(0), \quad (4.2b)$$

$$\mathfrak{d}_{\mathbf{g}}(\mathbf{s}, \mathbf{t}) = \sqrt{(\mathbf{s} - \mathbf{t})^\top \mathbf{g} (\mathbf{s} - \mathbf{t})}. \quad (4.2c)$$

Local positive curvature assumption We now recall the local positive curvature assumption LPC as presented in Poon et al. [2023]. The LPC property ensures that the kernel $K(\cdot, \cdot)$ behaves well in terms of its derivatives and curvature properties with respect to some Fisher-Rao distance, which is essential for the convergence of the BLASSO estimator.

Assumption 1 (LPC with parameters $s_0, \Delta_0, r_0, \bar{\varepsilon}_0$ and $\bar{\varepsilon}_2$). A real valued kernel $K(\cdot, \cdot)$ of positive type satisfies the local positive curvature assumption (LPC) if the following holds:

- Assume that, for all $0 \leq i, j \leq 2$ such that $i + j \leq 3$,

$$B_{ij} := \sup_{\mathbf{s}, \mathbf{t} \in \mathcal{X}} \|K^{(i,j)}(\mathbf{s}, \mathbf{t})\|_{\mathbf{s}, \mathbf{t}} < +\infty. \quad (4.3a)$$

and denote $B_i := 1 + B_{0i} + B_{1i}$ where $K^{(i,j)}(\mathbf{s}, \mathbf{t})$ is the covariant derivative² of order i with respect to the first variable $\mathbf{s}$ and of order j with respect to the second variable $\mathbf{t}$, and $\|\cdot\|_{\mathbf{s}, \mathbf{t}}$ denotes the operator norm with respect to the corresponding metric tensor given by $K(\cdot, \cdot)$ on the tangent spaces at $\mathbf{s}$ and $\mathbf{t}$.

- Assume that there exists $r_0 \in (0, 1/\sqrt{B_{02}})$ such that $K(\cdot, \cdot)$ has positive curvature constants defined as

$$\begin{aligned} \bar{\varepsilon}_0 &:= \frac{1}{2} \sup_{\varepsilon \geq 0} \left\{ \varepsilon : K(\mathbf{s}, \mathbf{t}) \leq 1 - \varepsilon, \forall \mathbf{s}, \mathbf{t} \in \mathcal{X} \text{ s.t. } \mathfrak{d}_{\mathbf{g}}(\mathbf{s}, \mathbf{t}) \geq r_0 \right\}, \\ \bar{\varepsilon}_2 &:= \frac{1}{4} \sup_{\varepsilon \geq 0} \left\{ \varepsilon : -K^{(0,2)}(\mathbf{s}, \mathbf{t})[\mathbf{v}, \mathbf{v}] \geq \varepsilon \|\mathbf{v}\|_{\mathbf{t}}^2, \forall \mathbf{v} \in \mathbb{T}_{\mathbf{t}}, \forall \mathbf{s}, \mathbf{t} \in \mathcal{X} \text{ s.t. } \mathfrak{d}_{\mathbf{g}}(\mathbf{s}, \mathbf{t}) < r_0 \right\}, \end{aligned} \quad (4.3b)$$

with $\mathfrak{d}_{\mathbf{g}}$ the Fisher-Rao distance associated with the metric tensor of $K(\cdot, \cdot)$ and $\mathbb{T}_{\mathbf{t}}$ the tangent space at point $\mathbf{t}$.

- Assume that there exists $s_0 \geq 2$ such that the set

$$\left\{ \Delta : 32 \sum_{l=2}^{s_0} \|K^{(i,j)}(\mathbf{x}_1, \mathbf{x}_l)\|_{\mathbf{x}_1, \mathbf{x}_l} \leq \min\left(\frac{\bar{\varepsilon}_0}{B_0}, \frac{2\bar{\varepsilon}_2}{B_2}\right), \forall (i, j) \in \{0, 1\} \times \{0, 2\}, \forall \{\mathbf{x}_l\}_{l=1}^{s_0} \in \mathcal{S}_{\Delta} \right\} \quad (4.3c)$$

is not empty, with $\mathcal{S}_{\Delta} := \{\{\mathbf{x}_l\}_{l=1}^{s_0} \in \mathcal{X}^{s_0} : \min_{k \neq l} \mathfrak{d}_{\mathbf{g}}(\mathbf{x}_k, \mathbf{x}_l) \geq \Delta\}$. Denote Δ_0 its infimum. This last condition controls the cumulative inter-spike interference through the tail sum $\sum_{l=2}^{s_0} \|K^{(i,j)}(\mathbf{x}_1, \mathbf{x}_l)\|$, and thus makes the minimal separation Δ_0 a structural feature of any LPC-based analysis: it is the technical embodiment of the well-separated regime, and not an artefact of a specific proof. In particular, the super-resolution regime $\Delta \ll \tau$ falls outside the scope of the LPC framework and requires fundamentally different techniques (see the references cited in the introduction).

In this case, the kernel $K(\cdot, \cdot)$ is said to satisfy the LPC with parameters $s_0, \Delta_0, r_0, \bar{\varepsilon}_0$ and $\bar{\varepsilon}_2$.

²We refer to [Poon et al., 2023, Section 4.1] for further details on covariant derivatives and their operator norm, and to Appendix B.1 which gives their expression for translation invariant kernels.

Pivot kernels Next, we introduce the concept of *pivot kernels*, which are a special class of kernels that can be used to analyze the BLASSO problem. Admissible model kernels are those such that the kernel switch condition (Assumption 2) defined below holds. This kernel switch allows us to leverage the properties of the pivot kernel to analyze the model kernel, which may be more complex.

The model kernel introduced in Definition 2.2 is a kernel of positive type associated to a unique Reproducing Kernel Hilbert Space (RKHS) which is denoted by $\mathcal{H}_{\text{mod}}$. One can prove that $\mathcal{H}_{\text{mod}}$ is separable and isometric to the closure in $\mathcal{F}$ of $\text{Im}(F)$, the range of F [De Castro et al., 2023, Appendix A.1],

$$(\mathcal{H}_{\text{mod}}, \|\cdot\|_{\mathcal{H}_{\text{mod}}}) \simeq (\overline{\text{Im}(F)}, \|\cdot\|_{\mathcal{F}}) \subseteq (\mathcal{F}, \|\cdot\|_{\mathcal{F}}),$$

where $\simeq$ denotes an isometry mapping.

Let $F^* : \mathcal{F} \rightarrow (\mathcal{M}(\mathcal{X}), \|\cdot\|_{\text{TV}})^*$ be the dual linear map of $F : \mathcal{M}(\mathcal{X}) \rightarrow \mathcal{F}$. One can prove [De Castro et al., 2023, Lemma A.2] that

$$F^* : c \in \mathcal{F} \mapsto \eta_c \in (\mathcal{C}(\mathcal{X}), \|\cdot\|_{\infty}) \quad (4.4a)$$

is a continuous linear operator, and

$$\langle c, F\nu \rangle_{\mathcal{F}} = \int_{\mathcal{X}} \langle c, F\delta_t \rangle_{\mathcal{F}} d\nu(t) = \langle F^*c, \nu \rangle_{\mathcal{C}(\mathcal{X}), \mathcal{M}(\mathcal{X})} = \int_{\mathcal{X}} \eta_c d\nu(t) \quad \text{for all } c \in \mathcal{F}, \nu \in \mathcal{M}(\mathcal{X}). \quad (4.4b)$$

Now, define

$$\forall c \in \mathcal{F}, \quad \eta_c : t \in \mathcal{X} \mapsto \langle c, F\delta_t \rangle_{\mathcal{F}} \in \mathbb{R}, \quad (4.4c)$$

we have the following useful proposition.

Proposition 4.1. *If (2.3–H_{mod}) holds then for all $\eta \in \mathcal{H}_{\text{mod}}$, there exists a unique $c \in \overline{\text{Im}(F)}$ such that $\|\eta\|_{\mathcal{H}_{\text{mod}}} = \|c\|_{\mathcal{F}}$ and $\eta = \eta_c$. Furthermore, it holds that $\eta = F^*c$.*

Proof. See Appendix B.3 □

We can now turn to the definition of a pivot kernel $K_{\text{pivot}}(\cdot, \cdot)$ with respect to the model kernel $K_{\text{mod}}(\cdot, \cdot)$.

Assumption 2 (Pivot kernels). *A real valued kernel $K_{\text{pivot}}(\cdot, \cdot)$ of positive type is a pivot kernel with respect to $K_{\text{mod}}(\cdot, \cdot)$ if a) its values at points $(\mathbf{t}, \mathbf{t})$ are normalized, namely*

$$\forall \mathbf{t} \in \mathcal{X}, \quad K_{\text{pivot}}(\mathbf{t}, \mathbf{t}) = 1; \quad (4.5a)$$

and b) its RKHS, denoted by $\mathcal{H}_{\text{pivot}}$, is such that $\mathcal{H}_{\text{pivot}} \subseteq \mathcal{H}_{\text{mod}}$, and it holds that the following identity map $\text{Id}_{\text{pivot}, \text{mod}} : (\mathcal{H}_{\text{pivot}}, \|\cdot\|_{\mathcal{H}_{\text{pivot}}}) \rightarrow (\mathcal{H}_{\text{mod}}, \|\cdot\|_{\mathcal{H}_{\text{mod}}})$ is continuous, namely

$$C_{\text{switch}}(K_{\text{pivot}}, K_{\text{mod}}) := \|\text{Id}_{\text{pivot}, \text{mod}}\|_{\text{op}} = \sup_{\eta \in \mathcal{H}_{\text{pivot}} \setminus \{0\}} \frac{\|\eta\|_{\mathcal{H}_{\text{mod}}}}{\|\eta\|_{\mathcal{H}_{\text{pivot}}}} < \infty. \quad (4.5b)$$

Tail bound assumption on the sketching function derivatives. The third and final assumption is a tail bound on the derivatives of the sketching function ψ_{ω} defined in Equation (2.12–H _{ψ_{ω}, Λ}), which also depends on the sketching distribution Λ . It ensures that the derivatives of the sketching function do not grow too large with respect to the Fisher-Rao distance, which is important to control the sketching error. The tail bounds are defined in terms of a so-called survival function $\mathbb{P}_j(t)$, which quantifies the probability that the derivatives of the sketching function exceed a certain threshold t .

Assumption 3 (Tail bound on the sketching function derivatives). *The function ψ_ω and the law $\omega \sim \Lambda$ satisfy the derivatives tail bounds assumption if: a) the kernel $K(\cdot, \cdot)$ defined by (2.12– $\mathbf{H}_{\psi_\omega, \Lambda}$) satisfies the local positive curvature assumption (Assumption 1) with some parameters $s_0, \Delta_0, r_0, \bar{\varepsilon}_0$ and $\bar{\varepsilon}_2$; and b) there exists a constant $C > 0$ which depends on $K(\cdot, \cdot)$ and polynomially on d such that, for all $\alpha \in (0, 1)$, there exist positive reals $L_0, L_1, L_2, L_3 > 0$ which may depend on α verifying:*

$$\sum_{j=0}^3 \bar{\mathbb{P}}_j(L_j) \leq \frac{\min(\bar{\varepsilon}_0, \bar{\varepsilon}_2, \alpha)}{m_0} \quad (4.6a)$$

$$\max_{j=0}^3 \left(L_j^2 \sum_{i=0}^3 \bar{\mathbb{P}}_i(L_i) + 6 \int_{L_j}^{\infty} t \bar{\mathbb{P}}_j(t) dt \right) \leq \frac{\min(\bar{\varepsilon}_0, \bar{\varepsilon}_2)}{m_0} \quad (4.6b)$$

$$m_0 := C s_0 \left(C_1 \log(s_0) \log\left(\frac{s_0}{\alpha}\right) + C_2 \log\left(\frac{(s_0 N)^d}{\alpha}\right) \right) \quad (4.6c)$$

where

$$\bar{\mathbb{P}}_j(t) := 1 - \mathbb{P}_{\omega \sim \Lambda} \left(\sup_{\mathbf{x} \in \mathcal{X}} \|\psi_\omega^{(j)}(\mathbf{x})\|_{\mathbf{x}} \leq t \right) \quad (4.6d)$$

$$N := \frac{|\mathcal{X}| L_1}{\bar{\varepsilon}_0} + \frac{r_0 L_3 L_0 + L_2}{\bar{\varepsilon}_2}$$

$$C_1 := (L_0^2 + L_1^2)(B_0/\bar{\varepsilon}_0^2 + B_2/\bar{\varepsilon}_2^2)$$

$$C_2 := L_0^2/\bar{\varepsilon}_0^2 + L_{01} L_0/\bar{\varepsilon}_0 + L_2^2/\bar{\varepsilon}_2^2 + L_{01} L_2/\bar{\varepsilon}_2$$

$$L_{ij} := \sqrt{L_i^2 + L_j^2}$$

$$|\mathcal{X}| := \sup_{\mathbf{s}, \mathbf{t} \in \mathcal{X}} \mathfrak{d}_{\mathfrak{g}}(\mathbf{s}, \mathbf{t})$$

with $\psi_\omega^{(j)}(\mathbf{x})$ the covariant derivative of order j at $\mathbf{x}$, $\mathfrak{d}_{\mathfrak{g}}(\mathbf{s}, \mathbf{t})$ the Fisher-Rao distance, $\|\cdot\|_{\mathbf{x}}$ the norm of the corresponding metric tensor given by $K(\cdot, \cdot)$ on the tangent space at $\mathbf{x}$.

4.2 Main theorem

Theorem 1 (Estimation error bounds for BLASSO). *Let $\mathcal{X}$ be a compact set of $\mathbb{R}^d$, let $\mathcal{F}$ be a separable Hilbert space, and let $K_{\text{pivot}}(\cdot, \cdot)$ be a kernel of positive type which satisfies Assumption 1 with parameters s_0 , Δ_0 , r_0 , $\bar{\varepsilon}_0$ and $\bar{\varepsilon}_2$, and Assumption 2 with constant $C_{\text{switch}} < +\infty$.*

Let $F : \mathcal{M}(\mathcal{X}) \rightarrow \mathcal{F}$ be a linear map and let $\mu^0 \in \mathcal{M}(\mathcal{X})$, $\mathbf{y} \in \mathcal{F}$ and $\gamma \geq 0$ be such that

$$\mu^0 \in \mathcal{M}_{s_0, \Delta_0, \mathfrak{d}_{\mathfrak{g}}} \quad \text{and} \quad \gamma \geq \|\mathbf{y} - F\mu^0\|_{\mathcal{F}},$$

with $\mathfrak{d}_{\mathfrak{g}}$ the Fisher-Rao distance associated to $K_{\text{pivot}}(\cdot, \cdot)$.

Consider J_{κ} given by (2.1- J_{κ}) where $\kappa = c_{\kappa}\gamma/\sqrt{s_0}$ with $c_{\kappa} > 0$ some constant, and consider $\mu \in \mathcal{M}(\mathcal{X})$ be such that

$$J_{\kappa}(\mu) \leq J_{\kappa}(\mu^0).$$

Then for each $r > 0$ such that $r < \min(r_0, \sqrt{\bar{\varepsilon}_0/\bar{\varepsilon}_2})$, we have:

- *Control of the far region:*

$$|\mu|(\mathcal{F}^{\text{reg}}(r)) \leq \bar{c}_{\kappa} \left(\frac{\gamma}{\bar{\varepsilon}_2 r^2} \right) \sqrt{s_0}, \quad (4.7a)$$

- *Control of all the near regions: for all $k \in [s_0]$,*

$$|\mu(\mathcal{N}_k^{\text{reg}}(r)) - a_k^0| \leq \tilde{c}_{\kappa} \left(\frac{\gamma}{\bar{\varepsilon}_2 r^2} \right) \sqrt{s_0} + \hat{c}_{\kappa} \gamma, \quad (4.7b)$$

- *Detection level: for all Borelian $A \subset \mathcal{X}$ such that $|\mu|(A) > \bar{c}_{\kappa} \left(\frac{\gamma}{\bar{\varepsilon}_2 r^2} \right) \sqrt{s_0}$, there exists $\mathbf{x}_k^0$ such that*

$$\min_{t \in A} \mathfrak{d}_{\mathfrak{g}}(t, \mathbf{x}_k^0) \leq r, \quad (4.7c)$$

where the far and near regions are defined by Definition 3 and

$$\bar{c}_{\kappa} := \frac{(1 + \sqrt{2} C_{\text{switch}} c_{\kappa})^2}{2c_{\kappa}} \geq 2\sqrt{2} C_{\text{switch}} \quad (4.7d)$$

$$\tilde{c}_{\kappa} = \bar{c}_{\kappa} \max(1, \bar{\varepsilon}_0) \quad (4.7e)$$

$$\hat{c}_{\kappa} = 2\sqrt{2} C_{\text{switch}} (1 + \sqrt{2} C_{\text{switch}} c_{\kappa}) \quad (4.7f)$$

The proof is given in Appendix A.5. In the statement above, one can see that the choice of the regularisation parameter is $\kappa \propto \gamma/\sqrt{s_0}$ up to a free constant c_{κ} . The selection of c_{κ} influences the constants in the error bounds and their dependency on s_0 . We discuss several choices below:

Remark 4.2 (Tuning the regularisation parameter $\kappa = c_{\kappa}\gamma/\sqrt{s_0}$). *The constant c_{κ} in the definition of the regularisation parameter κ affects the pre-factors in the error bounds (4.7a), (4.7b), and (4.7c).*

1. **Optimal scaling with $\sqrt{s_0}$:** *Choosing $c_{\kappa} = 1/(\sqrt{2}C_{\text{switch}})$ minimizes the constant $\bar{c}_{\kappa}$ to its lower bound $2\sqrt{2}C_{\text{switch}}$. This yields error bounds that scale as $\mathcal{O}(\gamma\sqrt{s_0}/(\bar{\varepsilon}_2 r^2))$ for the far region control and the leading term in the near region control. This is generally the preferred choice for optimal statistical rates.*
2. **Alternative scaling with s_0 :** *If one chooses c_{κ} such that κ becomes independent of s_0 (e.g., by setting $c_{\kappa} = \sqrt{s_0}$), then $\kappa \propto \gamma$. This leads to $\bar{c}_{\kappa} \propto s_0$, degrading the error bounds to scale as $\mathcal{O}(\gamma s_0/(\bar{\varepsilon}_2 r^2))$. While this might simplify the expression for κ , it thus results in suboptimal statistical guarantees.*

3. **Caution on noise level dependency:** It is crucial that κ maintains its proportionality to the noise level γ . For instance, attempting to make κ independent of γ (e.g., by setting $c_\kappa \propto 1/\gamma$) would lead to error bounds with a constant bias term, such as $\mathcal{O}((\text{constant} + C_{\text{switch}})\sqrt{s_0})$, which does not vanish as $\gamma \rightarrow 0$. This is undesirable in statistical applications where γ typically decreases with increasing sample size (e.g., mixture model estimation, see Section 3 and Proposition 3.1).

4.3 New guarantees for sketched BLASSO

We now consider a sketched BLASSO given by a forward operator F as in (2.10a) with sketching function $\varphi_\omega(\cdot)$, whose model kernel is given by (2.11b).

Theorem 2 (Estimation error bounds for sketched BLASSO). *Let $\mathcal{X}$ be a compact set of $\mathbb{R}^d$. Let $\varphi_\omega, \psi_\omega : \mathcal{X} \rightarrow \mathbb{C}$ be functions and let Λ be a law on $\mathbb{R}^d$. Consider*

$$\begin{aligned} K_{\text{mod}}(\mathbf{s}, \mathbf{t}) &= \mathbb{E}_{\omega \sim \Lambda} [\varphi_\omega(\mathbf{s}) \overline{\varphi_\omega(\mathbf{t})}], \\ K_{\text{pivot}}(\mathbf{s}, \mathbf{t}) &= \mathbb{E}_{\omega \sim \Lambda} [\psi_\omega(\mathbf{s}) \overline{\psi_\omega(\mathbf{t})}]. \end{aligned} \quad (\mathbf{H}_{\psi_\omega, \Lambda})$$

Assume that $K_{\text{pivot}}(\cdot, \cdot)$ satisfies Assumption 1 (with parameters $s_0, \Delta_0, r_0, \bar{\varepsilon}_0$ and $\bar{\varepsilon}_2$) and Assumption 2 with constant $C_{\text{switch}} > 0$. Assume that the sketching function ψ_ω satisfies Assumption 3 and that its derivatives are Λ -almost surely bounded up to order 3. Denote $C_{\text{sketch}} := 2C \max(C_1, C_2) > 0$.

Let $\alpha \in (0, 1)$ and m be such that

$$m \geq C_{\text{sketch}} \max(d, \log(s_0)) s_0 \log\left(\frac{\max(1, |\mathcal{X}|) s_0}{\alpha}\right), \quad (4.8)$$

where $|\mathcal{X}| = \sup_{\mathbf{s}, \mathbf{t} \in \mathcal{X}} \mathfrak{d}_{\mathbf{g}}(\mathbf{s}, \mathbf{t})$ is the diameter of $\mathcal{X}$ with respect to the Fisher-Rao distance $\mathfrak{d}_{\mathbf{g}}$ given by $K_{\text{pivot}}(\cdot, \cdot)$.

Let $F : \mathcal{M}(\mathcal{X}) \rightarrow \mathbb{C}^m$ be given by

$$(F\mu)_i = \frac{1}{\sqrt{m}} \int_{\mathcal{X}} \varphi_{\omega_i}(t) d\mu(t), \quad i = 1, \dots, m.$$

where the sequence of i.i.d. random vectors $(\omega_i)_i$ is drawn with respect to Λ . Let $\mu^0 \in \mathcal{M}(\mathcal{X})$, $\mathbf{y}_{\text{sketch}} \in \mathbb{C}^m$ and $\gamma \geq 0$ be such that

$$\mu^0 \in \mathcal{M}_{s_0, \Delta_0, \mathfrak{d}_{\mathbf{g}}} \quad \text{and} \quad \gamma \geq \|\mathbf{y}_{\text{sketch}} - F\mu^0\|_{\mathbb{C}^m},$$

Consider J_κ given by (2.1- J_κ) where $\kappa = c_\kappa \gamma / \sqrt{s_0}$ with $c_\kappa > 0$ some constant, and let $\mu \in \mathcal{M}(\mathcal{X})$ be such that

$$J_\kappa(\mu) \leq J_\kappa(\mu^0).$$

Then, there exists a constant $C'_{\text{pivot}} > 0$ which depends only on the kernel K_{pivot} such that, with probability at least $1 - \alpha$ on the draw of $(\omega_1, \dots, \omega_m)$, for any $r > 0$ such that $r < \min(r_0, \sqrt{\bar{\varepsilon}_0/6\bar{\varepsilon}_2})$, we have:

- Control of the far region:

$$|\mu|(\mathcal{F}^{\text{reg}}(r)) \leq \bar{c}_\kappa \left(\frac{2\gamma}{3\bar{\varepsilon}_2 r^2} \right) \sqrt{s_0}, \quad (4.9a)$$

- Control of all the near regions: for all $k \in [s_0]$,

$$|\mu(\mathcal{N}_k^{\text{reg}}(r)) - a_k^0| \leq \bar{c}_\kappa \left(\frac{2\gamma}{3\bar{\varepsilon}_2 r^2} \right) \sqrt{s_0} + \hat{c}_\kappa \gamma, \quad (4.9b)$$

- *Detection level:* for all Borelian $A \subset \mathcal{X}$ such that $|\mu|(A) > \bar{c}_\kappa(\frac{2\gamma}{3\bar{\varepsilon}_2 r^2})\sqrt{s_0}$, there exists $\mathbf{x}_k^0$ such that

$$\min_{t \in A} \mathfrak{d}_{\mathfrak{g}}(t, \mathbf{x}_k^0) \leq r, \quad (4.9c)$$

where the far and near regions are defined by Definition 3 and

$$\bar{c}_\kappa := \frac{(1 + C'_{\text{pivot}} C_{\text{switch}} c_\kappa)^2}{2c_\kappa} \geq 2C'_{\text{pivot}} C_{\text{switch}} \quad (4.9d)$$

$$\tilde{c}_\kappa = \bar{c}_\kappa \max(1, \bar{\varepsilon}_0/4) \quad (4.9e)$$

$$\hat{c}_\kappa = 2C'_{\text{pivot}} C_{\text{switch}} (1 + C'_{\text{pivot}} C_{\text{switch}} c_\kappa) \quad (4.9f)$$

The proof is given in Appendix A.6.

Remark 4.3. The sketch size (4.8) presented in the statement of Theorem 2 is given for pivot sketching functions ψ_ω with Λ -almost surely bounded derivatives up to the order 3. One could weaken this hypothesis and only ask that the survival function $\bar{\mathbb{P}}_j(t)$ defined in Assumption 3 decays exponentially, which would incur additional logarithmic terms in s_0 and d in the sketch size (4.8).

Remark 4.4. For any $m \geq 1$, one can set

$$\alpha = \max(1, |\mathcal{X}|) s_0 \exp\left(-\frac{m}{C_{\text{sketch}} \max(d, \log(s_0)) s_0}\right)$$

and the result of Theorem 2 holds with probability $1 - \min(1, \alpha)$. We uncover that $m = \mathcal{O}(s_0)$ (up to multiplicative log factors in s_0 and multiplicative polynomial factors in d) are sufficient to get accurate statistical estimation error bounds.

Remark 4.5. While our theoretical results are established for real-valued finite signed Radon measures, the analysis and LPC properties generalise straightforwardly to complex-valued measures via an explicit real and imaginary decomposition. Specifically, one can apply the LPC-based guarantees to both coordinates under the general condition that the underlying model kernel is conjugate-symmetric.

4.4 The sinc-4 pivot: LPC and sketching

We consider the sinc-4 pivot $\Psi_\tau(\mathbf{x} - \mathbf{y}) = \text{sinc}^4(\frac{\mathbf{x} - \mathbf{y}}{4\tau})$ defined in Equation (2.9d). This kernel has been studied in De Castro et al. [2021] without proving that the LPC holds. Since the theoretical analysis of the sketched BLASSO is built upon LPC [Poon et al., 2023], the statistical error bounds of sketched BLASSO with the sinc-4 pivot was an open question. The next theorem shows that the sinc-4 kernel Ψ_τ satisfies Assumption 1. Note that its Fisher-Rao distance is

$$\mathfrak{d}_{\mathfrak{g}, \tau}(\mathbf{x}, \mathbf{y}) = \frac{1}{2\sqrt{3}\tau} \|\mathbf{x} - \mathbf{y}\|_2, \quad (4.10a)$$

which scales as τ^{-1} .

Theorem 3. Given any $\tau > 0$ and $s_0 \geq 1$, the kernel $\Psi_\tau(\mathbf{x} - \mathbf{y}) = \text{sinc}^4(\frac{\mathbf{x} - \mathbf{y}}{4\tau})$ on $\mathcal{X} = \mathbb{R}^d$ satisfies Assumption 1 with parameters s_0 , $\Delta_0 = 42.66 s_0^{1/4} d^{7/4}$, $r_0 = 1/(4d)$, $\bar{\varepsilon}_0 \geq 1/(32d^3)$ and $\bar{\varepsilon}_2 \geq 23/128$.

The proof is given in Appendix A.3. Given a set of distinct points $\{\mathbf{x}_1^0, \dots, \mathbf{x}_{s_0}^0\} \subset \mathbb{R}^d$, if

$$\tau \leq \frac{\min_{k,l} \|\mathbf{x}_k^0 - \mathbf{x}_l^0\|_2}{147.77 s_0^{1/4} d^{7/4}} \quad (4.10b)$$

then any measure supported on these points is a sparse target measure (2.4) for which $\Delta_0 \leq 42.66 s_0^{1/4} d^{7/4}$ and hence the LPC holds with $r_0 = 1/(4d)$, $\bar{\varepsilon}_0 \geq 1/(32d^3)$ and $\bar{\varepsilon}_2 \geq 23/128$. Condition (4.10b) is exactly (3.3–H_τ).

We now turn to the proof of Assumption 3 for the sinc-4 kernel. As a translation invariant kernel, we denote $f_\tau^{(4)}$ its spectral measure, detailed in Section C.5 and such that

$$\Psi_\tau(\mathbf{x} - \mathbf{y}) = \int e^{i\omega^\top(\mathbf{x}-\mathbf{y})} f_\tau^{(4)}(\omega) d\omega.$$

Since $\Psi_\tau(\mathbf{0}) = 1$ and Ψ_τ is a positive type kernel, $f_\tau^{(4)}$ is a probability density function. Thus, for any sketching distribution Λ , we may write the latter as an expectation in the form of Equation (2.12–H_{ψ_ω,Λ})

$$\Psi_\tau(\mathbf{x} - \mathbf{y}) = \mathbb{E}_{\omega \sim \Lambda} [\psi_\omega(\mathbf{x}) \bar{\psi}_\omega(\mathbf{y})], \quad \text{with sketching function } \psi_\omega(\mathbf{t}) = e^{i\omega^\top \mathbf{t}} \sqrt{\frac{f_\tau^{(4)}}{\Lambda}}(\omega).$$

In order to prove Assumption 3 we need to exhibit $L_0, L_1, L_2, L_3 > 0$ such that Equations (4.6) is verified for the LPC constants of Theorem 3. This is the purpose of the following theorem.

Theorem 4. *For any width τ , the sinc-4 kernel $\Psi_\tau(\mathbf{x} - \mathbf{y}) = \text{sinc}^4(\frac{\mathbf{x}-\mathbf{y}}{4\tau})$ and its sketching function ψ_τ verify Assumption 3 with:*

$$m_0 = C s_0 \left(C_1 \log(s_0) \log\left(\frac{s_0}{\alpha}\right) + C_2 \log\left(\frac{(s_0 N)^d}{\alpha}\right) \right) \quad (4.10c)$$

$$N := |\mathcal{X}| 32\sqrt{12}\sqrt{C_\Lambda} d^{7/2} + \frac{128}{23} (12\sqrt{12} C_\Lambda d^{1/2} + \sqrt{C_\Lambda} 12d) \quad (4.10d)$$

$$C_1 := (1 + 12d) C_\Lambda \left(1024d^6 (2 + \sqrt{12d}) + \frac{128^2}{23^2} (1 + \sqrt{12d} + 12d) \right) = \mathcal{O}(d^{15/2} C_\Lambda)$$

$$C_2 := C_\Lambda \left(1024d^6 + 32d^3 \sqrt{1 + 12d} + \frac{128^2}{23^2} 12^2 d^2 + \frac{128}{23} 12d \sqrt{1 + 12d} \right) = \mathcal{O}(d^6 C_\Lambda)$$

$$C_\Lambda := \sup_{\omega \in [-\frac{1}{\tau}, \frac{1}{\tau}]^d} \frac{f_\tau^{(4)}}{\Lambda}(\omega)$$

where $|\mathcal{X}| = \sup_{\mathbf{s}, \mathbf{t} \in \mathcal{X}} \mathfrak{d}_g(\mathbf{s}, \mathbf{t})$ and C depends on Ψ_τ and polynomially on d . Moreover, derivatives of ψ_τ are almost surely bounded at any order.

The proof is given in Appendix A.4.

Lastly, the following proposition highlights the flexibility of the sinc-4 kernel as pivot for the analysis of BLASSO problems involving translation-invariant kernels with sufficiently slow-decaying spectral measure.

Proposition 4.2 (The sinc-4 pivot for translation-invariant kernels). *Let $K_{\text{mod}}(\mathbf{s}, \mathbf{t}) = \rho_{\text{mod}}(\mathbf{s} - \mathbf{t})$ be a translation-invariant reproducing kernel with a non-zero Fourier transform on low frequencies $\mathbb{B}_\tau = [-1/\tau, 1/\tau]^d$. Then, we have that*

$$C_{\text{switch}}(\Psi_\tau, \rho) = \text{ess sup}_{\omega \in \mathbb{B}_\tau} \sqrt{\frac{\mathfrak{F}[\Psi_\tau]}{\mathfrak{F}[\rho_{\text{mod}}]}}(\omega) \leq \frac{\text{ess sup}_{\omega \in \mathbb{B}_\tau} \sqrt{\mathfrak{F}[\Psi_\tau]}(\omega)}{\text{ess inf}_{\omega \in \mathbb{B}_\tau} \sqrt{\mathfrak{F}[\rho_{\text{mod}}]}(\omega)}.$$

and if $\mathfrak{F}[\rho_{\text{mod}}]$ is bounded away from 0 almost everywhere on low-frequencies, i.e. $\text{ess inf}_{\mathbb{B}_\tau} \mathfrak{F}[\rho_{\text{mod}}] > 0$, then $C_{\text{switch}}(\Psi_\tau, \rho) < +\infty$ and $\mathcal{H}_{\text{sinc-4}}$ continuously embeds into $\mathcal{H}_\rho$.

The proof is given in Section C.7 and is a consequence of Zhang and Zhao [2013, Corollary 3.2]. For the smoothed mixture models considered in Section 3, the model kernel is $\rho_\phi = \lambda_\tau \star \phi \star \check{\phi}$ whose Fourier transform $\mathfrak{F}[\rho_\phi] = \mathfrak{F}[\lambda_\tau]|\mathfrak{F}[\phi]|^2$ contains low-frequencies. This yields the kernel switch constant of Equation (3.6–H_ϕ)

$$C_{\text{switch}}(\Psi_\tau, \rho_\phi) = C_{\text{switch}}(\tau, \phi) \leq \frac{\text{ess sup}_{\omega \in \mathbb{B}_\tau} \sqrt{\mathfrak{F}[\Psi_\tau](\omega)}}{\text{ess inf}_{\omega \in \mathbb{B}_\tau} \sqrt{\mathfrak{F}[\lambda_\tau]|\mathfrak{F}[\phi]|^2(\omega)}}, \quad (4.11)$$

which is finite when $\text{ess inf } \mathfrak{F}[\rho_\phi] > 0$. In this work, we use the rescaled sinus cardinal as λ_τ , which has a *constant* Fourier transform $\mathfrak{F}[\lambda_\tau] \propto \mathbb{1}_{\mathbb{B}_\tau}$. Thus, any template distribution ϕ such that $|\mathfrak{F}[\phi]|$ is bounded away from 0 on $\mathbb{B}_\tau$ is amenable to the kernel switch in Proposition 3.1, and the scaling of the constant $C_{\text{switch}}(\tau, \phi)$ depends on the smoothness of $\mathfrak{F}[\phi]$.

5 Conclusion and perspectives

This paper advances the theory of continuous sparse regularisation on measures, providing novel theoretical guarantees for sketched problems, including mixture models. We introduce a key embedding constant, C_{switch} , which we explicitly characterize for translation-invariant kernels via a ratio of Fourier transforms. Our analysis further reveals how the support of the solution belongs to so-called effective near regions that scale with the noise level and establishes formal error bounds for near-optimal solutions, validating them as reliable estimators in practice.

These contributions open promising avenues for future research. Theoretically, a natural extension is to exhibit the kernel switch constant for non-translation invariant models, such as Gaussian mixtures with unknown covariance. Methodologically, our work motivates the development of practical conic particle gradient descent algorithms for the BLASSO. This pursuit will also require addressing the crucial challenge of selecting the regularisation parameter κ and the sketch size in a data-driven manner, especially when the true number of components is unknown.

Finally, we note that our results are not limited to mixture models but can be applied to any continuous regression problem where some pivot kernel satisfies the LPC. This includes a wide range of applications in machine learning and statistics, such as density estimation, clustering, and regression tasks on continuous data (*e.g.*, tensor regression [Azaïs et al., 2024], shallow neural networks [Chizat, 2019, Azaïs et al., 2024]).

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A Proofs of the main results

A.1 Existence of pivot non-degenerate dual certificates

The crux of the statistical analysis lies in the construction of a *non-degenerate dual certificate* η^0 , i.e. a sub-gradient of the TV-norm interpolating $\text{sign}(a_k^0) = \pm 1$ at the support point t^k of μ^0 , and with controlled behaviour in the near and far regions, see Figure 3. Both existence and construction of such objects have been the focus of a lot of attention in the continuous sparse recovery literature, see for example [Candès and Fernandez-Granda, 2014, Poon et al., 2023, De Castro et al., 2021].

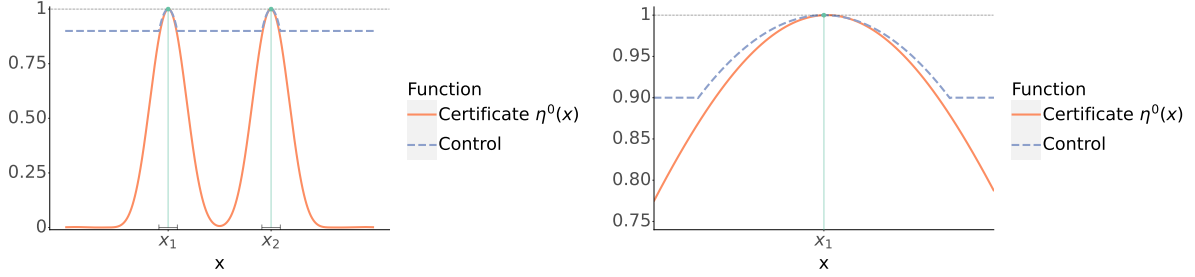


Figure 3: On the left, a one-dimensional illustration of the non-degenerate dual certificate η^0 in the near regions and far regions for some target μ^0 with two spikes x_1 and x_2 , shown in light green. On the right, a zoomed location of the curvature control in the near region $\mathcal{N}_1^{\text{reg}}(r)$ around t_1 . The certificates are drawn in orange while blue dotted lines illustrate the required control.

Definition 4 (Pivot non-degenerate certificates). *Let $K_{\text{pivot}}(\cdot, \cdot) : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ be a pivot kernel and let $\mu^0 \in \mathcal{M}_{s_0, \Delta_0, \mathfrak{d}_g}$ where the Fisher-Rao distance is given by $K_{\text{pivot}}(\cdot, \cdot)$.*

We say that η^0 is an $(\varepsilon_0, \varepsilon_2, r_0)$ -pivot non-degenerate dual certificate if

- *There exists $c^0 \in \mathcal{F}$ such that $\eta^0 = F^* c^0$,*
- *For all $i \in [s_0]$, $\eta^0(t_i^0) = \text{sign}(a_i^0)$,*
- *For all $x \in \mathcal{F}^{\text{reg}}(r_0)$, $|\eta^0(x)| \leq 1 - \varepsilon_0$,*
- *For all $i \in [s_0]$, for all $x \in \mathcal{N}_i^{\text{reg}}(r_0)$, $|\eta^0(x)| \leq 1 - \varepsilon_2 \mathfrak{d}_g(x, t_i^0)^2$,*

where $\mathfrak{d}_g$ is the Fisher-Rao distance associated to $K_{\text{pivot}}(\cdot, \cdot)$, and the near and far regions are those of $(t_i^0)_{i=1}^{s_0}$.

For all $i \in [s_0]$, we say that η_i^0 is an $(\varepsilon_0, \varepsilon_2, r_0)$ -pivot non-degenerate localizing certificate at point t_i^0 if

- *There exists $c_i^0 \in \mathcal{F}$ such that $\eta_i^0 = F^* c_i^0$,*
- *For all $x \in \mathcal{F}^{\text{reg}}(r_0)$, $|\eta_i^0(x)| \leq 1 - \varepsilon_0$,*
- *For all $j \in [s_0]$, for all $x \in \mathcal{N}_j^{\text{reg}}(r_0)$, $|\delta_{ij} - \eta_i^0(x)| \leq \varepsilon_2 \mathfrak{d}_g(x, t_j^0)^2$,*

where $\delta_{ij} = 1$ if $i = j$ and 0 otherwise.

Theorem 5 (Existence of pivot non-degenerate dual certificates). *Let $\mathcal{X}$ be a compact set of $\mathbb{R}^d$, let $\mathcal{F}$ be a separable Hilbert space, and let $K_{\text{pivot}}(\cdot, \cdot)$ be a positive type kernel that satisfies Assumption 1 with parameters $s_0, \Delta_0, r_0, \bar{\varepsilon}_0$ and $\bar{\varepsilon}_2$, and Assumption 2 with constant $C_{\text{switch}} > 0$ with respect to some model kernel $K_{\text{mod}}(\cdot, \cdot)$ which satisfies (2.3–H_{mod}).*

Then there exist an $(\bar{\varepsilon}_0, \bar{\varepsilon}_2, r_0)$ -pivot non-degenerate dual certificate such that $\|c^0\|_{\mathcal{F}} \leq \sqrt{2}C_{\text{switch}}\sqrt{s_0}$ and $(\bar{\varepsilon}_0, \bar{\varepsilon}_2, r_0)$ -pivot non-degenerate localizing certificates at points t_i^0 such that $\|c_i^0\|_{\mathcal{F}} \leq \sqrt{2}C_{\text{switch}}$.

Proof. This proof is an adaptation of the proof of [Poon et al., 2023, Theorem 2, Page 266] for the pivot kernel. Denote by $f_{\text{pivot}}(x) := K_{\text{pivot}}(x, \cdot)$ the canonical feature map of $\mathbb{H}_{\text{pivot}}$.

Consider some coefficients $(\alpha_1, \alpha_2) \in \mathbb{R}^{s_0} \times \mathbb{R}^{ds_0}$ such that

$$\eta^0(x) := \sum_{j=1}^{s_0} \alpha_{1,j} K_{\text{pivot}}(\mathbf{x}_j^0, x) + \langle \alpha_{2,j}, \nabla_1 K_{\text{pivot}}(\mathbf{x}_j^0, x) \rangle_{\mathbb{R}^d} \quad (\text{A.1})$$

satisfies $\nabla \eta^0(\mathbf{x}_j^0) = 0$ and $\eta^0(\mathbf{x}_j^0) = \text{sign}(a_j^0)$, for $j \in [s_0]$. These $s_0(d+1)$ constraints can be written as a linear system

$$\Upsilon \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} = \begin{pmatrix} (\text{sign}(a_j^0))_{j=1}^{s_0} \\ \mathbf{0}_{s_0 d} \end{pmatrix} =: \mathbf{u},$$

where $\Upsilon \in \mathbb{R}^{ds_0 \times ds_0}$ is a real symmetric matrix defined as

$$\Upsilon = \begin{pmatrix} (K_{\text{pivot}}(\mathbf{x}_i^0, \mathbf{x}_j^0))_{1 \leq i, j \leq s_0} & (\nabla_2 K_{\text{pivot}}(\mathbf{x}_i^0, \mathbf{x}_j^0)^\top)_{1 \leq i, j \leq s_0} \\ (\nabla_1 K_{\text{pivot}}(\mathbf{x}_i^0, \mathbf{x}_j^0))_{1 \leq i, j \leq s_0} & (\nabla_1 \nabla_2 K_{\text{pivot}}(\mathbf{x}_i^0, \mathbf{x}_j^0))_{1 \leq i, j \leq s_0} \end{pmatrix}.$$

Then, define

$$\mathbf{f}(x) := \left((K_{\text{pivot}}(\mathbf{x}_i^0, x))_{i=1}^{s_0}, (\nabla_1 K_{\text{pivot}}(\mathbf{x}_i^0, x)^\top)_{i=1}^{s_0} \right)^\top \in \mathbb{R}^{s_0(d+1)}$$

Since the mixed partial derivatives $\partial_i \partial_j K_{\text{pivot}}$ exist and are continuous, Steinwart and Christmann [2008, Lemma 4.34] ensures that the feature map $f_{\text{pivot}} : \mathbb{R}^d \rightarrow \mathcal{H}_{\text{pivot}}$ is continuously differentiable and we denote by $\nabla f_{\text{pivot}}(\mathbf{s}) = (\partial_i f_{\text{pivot}}(\mathbf{s}))_{i=1}^d \in \mathcal{H}_{\text{pivot}}^d$ the vector of partial derivatives, the latter verifying: $\langle \partial_i f_{\text{pivot}}(\mathbf{s}), \partial_j f_{\text{pivot}}(\mathbf{t}) \rangle_{\mathcal{H}_{\text{pivot}}} = \partial_i \partial_j K_{\text{pivot}}(\mathbf{s}, \mathbf{t})$.

We can thus define $\mathbf{f} = \left((f_{\text{pivot}}(\mathbf{x}_i^0))_{i=1}^{s_0}, (\nabla f_{\text{pivot}}(\mathbf{x}_i^0)^\top)_{i=1}^{s_0} \right)^\top \in (\mathbb{H}_{\text{pivot}})^{s_0(d+1)}$ and we introduce the following notation:

$$\forall p \geq 1, \forall q \geq 1, \forall (a_i)_{i=1}^p \in (\mathbb{H}_{\text{pivot}})^p, \forall (b_j)_{j=1}^q \in (\mathbb{H}_{\text{pivot}})^q, \quad \langle a \otimes b \rangle_{\mathbb{H}_{\text{pivot}}} := (\langle a_i, b_j \rangle_{\mathbb{H}_{\text{pivot}}})_{i \in [p], j \in [q]}.$$

Note that

$$\Upsilon = \begin{pmatrix} (\langle f_{\text{pivot}}(\mathbf{x}_i^0), f_{\text{pivot}}(\mathbf{x}_j^0) \rangle_{\mathbb{H}_{\text{pivot}}})_{1 \leq i, j \leq s_0} & (\langle f_{\text{pivot}}(\mathbf{x}_i^0) \otimes \nabla f_{\text{pivot}}(\mathbf{x}_j^0) \rangle_{\mathbb{H}_{\text{pivot}}})_{1 \leq i, j \leq s_0} \\ (\langle \nabla f_{\text{pivot}}(\mathbf{x}_i^0) \otimes f_{\text{pivot}}(\mathbf{x}_j^0) \rangle_{\mathbb{H}_{\text{pivot}}})_{1 \leq i, j \leq s_0} & (\langle \nabla f_{\text{pivot}}(\mathbf{x}_i^0) \otimes \nabla f_{\text{pivot}}(\mathbf{x}_j^0) \rangle_{\mathbb{H}_{\text{pivot}}})_{1 \leq i, j \leq s_0} \end{pmatrix}$$

$$\mathbf{f}(x) = \langle \mathbf{f}, f_{\text{pivot}}(x) \rangle_{\mathbb{H}_{\text{pivot}}}.$$

We deduce that Υ is the Gram matrix of $\mathbf{f}$, with respect to the dot product of $\mathbb{H}_{\text{pivot}}$, namely $\Upsilon = \langle \mathbf{f} \otimes \mathbf{f} \rangle_{\mathbb{H}_{\text{pivot}}}$.

Assuming that Υ is invertible, we can rewrite (A.1) as

$$\eta^0(x) = (\Upsilon^{-1} \mathbf{u})^\top \mathbf{f}(x) \quad (\text{A.2a})$$

$$\eta^0(x) = \langle \mathbf{f}^\top \Upsilon^{-1} \mathbf{u}, f_{\text{pivot}}(x) \rangle_{\mathbb{H}_{\text{pivot}}} \quad (\text{A.2b})$$

and we deduce that $\eta^0 = \mathbf{f}^\top \Upsilon^{-1} \mathbf{u}$ where the equality holds in $\mathbb{H}_{\text{pivot}}$, by the reproducing property.

Hence

$$\|\eta^0\|_{\mathbb{H}_{\text{pivot}}}^2 = \langle \mathbf{f}^\top \Upsilon^{-1} \mathbf{u}, \mathbf{f}^\top \Upsilon^{-1} \mathbf{u} \rangle_{\mathbb{H}_{\text{pivot}}} = \mathbf{u}^\top \Upsilon^{-1} \mathbf{u},$$

using that $\langle \mathbf{f} \otimes \mathbf{f} \rangle_{\mathbb{H}_{\text{pivot}}} = \Upsilon$.

We also define the block diagonal normalisation matrix $D_{\mathbf{g}} \in \mathbb{R}^{s_0(d+1) \times s_0(d+1)}$ as

$$D_{\mathbf{g}} := \begin{pmatrix} \text{Id}_s & & & \\ & \mathbf{g}_{x_1}^{-\frac{1}{2}} & & \\ & & \ddots & \\ & & & \mathbf{g}_{x_s}^{-\frac{1}{2}} \end{pmatrix} \quad (\text{A.2c})$$

and $\tilde{\Upsilon} := D_{\mathbf{g}} \Upsilon D_{\mathbf{g}}$ which has constant value 1 along its diagonal.

We deduce that

$$\|\eta^0\|_{\mathbb{H}_{\text{pivot}}}^2 \leq \|\tilde{\Upsilon}^{-1}\| \|D_{\mathbf{g}} \mathbf{u}\|_2^2 \leq 2s_0. \quad (\text{A.2d})$$

using that $\|\tilde{\Upsilon}^{-1}\| \leq 2$ by [Poon et al., 2023, Lemma 3].

Invoke Proposition 4.1, (A.2d) and (4.5b) to get that

$$\|c^0\|_{\mathcal{F}} = \|\eta^0\|_{\mathbb{H}_{\text{mod}}} \leq C_{\text{switch}} \|\eta^0\|_{\mathbb{H}_{\text{pivot}}} \leq C_{\text{switch}} \sqrt{2s_0} \quad (\text{A.2e})$$

By [Poon et al., 2023, Theorem 2], η^0 is a $(\varepsilon_0, \varepsilon_2, r_0)$ -pivot non-degenerate dual certificate (note that we slightly changed the values of $\varepsilon_0, \varepsilon_2$ of [Poon et al., 2023, Assumption] in Assumption 1).

The existence of $(\varepsilon_0, \varepsilon_2, r_0)$ -pivot non-degenerate localizing certificates follows the same line. In this case $\mathbf{u}^\top = ((1_{\{i=j\}} \text{sign}(a_j^0))_{j \in [s_0]}, \mathbf{0}_{s_0 d}^\top)$ and hence $\|D_{\mathbf{g}} \mathbf{u}\|_2 = 1$. $\square$

A.2 Existence of sketch pivot non-degenerate dual certificates

Definition 5 (Sketch pivot non-degenerate certificates). *Let $\psi_\omega : \mathcal{X} \rightarrow \mathbb{C}$ be a function and let Λ be a law on $\mathbb{R}^d$. Consider*

$$K_{\text{pivot}}(\mathbf{s}, \mathbf{t}) = \mathbb{E}_{\omega \sim \Lambda} [\psi_\omega(\mathbf{s}) \overline{\psi_\omega(\mathbf{t})}], \quad (\mathcal{H}_{\psi_\omega, \Lambda})$$

and let $\mu^0 \in \mathcal{M}_{s_0, \Delta_0, \mathfrak{d}_{\mathbf{g}}}$ where the Fisher-Rao distance is given by $K_{\text{pivot}}(\cdot, \cdot)$. Let $m \geq 1$ and let $(\omega_i)_i$ be i.i.d. vectors drawn with respect to Λ .

We say that η^0 is an $(\varepsilon_0, \varepsilon_2, r_0)$ -sketch pivot non-degenerate dual certificate if

- There exists $c^0 \in \mathbb{C}^m$ such that $\eta^0 = \frac{1}{\sqrt{m}} \sum_{j=1}^m c_j^0 \varphi_{\omega_j}$,
- For all $i \in [s_0]$, $\eta^0(t_i^0) = \text{sign}(a_i^0)$,
- For all $x \in \mathcal{F}^{\text{reg}}(r_0)$, $|\eta^0(x)| \leq 1 - \varepsilon_0$,
- For all $i \in [s_0]$, for all $x \in \mathcal{N}_i^{\text{reg}}(r_0)$, $|\eta^0(x)| \leq 1 - \varepsilon_2 \mathfrak{d}_{\mathbf{g}}(x, t_i^0)^2$,

where $\mathfrak{d}_{\mathbf{g}}$ is the Fisher-Rao distance associated to $K_{\text{pivot}}(\cdot, \cdot)$, and the near and far regions are those of $(t_i^0)_{i=1}^{s_0}$.

For all $i \in [s_0]$, we say that η_i^0 is an $(\varepsilon_0, \varepsilon_2, r_0)$ -sketch pivot non-degenerate localizing certificate at point t_i^0 if

- There exists $c_i^0 \in \mathbb{C}^m$ such that $\eta_i^0 = \frac{1}{\sqrt{m}} \sum_{j=1}^m c_{i,j}^0 \varphi_{\omega_j}$,
- For all $x \in \mathcal{F}^{\text{reg}}(r_0)$, $|\eta_i^0(x)| \leq 1 - \varepsilon_0$,
- For all $j \in [s_0]$, for all $x \in \mathcal{N}_j^{\text{reg}}(r_0)$, $|1_{\{i=j\}} - \eta_i^0(x)| \leq \varepsilon_2 \mathfrak{d}_{\mathbf{g}}(x, t_j^0)^2$,

where $1_{\{i=j\}}$ equals one if $i = j$ and 0 otherwise.

Theorem 6 (Existence of sketch pivot non-degenerate dual certificates). *Let $\mathcal{X}$ be a compact set of $\mathbb{R}^d$. Let $\varphi_\omega, \psi_\omega : \mathcal{X} \rightarrow \mathbb{C}$ be functions and let Λ be a law on $\mathbb{R}^d$. Consider*

$$\begin{aligned} K_{\text{mod}}(\mathbf{s}, \mathbf{t}) &= \mathbb{E}_{\omega \sim \Lambda} [\varphi_\omega(\mathbf{s}) \overline{\varphi_\omega(\mathbf{t})}], \\ K_{\text{pivot}}(\mathbf{s}, \mathbf{t}) &= \mathbb{E}_{\omega \sim \Lambda} [\psi_\omega(\mathbf{s}) \overline{\psi_\omega(\mathbf{t})}]. \end{aligned} \quad (\mathcal{H}_{\psi_\omega, \Lambda})$$

Assume that ψ_ω satisfies Assumption 3, and assume that $K_{\text{pivot}}(\cdot, \cdot)$ satisfies Assumption 1 (with parameters $s_0, \Delta_0, r_0, \bar{\varepsilon}_0$ and $\bar{\varepsilon}_2$) and Assumption 2 with constant $C_{\text{switch}} > 0$.

Let $\alpha \in (0, 1)$ and m be such that

$$m \geq C_{\text{sketch}} \max(d, \log(s_0)) s_0 \log \left(\frac{\max(1, |\mathcal{X}|) s_0}{\alpha} \right), \quad (\text{A.3})$$

where $C_{\text{sketch}} > 0$ depends on K_{pivot} and polynomially on $d, |\mathcal{X}| = \sup_{\mathbf{s}, \mathbf{t} \in \mathcal{X}} \mathfrak{d}_{\mathfrak{g}}(\mathbf{s}, \mathbf{t})$ and the Fisher-Rao distance $\mathfrak{d}_{\mathfrak{g}}$ is given by $K_{\text{pivot}}(\cdot, \cdot)$.

Then, there exists a constant $C'_{\text{pivot}} > 0$ which depends only on the kernel K_{pivot} such that, with probability at least $1 - \alpha$, for all $m \geq m_0$, there exist a $(\bar{\varepsilon}_0/4, 3\bar{\varepsilon}_2/2, r_0)$ -sketch pivot non-degenerate dual certificate such that $\|c^0\|_{\mathbb{C}^m} \leq C'_{\text{pivot}} C_{\text{switch}} \sqrt{s_0}$ and $(\bar{\varepsilon}_0/4, 3\bar{\varepsilon}_2/2, r_0)$ -sketch pivot non-degenerate localizing certificates at points t_i^0 such that $\|c_i^0\|_{\mathbb{C}^m} \leq C'_{\text{pivot}} C_{\text{switch}}$.

Proof. This result is a consequence of [Poon et al., 2023, Theorem 4, Page 274] and Theorem 5. The only subtlety lies in [Poon et al., 2023, Section 6.5, Step 3, Page 284] where one has to invoke Proposition 4.1 and (4.5b) to get the bound $\|c^0\|_{\mathbb{C}^m} \leq C'_{\text{pivot}} C_{\text{switch}} \sqrt{s_0}$. Except this latter point, the reasoning is identical. $\square$

A.3 Proof of Theorem 3

Consider the sinc-4 kernel Ψ_τ as defined by (2.9d). We want to prove that this kernel satisfies the local positive curvature assumption, Assumption 1. We split the proof into 5 parts.

A.3.1 Useful geometric object

For $\tau > 0$ and $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$, we define:

- The random features $\psi_\omega(\mathbf{x}) = \sqrt{\frac{f_\tau^{(4)}}{\Lambda}}(\omega) e^{-i\omega^\top \mathbf{x}}$ with the spectral measure $f_\tau^{(4)} = (2\pi)^{-d} \mathfrak{F}[\Psi_\tau]$ detailed in Section C.5
- The kernel $K_\tau(\mathbf{x}, \mathbf{y}) = \mathbb{E}_{\omega \sim \Lambda_\tau} [\overline{\psi_\omega(\mathbf{x})} \psi_\omega(\mathbf{y})] = \Psi_\tau(\mathbf{x} - \mathbf{y}) = \Psi_1(\frac{\mathbf{x} - \mathbf{y}}{\tau}) = K_1(\frac{\mathbf{x}}{\tau}, \frac{\mathbf{y}}{\tau})$.
- The metric $\mathfrak{g}_{\mathbf{x}, \tau} = \nabla_1 \nabla_2 K_\tau(\mathbf{x}, \mathbf{x}) = -\nabla^2 \Psi_\tau(\mathbf{0}) = \frac{1}{12\tau^2} \text{Id}$. It is independent of the point $\mathbf{x}$ in the case of translation invariant kernel.
- The norm at point $\mathbf{x}$: $\|\mathbf{z}\|_{\mathbf{x}, \tau}^2 = \mathbf{z}^\top \mathfrak{g}_{\mathbf{x}, \tau} \mathbf{z} = \frac{1}{12\tau^2} \|\mathbf{z}\|_2^2$
- The geodesic distance from $\mathbf{x}$ to $\mathbf{y}$ which is proportional to the Euclidean norm since the kernel is translation invariant:

$$\mathfrak{d}_{\mathfrak{g}, \tau}(\mathbf{x}, \mathbf{y}) := \inf \left\{ \int_0^1 \|g'(t)\|_{g(t)} : g(0) = \mathbf{x}, g(1) = \mathbf{y} \right\} = \frac{1}{2\sqrt{3}\tau} \|\mathbf{x} - \mathbf{y}\|_2$$

Moreover, the metric has the following re-scaling properties

$$\mathfrak{g}_{\mathbf{x},\tau} = \frac{1}{\tau^2} \mathfrak{g}_{\mathbf{u},1}, \quad \forall \mathbf{x}, \mathbf{u} \in \mathbb{R}^d, \quad (\text{A.4a})$$

$$\mathfrak{d}_{\mathfrak{g},\tau}(\mathbf{x}, \mathbf{y}) = \mathfrak{d}_{\mathfrak{g},1}\left(\frac{\mathbf{x}}{\tau}, \frac{\mathbf{y}}{\tau}\right). \quad (\text{A.4b})$$

Finally, notice that this choice of K_τ defines a flat Euclidean geometry so that Euclidean and Riemannian gradients (and Hessians) coincide. With a slight abuse of notations, we will use $\Psi_\tau(\mathbf{x}, \mathbf{y})$ in place of $K_\tau(\mathbf{x}, \mathbf{y}) = \Psi_\tau(\mathbf{x} - \mathbf{y})$.

A.3.2 Reduction to width one

• **Covariant derivatives:** First, we begin by proving that it suffices to prove the result for $\tau = 1$. The controls we look after, see Equations (4.3), involve the covariant derivatives of the kernel of order (i, j) seen as multi-linear maps $\Psi^{(ij)}(\mathbf{x}, \mathbf{y}) : (\mathbb{C}^d)^{i+j} \rightarrow \mathbb{C}$:

$$\Psi^{(ij)}(\mathbf{x}, \mathbf{y})[\mathbf{q}_1, \dots, \mathbf{q}_{i+j}] := \mathbb{E}_\omega \left[\overline{D_i[\psi_\omega](\mathbf{x})[\mathbf{q}_1, \dots, \mathbf{q}_i]} D_j[\psi_\omega](\mathbf{y})[\mathbf{q}_{i+1}, \dots, \mathbf{q}_{i+j}] \right] \quad (\text{A.5a})$$

In the particular case of ψ_ω yielding the Ψ_τ kernel, we have simple expression of the partial derivatives of order r as $\partial_{i_1, \dots, i_r} \psi_\omega(\mathbf{x}) = (-i)^r \omega_{i_1} \dots \omega_{i_r} \psi_\omega(\mathbf{x})$ so that the differential of order r can be written as a rank-1 tensor in the canonical basis of $(\mathbb{C}^d)^r$

$$D_r[\psi_\omega](\mathbf{x})[\cdot] = (-i)^r \psi_\omega(\mathbf{x}) (\omega \otimes \dots \otimes \omega)[\cdot] = (-i)^r \psi_\omega(\mathbf{x}) \omega^{\otimes r}[\cdot]. \quad (\text{A.5b})$$

This leads to the following simplification of Equation (A.5a) putting $Q = (\mathbf{q}_1, \dots, \mathbf{q}_{i+j})$

$$\Psi^{(ij)}(\mathbf{x}, \mathbf{y})[\mathbf{q}_1, \dots, \mathbf{q}_{i+j}] = (-1)^j \mathbb{E}_\omega \left[\overline{D_{i+j}[\psi_\omega](\mathbf{x})[Q]} \psi_\omega(\mathbf{y}) \right] \quad (\text{A.5c})$$

The operator norm at $\mathbf{x}, \mathbf{y}$ is then defined as Poon et al. [2023, Eq. (26)]

$$\|\Psi^{(ij)}(\mathbf{x}, \mathbf{y})\|_{\mathbf{x}, \mathbf{y}} := \sup_{\substack{\|\mathbf{q}_l\|_{\mathbf{x}} \leq 1, l=1, \dots, i \\ \|\mathbf{q}_{l+i}\|_{\mathbf{y}} \leq 1, l=1, \dots, j}} \left| \mathbb{E}_\omega \left[\overline{D_i[\psi_\omega](\mathbf{x})[\mathbf{q}_1, \dots, \mathbf{q}_i]} D_j[\psi_\omega](\mathbf{y})[\mathbf{q}_{i+1}, \dots, \mathbf{q}_{i+j}] \right] \right|. \quad (\text{A.5d})$$

The first fact is that Ψ_τ and Ψ_1 (and their covariant derivatives) have the same operator norm provided we rescale the coordinate by $1/\tau$, as stated by the following lemma.

Lemma A.1 (Operator norm invariance by scaling $\mathbf{x}$ onto $\mathbf{x}/\tau$). *Let $\tau > 0$ and $\mathbf{x}, \mathbf{y} \in \mathcal{X}$, then, putting $\mathbf{u} = \mathbf{x}/\tau$ and $\mathbf{v} = \mathbf{y}/\tau$ for all $0 \leq i, j \leq 2$ we have*

$$\|K_\tau^{(ij)}(\mathbf{x}, \mathbf{y})\|_{\mathbf{x}, \mathbf{y}} = \|K_1^{(ij)}(\mathbf{u}, \mathbf{v})\|_{\mathbf{u}, \mathbf{v}}$$

Proof. We begin by noting that the norm $\|\mathbf{z}\|_{\mathbf{x}} = \|\mathbf{z}\|_{\mathbf{y}} \propto \frac{1}{\tau} \|\mathbf{z}\|_2$ is a rescaled Euclidean metric, and is thus independent of the points $\mathbf{x}$ or $\mathbf{y}$. Thus, the operator norm defined in Equation (A.5d) can be written as

$$\begin{aligned} \|K_\tau^{(ij)}(\mathbf{x}, \mathbf{y})\|_{\mathbf{x}, \mathbf{y}} &= \sup_{\substack{\|\mathbf{q}_l\|_{\tau} \leq 1, \\ l=1, \dots, i+j}} \left| \mathbb{E}_\omega \left[\overline{D_i[\psi_\omega](\mathbf{x})[\mathbf{q}_1, \dots, \mathbf{q}_i]} D_j[\psi_\omega](\mathbf{y})[\mathbf{q}_{i+1}, \dots, \mathbf{q}_{i+j}] \right] \right|, \\ &= \sup_{\substack{\|\mathbf{q}_l\|_{\tau} \leq 1, \\ l=1, \dots, i+j}} \left| (-1)^j \mathbb{E}_\omega \left[\overline{D_{i+j}[\psi_\omega](\mathbf{x})[Q]} \psi_\omega(\mathbf{y}) \right] \right| \end{aligned}$$

Using this symmetry, it suffices to prove the desired identity for $j = 0$ and $i = 0, \dots, 4$

- $i = j = 0$, $\|K_\tau^{(00)}(\mathbf{x}, \mathbf{y})\|_{\mathbf{x}} = K_\tau(\mathbf{x}, \mathbf{y}) = K_1(\frac{\mathbf{x}}{\tau}, \frac{\mathbf{y}}{\tau}) = K_1(\mathbf{u}, \mathbf{v})$.

- $i = 1, j = 0$: using that $\tau^2 \mathfrak{g}_{x,\tau} = \mathfrak{g}_{u,1}$ when $\mathbf{u} = \mathbf{x}/\tau$

$$\begin{aligned}
\|K_\tau^{(10)}(\mathbf{x}, \mathbf{y})\|_x &= \|\mathfrak{g}_{x,\tau}^{-1/2} \nabla_1 K_\tau(\mathbf{x}, \mathbf{y})\|_2 \\
&= \|\mathfrak{g}_{x,\tau}^{-1/2} \frac{1}{\tau} \nabla_1 K_1(\mathbf{u}, \mathbf{v})\|_2 \\
&= \|\mathfrak{g}_{u,1}^{-1/2} \nabla_1 K_1(\mathbf{u}, \mathbf{v})\|_2 \\
&= \|K_1^{(10)}(\mathbf{u}, \mathbf{v})\|_u.
\end{aligned}$$

- $i = 2, j = 0$: similarly

$$\begin{aligned}
\|K_\tau^{(20)}(\mathbf{x}, \mathbf{y})\|_x &= \|\mathfrak{g}_{x,\tau}^{-1/2} \nabla^2 \Psi_\tau(\mathbf{x} - \mathbf{y}) \mathfrak{g}_{x,\tau}^{-1/2}\|_2, \\
&= \|\mathfrak{g}_{x,\tau}^{-1/2} \frac{1}{\tau^2} \nabla^2 \Psi_1(\mathbf{u} - \mathbf{v}) \mathfrak{g}_{x,\tau}^{-1/2}\|_2, \\
&= \|\mathfrak{g}_{u,1}^{-1/2} \nabla^2 \Psi_1(\mathbf{u} - \mathbf{v}) \mathfrak{g}_{u,1}^{-1/2}\|_2, \\
&= \|K_1^{(20)}(\mathbf{u}, \mathbf{v})\|_u.
\end{aligned}$$

The same follows for higher order derivatives. □

• **Local curvature:** Poon et al. [2023] define geometric quantities related to the local curvature of the kernel K_τ (and hence of η^0) on near and far regions.

Definition 6 ($\bar{\varepsilon}_0$ and $\bar{\varepsilon}_2$). Let $r \geq 0$,

$$\begin{aligned}
\bar{\varepsilon}_0(r, \tau) &:= \sup \{ \varepsilon ; K_\tau(\mathbf{x}, \mathbf{y}) \leq 1 - \varepsilon, \forall \mathbf{x}, \mathbf{y} \text{ s.t. } \mathfrak{d}_{\mathfrak{g},\tau}(\mathbf{x}, \mathbf{y}) \geq r \}, \\
\bar{\varepsilon}_2(r, \tau) &:= \sup \left\{ \varepsilon ; \Psi_\tau^{(02)}(\mathbf{x}, \mathbf{y})[\mathbf{z}, \mathbf{z}] \geq \varepsilon \|\mathbf{z}\|_{x,\tau}^2, \forall \mathbf{z} \in \mathbb{R}^d, \forall \mathbf{x}, \mathbf{y} \text{ s.t. } \mathfrak{d}_{\mathfrak{g},\tau}(\mathbf{x}, \mathbf{y}) < r \right\}.
\end{aligned}$$

These controls depend both on the kernel and the radius $r = r_{near}$ of the regions considered. However, the following lemmas show that these controls do not depend on the width τ .

Lemma A.2 (Isometry between K_τ and K_1). Let $\tau > 0$ and

$$\mathcal{H}_\tau := \left\{ g \in L^2(\mathbb{R}^d) : \|g\|_{\mathcal{H}_\tau}^2 := \frac{1}{(2\pi)^d} \int \frac{|\mathfrak{F}[g]|^2}{\mathfrak{F}[\Psi_\tau]} < +\infty \right\},$$

then the rescaling map $iso_\tau : g \in \mathcal{H}_1 \rightarrow g(\frac{\cdot}{\tau}) \in \mathcal{H}_\tau$ is an isometry between $(\mathcal{H}_1, \|\cdot\|_{\tau=1})$ and $(\mathcal{H}_\tau, \|\cdot\|_\tau)$. Moreover, for any $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$, we have:

1. $K_1(\frac{\mathbf{x}}{\tau}, \frac{\mathbf{x}}{\tau}) = K_\tau(\mathbf{x}, \mathbf{y})$
2. $\nabla_1 K_1(\frac{\mathbf{x}}{\tau}, \frac{\mathbf{y}}{\tau}) = \tau \nabla_1 K_\tau(\mathbf{x}, \mathbf{y})$
3. $\nabla_{12} K_1(\frac{\mathbf{x}}{\tau}, \frac{\mathbf{y}}{\tau}) = \tau^2 \nabla_{12} K_\tau(\mathbf{x}, \mathbf{y})$

Proof. Let $\tau > 0$,

1. By definition of K_τ and Ψ_τ .
2. $\nabla_1 K_1(\frac{\mathbf{x}}{\tau}, \frac{\mathbf{y}}{\tau}) = \nabla \Psi_1(\frac{\mathbf{x}-\mathbf{y}}{\tau}) = \tau \frac{1}{\tau} \nabla \Psi_1(\frac{\mathbf{x}-\mathbf{y}}{\tau}) = \tau \nabla \Psi_\tau(\mathbf{x} - \mathbf{y}) = \tau \nabla_1 K_\tau(\mathbf{x}, \mathbf{y})$
3. $\nabla_{12} K_1(\frac{\mathbf{x}}{\tau}, \frac{\mathbf{y}}{\tau}) = -\nabla^2 \Psi_1(\frac{\mathbf{x}-\mathbf{y}}{\tau}) = -\tau^2 \frac{1}{\tau^2} \nabla^2 \Psi_1(\frac{\mathbf{x}-\mathbf{y}}{\tau}) = -\tau^2 \nabla^2 \Psi_\tau(\mathbf{x} - \mathbf{y}) = +\tau^2 \nabla_{12} K_\tau(\mathbf{x}, \mathbf{y})$

For the isometry result, we simply use the change of variable $\omega' = \omega/\tau$ in the RKHS norm definition, with diagonal Jacobian yielding $d\omega' = \tau^{-d} d\omega$.

$$\begin{aligned}
\|g\|_{\mathcal{H}_1} &= \frac{1}{(2\pi)^d} \int \frac{|\mathfrak{F}[g](\omega)|^2}{\mathfrak{F}[\Psi_1](\omega)} d\omega, \\
&= \frac{1}{(2\pi)^d} \int \frac{|\mathfrak{F}[g](\tau\omega')|^2}{\mathfrak{F}[\Psi_1](\tau\omega')} \tau^d d\omega', & (\text{change of variables : } \omega = \tau\omega') \\
&= \frac{1}{(2\pi)^d} \int \frac{\tau^{-2d} |\mathfrak{F}[g(\cdot/\tau)](\omega')|^2}{\mathfrak{F}[\Psi_1](\omega')} \tau^d d\omega', & (\mathfrak{F}[g](\tau\omega') = \tau^{-d} \mathfrak{F}[g(\cdot/\tau)](\omega')), \\
&= \frac{1}{(2\pi)^d} \int \frac{|\mathfrak{F}[g(\cdot/\tau)](\omega')|^2}{\tau^d \mathfrak{F}[\Psi_1](\tau\omega')} d\omega', \\
&= \frac{1}{(2\pi)^d} \int \frac{|\mathfrak{F}[g(\cdot/\tau)](\omega')|^2}{\mathfrak{F}[\Psi_\tau](\omega')} d\omega', & (\text{same argument on } g = \Psi_1), \\
&= \|\text{iso}_\tau(g)\|_{\mathcal{H}_\tau},
\end{aligned}$$

which gives the result. □

Lemma A.3 ($\bar{\varepsilon}_0$ and $\bar{\varepsilon}_2$). *Let $\tau > 0$ and $r > 0$, then for*

$$j \in \{0, 2\}, \quad \bar{\varepsilon}_j(r, \tau) = \bar{\varepsilon}_j(r, 1)$$

Proof. Let $r > 0$. First, notice that due to the scaled Euclidean metric induced by K_τ ,

$$\mathfrak{d}_{\mathfrak{g}, \tau}(\mathbf{x}, \mathbf{y}) \geq r \iff \mathfrak{d}_{\mathfrak{g}, 1}\left(\frac{\mathbf{x}}{\tau}, \frac{\mathbf{y}}{\tau}\right) \geq r$$

For $j = 0$, it suffices to use Lemma A.2 (i) together with the above equivalence to show that

$$\begin{aligned}
\bar{\varepsilon}_0(r, \tau) &= \sup \{ \varepsilon : K_\tau(\mathbf{x}, \mathbf{y}) \leq 1 - \varepsilon, \forall \mathbf{x}, \mathbf{y} \text{ s.t. } \mathfrak{d}_{\mathfrak{g}, \tau}(\mathbf{x}, \mathbf{y}) \geq r \}, \\
&= \sup \left\{ \varepsilon : K_1\left(\frac{\mathbf{x}}{\tau}, \frac{\mathbf{y}}{\tau}\right) \leq 1 - \varepsilon, \forall \mathbf{x}, \mathbf{y} \text{ s.t. } \mathfrak{d}_{\mathfrak{g}, 1}\left(\frac{\mathbf{x}}{\tau}, \frac{\mathbf{y}}{\tau}\right) \geq r \right\}, \\
&= \sup \{ \varepsilon : K_1(\mathbf{u}, \mathbf{v}) \leq 1 - \varepsilon, \forall \mathbf{u}, \mathbf{v} \text{ s.t. } \mathfrak{d}_{\mathfrak{g}, 1}(\mathbf{u}, \mathbf{v}) \geq r \}, \\
&= \bar{\varepsilon}_0(r, 1).
\end{aligned}$$

For $j = 2$, we begin by noting that $\psi_\omega(\mathbf{x}) = \sqrt{f_\tau^{(4)}}(\omega) e^{-i\omega^\top \mathbf{x}}$ so that, in the same lines as Lemma A.2 (iii):

$$\begin{aligned}
K_\tau^{(02)}(\mathbf{x}, \mathbf{y})[z, z] &= \mathbb{E}_\omega \left[\overline{\psi_\omega\left(\frac{\mathbf{x}}{\tau}\right)} z^\top \nabla^2 \psi_\omega\left(\frac{\mathbf{y}}{\tau}\right) z \right], \\
&= \mathbb{E}_\omega \left[\overline{\psi_\omega\left(\frac{\mathbf{x}}{\tau}\right)} \psi_\omega\left(\frac{\mathbf{y}}{\tau}\right) z^\top (-i)^2 \omega \omega^\top z \right], \\
&= -z^\top \mathbb{E}_\omega \left[\omega \omega^\top \overline{\psi_\omega\left(\frac{\mathbf{x}}{\tau}\right)} \psi_\omega\left(\frac{\mathbf{y}}{\tau}\right) \right] z, \\
&= -z^\top \nabla^2 K_\tau(\mathbf{x} - \mathbf{y}) z, \\
&= z^\top \frac{1}{\tau^2} (-\tau^2) \nabla^2 \Psi_\tau(\mathbf{x} - \mathbf{y}) z, \\
&= \frac{1}{\tau^2} z^\top \left[-\nabla^2 \Psi_1\left(\frac{\mathbf{x} - \mathbf{y}}{\tau}\right) \right] z, \\
&= \frac{1}{\tau^2} \Psi_1^{(02)}\left(\frac{\mathbf{x}}{\tau}, \frac{\mathbf{y}}{\tau}\right) [z, z].
\end{aligned}$$

Moreover, for all $\mathbf{z}$ and $\mathbf{x}$, $\|\mathbf{z}\|_{\mathbf{x},\tau}^2 = \frac{1}{\tau^2} \|\mathbf{z}\|_{\mathbf{x}/\tau,1}^2$. Thus, we again have equality of the ε -sets over which the supremum is taken.

$$\begin{aligned}\bar{\varepsilon}_2(r, \tau) &= \sup \left\{ \varepsilon; K_\tau^{(02)}(\mathbf{x}, \mathbf{y})[\mathbf{z}, \mathbf{z}] \geq \varepsilon \|\mathbf{z}\|_{\mathbf{x},\tau}^2, \forall \mathbf{z} \in \mathbb{R}^d, \forall \mathbf{x}, \mathbf{y} \text{ s.t. } \mathfrak{d}_{\mathbf{g},\tau}(\mathbf{x}, \mathbf{y}) < r \right\}, \\ &= \sup \left\{ \varepsilon; \frac{1}{\tau^2} K_1^{(02)}\left(\frac{\mathbf{x}}{\tau}, \frac{\mathbf{y}}{\tau}\right)[\mathbf{z}, \mathbf{z}] \geq \varepsilon \frac{1}{\tau^2} \|\mathbf{z}\|_{\mathbf{x}/\tau,1}^2, \forall \mathbf{z} \in \mathbb{R}^d, \forall \mathbf{x}, \mathbf{y} \text{ s.t. } \mathfrak{d}_{\mathbf{g},1}\left(\frac{\mathbf{x}}{\tau}, \frac{\mathbf{y}}{\tau}\right) < r \right\}, \\ &= \sup \left\{ \varepsilon; K_1^{(02)}\left(\frac{\mathbf{x}}{\tau}, \frac{\mathbf{y}}{\tau}\right)[\mathbf{z}, \mathbf{z}] \geq \varepsilon \|\mathbf{z}\|_{\mathbf{x}/\tau,1}^2, \forall \mathbf{z} \in \mathbb{R}^d, \forall \mathbf{x}, \mathbf{y} \text{ s.t. } \mathfrak{d}_{\mathbf{g},1}\left(\frac{\mathbf{x}}{\tau}, \frac{\mathbf{y}}{\tau}\right) < r \right\}, \\ &= \sup \left\{ \varepsilon; K_1^{(02)}(\mathbf{u}, \mathbf{v})[\mathbf{z}, \mathbf{z}] \geq \varepsilon \|\mathbf{z}\|_{\mathbf{u},1}^2, \forall \mathbf{z} \in \mathbb{R}^d, \forall \mathbf{u}, \mathbf{v} \text{ s.t. } \mathfrak{d}_{\mathbf{g},1}(\mathbf{u}, \mathbf{v}) < r \right\}, \\ &= \bar{\varepsilon}_2(r, 1).\end{aligned}$$

□

• **Separation:** Then, for a number of components $s \in \mathbb{N}$ and $h > 0$, the kernel width of K_τ is defined by Poon et al. [2023] as

$$\Delta(h, s) := \inf \left\{ \Delta; \sum_{k=2}^s \left\| K_\tau^{(ij)}(\mathbf{x}_1^0, \mathbf{x}_k^0) \right\|_{\mathbf{x}_1^0, \mathbf{x}_k^0} \leq h, \quad (i, j) \in \{0, 1\} \times \{0, 2\}, \quad \{\mathbf{x}_k^0\}_{k=1}^s \in \mathcal{S}_{\Delta, \tau} \right\}, \quad (\text{A.6})$$

where $\mathcal{S}_{\Delta, \tau} := \{(\mathbf{x}_k)_{k=1}^s \in \mathcal{X}; \mathfrak{d}_\tau(\mathbf{x}_k, \mathbf{x}_l) \geq \Delta, \forall k \neq l\}$ is the set of s -tuples of Δ -separated points. This notion is somewhat related to the coherence of the kernel K meaning that the interference is small enough so that functions $K_\tau(\mathbf{x}_k^0, \cdot)$ can interpolate between parameters.

Remark A.1. The particular form of the metric $\mathfrak{d}_\tau$ gives $\mathcal{S}_{\Delta, \tau} = \tau \cdot \mathcal{S}_{\Delta, 1}$. Together with Lemma A.1, this implies that $\Delta(h, s)$ does not depend on τ .

A.3.3 First part of LPC

This leads to tracking a uniform bound on the kernel and its derivatives

$$B_{ij} := \sup_{\mathbf{x}, \mathbf{y}} \|\Psi_\tau^{(ij)}(\mathbf{x} - \mathbf{y})\|_{\mathbf{x}, \mathbf{y}} \quad B_i := 1 + B_{0i} + B_{1i}, \quad (\text{A.7})$$

whose regimes w.r.t. the dimension d are given by the following proposition.

Proposition A.1. One has

$$\forall i, j, \quad B_{ij} \leq (12d)^{\frac{i+j}{2}},$$

and hence $1/\sqrt{B_{02}} \geq 1/\sqrt{12d}$ and $B_2 \leq 1 + \sqrt{12d} + 12d$.

Proof. Again, the rate does not depend on τ as Lemma A.1 gives

$$\sup_{\mathbf{x}, \mathbf{y}} \|K_\tau^{(ij)}(\mathbf{x}, \mathbf{y})\|_{\mathbf{x}, \mathbf{y}} = \sup_{\mathbf{u}, \mathbf{v}} \|K_1^{(ij)}(\mathbf{u}, \mathbf{v})\|_{\mathbf{u}, \mathbf{v}}.$$

For the rest of the proof, τ is fixed equal to 1.

Trivially, for $i = j = 0$, a triangle inequality yields a uniform control on $(\mathbf{u}, \mathbf{v})$ as:

$$\sup_{\mathbf{u}, \mathbf{v}} |K_1(\mathbf{u}, \mathbf{v})| \leq \sup_{\mathbf{u}, \mathbf{v}} \int \left| e^{i\omega^\top(\mathbf{u}-\mathbf{v})} \right| f_\tau^{(4)}(\omega) d\omega = \int f_\tau^{(4)}(\omega) d\omega = 1. \quad (\text{A.8a})$$

For higher order derivative, the proof relies on Section B.1 and the fact that the covariant derivatives of the kernel involves differential $\nabla^r \psi_\omega(\mathbf{u}) : (\mathbb{C}^d)^r \rightarrow \mathbb{C}$ for which we can control the supremum uniformly over all $\mathbf{u} \in \mathcal{X}$:

$$\sup_{\|\mathbf{q}_l\|_2^2 \leq 12, l=1, \dots, r} |\nabla^r \psi_\omega(\mathbf{u})[\mathbf{q}_1, \dots, \mathbf{q}_r]| = \sqrt{\frac{f_\tau^{(4)}(\omega)}{\Lambda}} \times \sup_{\|\mathbf{q}_l\|_2^2 \leq 12, l=1, \dots, r} \prod_{l=1}^r |\langle \omega, \mathbf{q}_l \rangle|. \quad (\text{A.8b})$$

Hence, the covariant derivative of the kernel can be uniformly upper bounded for all $\mathbf{u}, \mathbf{v} \in \mathcal{X}$ by

$$\begin{aligned}
|K_1^{(ij)}(\mathbf{u}, \mathbf{v})[\mathbf{q}_1, \dots, \mathbf{q}_{i+j}]| &\leq \int_{[-1,1]^d} \left(\prod_{l=1}^{i+j} \sup_{\|\mathbf{q}_l\|_2^2 \leq 12} | \langle \boldsymbol{\omega}, \mathbf{q}_l \rangle | \right) f_\tau^{(4)}(\boldsymbol{\omega}) |e^{-i\boldsymbol{\omega}^T(\mathbf{u}-\mathbf{v})}| d\boldsymbol{\omega}, \\
&\leq \sup_{\boldsymbol{\omega} \in [-1,1]^d} \prod_{l=1}^{i+j} \sup_{\|\mathbf{q}_l\|_2^2 \leq 12} | \langle \boldsymbol{\omega}, \mathbf{q}_l \rangle |, \\
&\hspace{15em} (f_\tau^{(4)} \text{ p.d.f on the unit hypercube}) \\
&\leq \sup_{\boldsymbol{\omega} \in [-1,1]^d} (2\sqrt{3}\|\boldsymbol{\omega}\|_2)^{i+j}, \hspace{5em} (\text{Cauchy-Schwarz}) \\
&\leq \sqrt{12d}^{i+j}.
\end{aligned}$$

Taking the sup of the latter on $\mathbf{u}, \mathbf{v}$ terminates the proof. $\square$

A.3.4 Second part of LPC

Then, we focus on the triplets r_0 , $\bar{\varepsilon}_0$ and $\bar{\varepsilon}_2$ defined in Equation (4.3b) which, in turn, define the minimal separation Δ_0 required between the parameters. Specifically, we need to exhibit some radius $r_0 \in (0, 1/\sqrt{B_{02}})$ for the near regions such that the kernel K_τ has positive curvature constants $\bar{\varepsilon}_0$ and $\bar{\varepsilon}_2$.

$$\bar{\varepsilon}_0 := \frac{1}{2} \sup_{\varepsilon \geq 0} \left\{ \varepsilon : K_1(\mathbf{s}, \mathbf{t}) \leq 1 - \varepsilon, \forall \mathbf{s}, \mathbf{t} \in \mathcal{X} \text{ s.t. } \mathfrak{d}_{\mathfrak{g}}(\mathbf{s}, \mathbf{t}) \geq r_0 \right\}, \quad (\text{A.9a})$$

$$\bar{\varepsilon}_2 := \frac{1}{4} \sup_{\varepsilon \geq 0} \left\{ \varepsilon : -K_1^{(0,2)}(\mathbf{s}, \mathbf{t})[\mathbf{v}, \mathbf{v}] \geq \varepsilon \|\mathbf{v}\|_{\mathbf{t}}^2, \forall \mathbf{v} \in \mathbb{T}_{\mathbf{t}}, \forall \mathbf{s}, \mathbf{t} \in \mathcal{X} \text{ s.t. } \mathfrak{d}_{\mathfrak{g}}(\mathbf{s}, \mathbf{t}) < r_0 \right\}, \quad (\text{A.9b})$$

$$\Delta_0 := \inf \left\{ \Delta : 32 \sum_{l=2}^{s_0} \|K_1^{(i,j)}(\mathbf{x}_1, \mathbf{x}_l)\|_{\mathbf{x}_1, \mathbf{x}_l} \leq \min \left(\frac{\bar{\varepsilon}_0}{B_0}, \frac{2\bar{\varepsilon}_2}{B_2} \right), (i, j) \in \{0, 1\} \times \{0, 2\}, \{\mathbf{x}_l\}_{l=1}^{s_0} \in \mathcal{S}_\Delta \right\}. \quad (\text{A.9c})$$

This is the purpose of the following proposition, which also exhibits the scaling of the last quantities with respect to the dimension d .

Proposition A.2. *Let r_0 be such that $r_0 = 1/(4d)$ then $\bar{\varepsilon}_0 \geq 1/(32d^3)$ and $\bar{\varepsilon}_2 \geq 23/128$ in Equations (A.9).*

Proof. One can check the following calculus:

$$\forall z \in \mathbb{R}, \quad |\text{sinc } z| \leq \left(1 - \frac{z^2}{12}\right) \mathbb{1}_{|z| \leq 2} + \frac{1}{2} \mathbb{1}_{|z| > 2}. \quad (\text{A.10a})$$

Now, let $c > 0$ and $\mathbf{t} \in \mathbb{R}^d$ such that $(1/\sqrt{12})\|\mathbf{t}\|_2 \geq c/d$, then there exists a coordinate z of the vector $\mathbf{t}$ such that $|z| \geq c\sqrt{12}d^{-3/2}$ and we deduce by (A.10a) that $\bar{\varepsilon}_0 \geq c^2d^{-3}/2$. The result is given for $c = 1/4$.

We turn to $\bar{\varepsilon}_2$. Let $\mathbf{s}, \mathbf{t}, \mathbf{v} \in \mathbb{R}^d$ be fixed. Writing the exact Taylor expansion of $\mathbf{u} \mapsto -K_1^{(0,2)}(\mathbf{s}, \mathbf{u})[\mathbf{v}, \mathbf{v}]$ at $\mathbf{s}$ and using (B.1e), there exists $\boldsymbol{\xi} \in [\mathbf{s}, \mathbf{t}]$ (which is the geodesic between these points) such that

$$\begin{aligned}
-K_1^{(0,2)}(\mathbf{s}, \mathbf{t})[\mathbf{v}, \mathbf{v}] &= -K_1^{(0,2)}(\mathbf{s}, \mathbf{s})[\mathbf{v}, \mathbf{v}] - K_1^{(0,3)}(\mathbf{s}, \mathbf{s})[\mathbf{v}, \mathbf{v}, \mathbf{s} - \mathbf{t}] - \frac{1}{2} K_1^{(0,4)}(\mathbf{s}, \boldsymbol{\xi})[\mathbf{v}, \mathbf{v}, \mathbf{s} - \mathbf{t}, \mathbf{s} - \mathbf{t}] \\
&= \|\mathbf{v}\|_{\mathbf{t}}^2 + 0 + \frac{1}{2} \mathbf{v}^\top (A(\mathbf{s}, \boldsymbol{\xi})[\mathbf{s} - \mathbf{t}, \mathbf{s} - \mathbf{t}]) \mathbf{v} \\
&\geq (1 - 6\rho(A(\mathbf{s}, \boldsymbol{\xi})[\mathbf{s} - \mathbf{t}, \mathbf{s} - \mathbf{t}])) \|\mathbf{v}\|_{\mathbf{t}}^2
\end{aligned}$$

where the quadratic form $K_1^{(0,4)}(\mathbf{s}, \boldsymbol{\xi})[\mathbf{s} - \mathbf{t}, \mathbf{s} - \mathbf{t}]$ is identified to the $d \times d$ matrix in the canonical basis:

$$\forall i_1, i_2 \in [d], \quad (A(\mathbf{s}, \boldsymbol{\xi})[\mathbf{s} - \mathbf{t}, \mathbf{s} - \mathbf{t}])_{i_1, i_2} = \sum_{i_3=1}^d \sum_{i_4=1}^d (s - t)_{i_3} (s - t)_{i_4} \partial_{i_1, i_2, i_3, i_4} \Psi_1(\mathbf{s} - \boldsymbol{\xi}), \quad (\text{A.10b})$$

and $\rho(A)$ denotes its spectral radius.

Then, notice that $f = \text{sinc}$ has all derivatives upper bounded by $1/2$ in absolute value so that

$$\begin{aligned} \frac{d^4}{dx^4} f^4(x) &= 4f^3(x) \frac{d^4}{dx^4} f(x) + 48f^2(x) \frac{d}{dx} f(x) \frac{d^3}{dx^3} f(x) \\ &\quad + 36f^2(x) \left(\frac{d^2}{dx^2} f(x) \right)^2 + 144f(x) \left(\frac{d}{dx} f(x) \right)^2 \frac{d^2}{dx^2} f(x) + 24 \left(\frac{d}{dx} f(x) \right)^4 \\ &\leq \frac{256}{16} = 16. \end{aligned}$$

Remembering that $\Psi_1 = \text{sinc}^4(\cdot/4)$, we deduce from the previous calculus that

$$\forall i_1, i_2, i_3, i_4 \in [d], \quad |\partial_{i_1, i_2, i_3, i_4} \Psi_1(\mathbf{s} - \boldsymbol{\xi})| \leq 16 \times \frac{1}{4^4} = \frac{1}{16}, \quad (\text{A.10c})$$

$$\forall i_1, i_2 \in [d], \quad |(A(\mathbf{s}, \boldsymbol{\xi})[\mathbf{s} - \mathbf{t}, \mathbf{s} - \mathbf{t}])_{i_1, i_2}| \leq \frac{d \|\mathbf{s} - \mathbf{t}\|_2^2}{16}. \quad (\text{A.10d})$$

Finally, using that $\rho(A) \leq d \sup_{i_1, i_2} |A_{i_1, i_2}|$, one has

$$6\rho(A(\mathbf{s}, \boldsymbol{\xi})) \leq \frac{6d^2 \|\mathbf{s} - \mathbf{t}\|_2^2}{16} = \frac{9d^2}{2} \mathfrak{d}_{\mathfrak{g}}(\mathbf{s}, \mathbf{t})^2,$$

Hence for $r_0 = 1/(4d)$ one has $-K_1^{(0,2)}(\mathbf{s}, \mathbf{t})[\mathbf{v}, \mathbf{v}] \geq (1 - 9/32) \|\mathbf{v}\|_{\mathbf{t}}^2$. $\square$

Remark A.2. Note that $r_0 < B_{02}^{-1/2}$ since $r_0 = 1/(4d)$ and $B_{02}^{-1/2} \geq 1/\sqrt{12d}$.

A.3.5 Last part of LPC

Note that $B_0 \leq 1 + \sqrt{12d} + 1 = 2 + \sqrt{12d}$ and $\bar{\varepsilon}_0/B_0 \geq 1/(32d^3(2 + \sqrt{12d})) \geq 1/(175d^{7/2})$. Also, it holds $B_2 \leq 12d + (12d)^{3/2} + 1$ and $2\bar{\varepsilon}_2/B_2 \geq 23/(64 + 768d + 64(12d)^{3/2}) \geq 23/(1088d^{3/2})$. Hence

$$\frac{1}{32} \min \left(\frac{\bar{\varepsilon}_0}{B_0}, \frac{2\bar{\varepsilon}_2}{B_2} \right) \geq \frac{1}{5600d^{7/2}}.$$

Using Lemma A.4 below, we deduce that

$$\sum_{k=2}^s \|K_1^{(ij)}(\mathbf{u}_k, \mathbf{u}_1)\|_{\mathbf{u}_k, \mathbf{u}_1} \leq \left(\frac{4}{3}\right)^2 \sqrt{48d}^{i+j} \frac{sd^2}{\Delta^4} \leq \frac{1}{5600d^{7/2}} \leq \frac{1}{32} \min \left(\frac{\bar{\varepsilon}_0}{B_0}, \frac{2\bar{\varepsilon}_2}{B_2} \right)$$

as soon as $\Delta \geq \Delta_0 := 42.66s^{1/4}d^{7/4}$.

Lemma A.4. It holds that

$$\|K_1^{(ij)}(\mathbf{s}, \mathbf{t})\|_{\mathbf{s}, \mathbf{t}} \leq \left(\frac{4}{3}\right)^2 \sqrt{48d}^{i+j} \frac{d^2}{\mathfrak{d}_{\mathfrak{g}}(\mathbf{s}, \mathbf{t})^4}.$$

Proof. We bound each entry of $\Psi_1^{(i+j)}(\mathbf{s} - \mathbf{t})$ using that $|\text{sinc}^{(i)}(x/4)| \leq (1+i)4/|x|$ for all $x \neq 0$. One has

$$\Psi_1(\mathbf{s} - \mathbf{t}) = \prod_{\ell=1}^d \text{sinc}^4\left(\frac{s_\ell - t_\ell}{4}\right) \leq \frac{4^4}{(s_{\ell_0} - t_{\ell_0})^4} \prod_{\ell \neq \ell_0}^d \text{sinc}^4\left(\frac{s_\ell - t_\ell}{4}\right) \leq \frac{4^4 d^2}{\|\mathbf{s} - \mathbf{t}\|_2^4} = \left(\frac{4}{3}\right)^2 \frac{d^2}{\mathfrak{d}_{\mathbf{g}}(\mathbf{s}, \mathbf{t})^4},$$

because one coordinate ℓ_0 exists such that $(s_{\ell_0} - t_{\ell_0})^2 \geq \|\mathbf{s} - \mathbf{t}\|_2^2/d$. Let $h := \mathbf{s} - \mathbf{t}$ and ∂_u the derivative with respect to coordinate u , one has

$$\partial_u \Psi_1(h) = \text{sinc}^{(1)}\left(\frac{h_u}{4}\right) \text{sinc}^3\left(\frac{h_u}{4}\right) \prod_{\ell \neq u} \text{sinc}^4\left(\frac{h_\ell}{4}\right) \leq \frac{2 \cdot 4^4}{(h_{\ell_0})^4} \leq 2 \left(\frac{4}{3}\right)^2 \frac{d^2}{\mathfrak{d}_{\mathbf{g}}(\mathbf{s}, \mathbf{t})^4}.$$

because one coordinate ℓ_0 exists such that $h_{\ell_0}^2 \geq \|h\|_2^2/d$. For the second order derivatives, if $u = v$ one has

$$\begin{aligned} \partial_{u,v}^2 \Psi_1(h) &= \left(\frac{1}{4}\right) \text{sinc}^{(2)}\left(\frac{h_u}{4}\right) \text{sinc}^3\left(\frac{h_u}{4}\right) \prod_{\ell \neq u} \text{sinc}^4\left(\frac{h_\ell}{4}\right) + \left(\frac{3}{4}\right) \left(\text{sinc}^{(1)}\left(\frac{h_u}{4}\right)\right)^2 \text{sinc}^2\left(\frac{h_u}{4}\right) \prod_{\ell \neq u} \text{sinc}^4\left(\frac{h_\ell}{4}\right) \\ &\leq 4 \left(\frac{4}{3}\right)^2 \frac{d^2}{\mathfrak{d}_{\mathbf{g}}(\mathbf{s}, \mathbf{t})^4}, \end{aligned}$$

and otherwise if $u \neq v$

$$\partial_{u,v}^2 \Psi_1(h) = \text{sinc}^{(1)}\left(\frac{h_u}{4}\right) \text{sinc}^3\left(\frac{h_u}{4}\right) \text{sinc}^{(1)}\left(\frac{h_v}{4}\right) \text{sinc}^3\left(\frac{h_v}{4}\right) \prod_{\ell \neq u,v} \text{sinc}^4\left(\frac{h_\ell}{4}\right) \leq 4 \left(\frac{4}{3}\right)^2 \frac{d^2}{\mathfrak{d}_{\mathbf{g}}(\mathbf{s}, \mathbf{t})^4}.$$

Pursuing the reasoning, one gets

$$|\partial^{i+j} \Psi_1(h)| \leq 2^{i+j} \left(\frac{4}{3}\right)^2 \frac{d^2}{\mathfrak{d}_{\mathbf{g}}(\mathbf{s}, \mathbf{t})^4}.$$

Now, observe that

$$\|K_1^{(ij)}(\mathbf{s}, \mathbf{t})\|_{\mathbf{s}, \mathbf{t}} = \sqrt{12^{i+j}} \|K_1^{(ij)}(\mathbf{s}, \mathbf{t})\|_2 \leq (12d)^{\frac{i+j}{2}} \max |\partial^{i+j} \Psi_1| \leq (48d)^{\frac{i+j}{2}} \left(\frac{4}{3}\right)^2 \frac{d^2}{\mathfrak{d}_{\mathbf{g}}(\mathbf{s}, \mathbf{t})^4},$$

which gives the result. $\square$

A.4 Proof of Theorem 4

We refer the reader to the definitions of N , C_1 , C_2 and L_{ij} in Assumption 3. Since $\bar{\varepsilon}_0$ and $\bar{\varepsilon}_2$ are already treated in Theorem 3, we only need to find the scaling of L_1 , L_2 , and L_3 . Concretely, for $r \in \{0, \dots, 3\}$, we need to study the tail of the distribution of the random variable

$$Y_r(\omega) = \sup_{\mathbf{x} \in \mathcal{X}} \|D_r[\psi_\omega](\mathbf{x})\|_{\mathbf{x}}, \quad (\text{A.11})$$

where $\psi_\omega(\mathbf{x}) = e^{i\omega^\top \mathbf{x}} \sqrt{\frac{f_\tau^{(4)}}{\Lambda}}(\omega)$ and the convention $\|D_0[\psi_\omega](\mathbf{x})\|_{\mathbf{x}} = |\psi_\omega(\mathbf{x})|$. Specifically, we want to exhibit $L_r \in \mathbb{R}$ such that

$$\sum_{r=0}^3 1 - \mathbb{P}_{\omega \sim \Lambda}(Y_r(\omega) \leq L_r) \leq \frac{\min(\bar{\varepsilon}_0, \bar{\varepsilon}_2, \alpha)}{m}.$$

In the sinc-4 case, this is given by the following Lemma, independently of α , r_0 , $\bar{\varepsilon}_0$, $\bar{\varepsilon}_2$ and m

Lemma A.5. *For any $\tau > 0$, and any $r \in \{0, \dots, 3\}$, the covariant derivatives are almost surely bounded and we have*

$$L_r = (\sqrt{12d})^r \sup_{\omega} \sqrt{\frac{f_{\tau}^{(4)}}{\Lambda}}(\omega). \quad (\text{A.12})$$

Proof. Since the support of Λ is bounded and $\psi_{\omega}(\cdot)$ is a weighted Fourier feature, we don't need refined probabilistic control on the tail bound of $Y_r(\omega)$. Indeed, using Section B.1 we have

$$Y_0(\omega) = \sup_x |\psi_{\omega}(x)| = \sqrt{\frac{f_{\tau}^{(4)}}{\Lambda}}(\omega), \quad (\text{A.13a})$$

$$Y_1(\omega) = \sup_x \|\mathfrak{g}_{x,\tau}^{-1/2} \nabla \psi_{\omega}(x)\|_2 = \sqrt{\frac{f_{\tau}^{(4)}}{\Lambda}}(\omega) \sqrt{12\tau} \|\omega\|_2, \quad (\text{A.13b})$$

$$Y_2(\omega) = \sup_x \|\mathfrak{g}_{x,\tau}^{-1/2} \nabla^2 \psi_{\omega}(x) \mathfrak{g}_{x,\tau}^{-1/2}\|_{op} = \sqrt{\frac{f_{\tau}^{(4)}}{\Lambda}}(\omega) (\sqrt{12\tau})^2 \|\omega\|_2^2, \quad (\text{A.13c})$$

$$Y_3(\omega) = \sup_x \|\nabla^3 \psi_{\omega}(x) [\mathfrak{g}_{x,\tau}^{-1/2}, \mathfrak{g}_{x,\tau}^{-1/2}, \mathfrak{g}_{x,\tau}^{-1/2}]\|_{op} = \sqrt{\frac{f_{\tau}^{(4)}}{\Lambda}}(\omega) (\sqrt{12\tau})^3 \|\omega\|_2^3. \quad (\text{A.13d})$$

Thus, since Λ has a bounded support on $\Omega = [-1/\tau, 1/\tau]^d$, and using $\|\omega\|_2 \leq \sqrt{d} \|\omega\|_{\infty}$, the random variables $Y_r(\omega)$ are a.s. bounded by

$$Y_r(\omega) \leq \sup_{\omega \in \Omega} \sqrt{\frac{f_{\tau}^{(4)}}{\Lambda}}(\omega) (\sqrt{12\tau})^r \left(\frac{\sqrt{d}}{\tau}\right)^r = \sup_{\omega \in \Omega} \sqrt{\frac{f_{\tau}^{(4)}}{\Lambda}}(\omega) (\sqrt{12}\sqrt{d})^r.$$

Denoting $C_{\Lambda} = \sup_{\omega \in \Omega} \sqrt{\frac{f_{\tau}^{(4)}}{\Lambda}}(\omega)$, we get that

$$\mathbb{P}_{\omega \sim \Lambda} \left(Y_r(\omega) \leq \sqrt{C_{\Lambda}} (\sqrt{12}\sqrt{d})^r \right) = 1,$$

so that ψ_{ω} satisfies Assumption 3 with $L_r := \sqrt{C_{\Lambda}} (\sqrt{12}\sqrt{d})^r$. The proof ends by plugging these values in N, C_1, C_2 and L_{ij} . $\square$

A.5 Proof of Theorem 1

By Theorem 5, we know that there exist pivot non-degenerate certificates. Let η^0 be an $(\bar{\varepsilon}_0, \bar{\varepsilon}_2, r_0)$ -pivot non-degenerate dual certificate. Note that

$$\frac{1}{2} \|\mathbf{y} - F\mu\|_{\mathcal{F}}^2 + \kappa \|\mu\|_{\text{TV}} \leq \frac{1}{2} \|\mathbf{y} - F\mu^0\|_{\mathcal{F}}^2 + \kappa \|\mu^0\|_{\text{TV}} \leq \frac{\gamma^2}{2} + \kappa \|\mu\|_{\text{TV}} \quad (\text{A.14a})$$

Using $\eta^0 = F^* c^0$ and (B.3), we obtain

$$\begin{aligned} \kappa \mathcal{D}_{\eta^0}(\mu \parallel \mu^0) + \kappa \langle \eta^0, \mu - \mu^0 \rangle_{\mathcal{C}(\mathcal{X}) \times \mathcal{M}(\mathcal{X})} + \frac{1}{2} \|\mathbf{y} - F\mu\|_{\mathcal{F}}^2 &\leq \frac{\gamma^2}{2} \\ \kappa \mathcal{D}_{\eta^0}(\mu \parallel \mu^0) + \frac{1}{2} \|\kappa c^0 + F\mu - \mathbf{y}\|_{\mathcal{F}}^2 &\leq \frac{\gamma^2}{2} + \frac{\kappa^2 \|c^0\|_{\mathcal{F}}^2}{2} - \langle \kappa c^0, \Gamma \rangle_{\mathcal{F}} \quad (\text{A.14b}) \\ \mathcal{D}_{\eta^0}(\mu \parallel \mu^0) &\leq \frac{(\gamma + \kappa \|c^0\|_{\mathcal{F}})^2}{2\kappa} \\ &\leq \frac{(\gamma + C_{\text{switch}} \kappa \sqrt{2s_0})^2}{2\kappa} \end{aligned}$$

where we recall that

$$\mathcal{D}_{\eta^0}(\mu \parallel \mu^0) := \|\mu\|_{\text{TV}} - \|\mu^0\|_{\text{TV}} - \langle \eta^0, \mu - \mu^0 \rangle_{\mathcal{C}(\mathcal{X}) \times \mathcal{M}(\mathcal{X})},$$

see (2.8), and $\Gamma := \mathbf{y} - F\mu^0$. Choosing $\kappa = c_\kappa\gamma/\sqrt{s_0}$ one gets

$$\mathcal{D}_{\eta^0}(\mu || \mu^0) \leq \bar{c}_\kappa \gamma \sqrt{s_0} \quad \text{where} \quad \bar{c}_\kappa := \frac{(1 + \sqrt{2} C_{\text{switch}} c_\kappa)^2}{2c_\kappa} \geq 2\sqrt{2} C_{\text{switch}}. \quad (\text{A.14c})$$

On the other hand, using that η^0 is an $(\bar{\varepsilon}_0, \bar{\varepsilon}_2, r_0)$ -pivot non-degenerate dual certificate, for $r \in (0, r_0]$,

$$\begin{aligned} \mathcal{D}_{\eta^0}(\mu || \mu^0) &= \|\mu\|_{\text{TV}} - \langle \eta^0, \mu \rangle_{\mathcal{C}(\mathcal{X}) \times \mathcal{M}(\mathcal{X})} \\ &\geq \|\mu\|_{\text{TV}} - \sum_{l=1}^{s_0} \int_{\mathcal{N}_l^{\text{reg}}(r)} |\eta^0| d|\mu| - \int_{\mathcal{F}_l^{\text{reg}}(r)} |\eta^0| d|\mu| \end{aligned}$$

Now observe that, for $r \in (0, r_0]$,

- For all $i \in [s_0]$, $\eta^0(t_i^0) = \text{sign}(a_i^0)$,
- For all $x \in \mathcal{F}^{\text{reg}}(r)$,
$$|\eta^0(x)| \leq 1 - \min(\bar{\varepsilon}_0, \bar{\varepsilon}_2 r^2) = 1 - \bar{\varepsilon}_2 r^2, \quad (\text{A.14d})$$
- For all $i \in [s_0]$, for all $x \in \mathcal{N}_i^{\text{reg}}(r)$, $|\eta^0(x)| \leq 1 - \bar{\varepsilon}_2 \mathfrak{d}_{\mathfrak{g}}(x, t_i^0)^2$.

Hence,

$$\mathcal{D}_{\eta^0}(\mu || \mu^0) \geq \bar{\varepsilon}_2 r^2 |\mu|(\mathcal{F}^{\text{reg}}(r)) + \bar{\varepsilon}_2 \sum_{l=1}^{s_0} \int_{\mathcal{N}_l^{\text{reg}}(r)} \mathfrak{d}_{\mathfrak{g}}(x, t_i^0)^2 d|\mu|(x). \quad (\text{A.14e})$$

Using (A.14c) and (A.14e), one gets (4.7a) and, by way of contradiction, (4.7c).

It remains to prove (4.7b). Observe that, using (A.14c) and (A.14d) and (A.14e),

$$\begin{aligned} |\mu(\mathcal{N}_j^{\text{reg}}(r)) - a_j^0| &= \left| \int_{\mathcal{N}_j^{\text{reg}}(r)} d(\mu - \mu^0) \right| \\ &= \left| \int_{\mathcal{X}} \eta_j^0 d(\mu - \mu^0) + \int_{\mathcal{N}_j^{\text{reg}}(r)} (1 - \eta_j^0) d(\mu - \mu^0) \right. \\ &\quad \left. - \sum_{l \neq j} \int_{\mathcal{N}_l^{\text{reg}}(r)} \eta_j^0 d(\mu - \mu^0) - \int_{\mathcal{F}^{\text{reg}}(r)} \eta_j^0 d(\mu - \mu^0) \right| \\ &\leq \left| \int_{\mathcal{X}} \eta_j^0 d(\mu - \mu^0) \right| + \bar{\varepsilon}_2 \sum_{l=1}^{s_0} \int_{\mathcal{N}_l^{\text{reg}}(r)} \mathfrak{d}_{\mathfrak{g}}(x, t_j^0)^2 d|\mu|(x) + (1 - \bar{\varepsilon}_2 r^2) |\mu|(\mathcal{F}^{\text{reg}}(r)) \\ &\leq \left| \int_{\mathcal{X}} \eta_j^0 d(\mu - \mu^0) \right| + \frac{c_r}{r^2} \left(\bar{\varepsilon}_2 r^2 |\mu|(\mathcal{F}^{\text{reg}}(r)) + \bar{\varepsilon}_2 \sum_{l=1}^{s_0} \int_{\mathcal{N}_l^{\text{reg}}(r)} \mathfrak{d}_{\mathfrak{g}}(x, t_i^0)^2 d|\mu|(x) \right) \\ &\leq \left| \int_{\mathcal{X}} \eta_j^0 d(\mu - \mu^0) \right| + c_r \bar{c}_\kappa \left(\frac{\gamma}{r^2} \right) \sqrt{s_0} \end{aligned} \quad (\text{A.14f})$$

where

$$c_r := \max(r^2, \frac{1 - \bar{\varepsilon}_2 r^2}{\bar{\varepsilon}_2}) \leq \frac{\max(1, \bar{\varepsilon}_0)}{\bar{\varepsilon}_2}.$$

Now, note that

$$\begin{aligned} \left| \int_{\mathcal{X}} \eta_j^0 d(\mu - f^0) \right| &= \left| \langle \eta_j^0, \mu - \mu^0 \rangle_{\mathcal{C}(\mathcal{X}) \times \mathcal{M}(\mathcal{X})} \right| \\ &= \left| \langle c_j^0, F\mu - F\mu^0 \rangle_{\mathcal{F}} \right| \\ &\leq \|c_j^0\|_{\mathcal{F}} (\gamma + \|\mathbf{y} - F\mu\|_{\mathcal{F}}) \end{aligned} \quad (\text{A.14g})$$

and, using (A.14b), one has

$$\|\kappa c^0 + F\mu - \mathbf{y}\|_{\mathcal{F}} \leq \gamma + \kappa \|c^0\|_{\mathcal{F}},$$

which leads to

$$\|F\mu - \mathbf{y}\|_{\mathcal{F}} \leq \gamma + 2\kappa \|c^0\|_{\mathcal{F}}. \quad (\text{A.14h})$$

By (A.14g) and (A.14h), one has

$$\left| \int_{\mathcal{X}} \eta_j^0 d(\mu - \mu^0) \right| \leq 2\|c_j^0\|_{\mathcal{F}}(\gamma + \kappa \|c^0\|_{\mathcal{F}}) \leq 2\sqrt{2} C_{\text{switch}}(1 + \sqrt{2} C_{\text{switch}} c_{\kappa})\gamma \quad (\text{A.14i})$$

using Theorem 5. Combining (A.14f) and (A.14i), one gets the result.

A.6 Proof of Theorem 2

The choice $C_{\text{sketch}} = 2C \max(C_1, C_2)$ together with Equation (4.8) ensures that $m \geq m_0$ from Assumption 3. First invoke Theorem 6 to get that there exists a constant $C'_{\text{pivot}} > 0$ which depends only on the kernel K_{pivot} such that, with probability at least $1 - \alpha$, for all $m \geq m_0$, there exist a $(\bar{\varepsilon}_0/4, 3\bar{\varepsilon}_2/2, r_0)$ -sketch pivot non-degenerate dual certificate such that $\|c^0\|_{\mathbb{C}^m} \leq C'_{\text{pivot}} C_{\text{switch}} \sqrt{s_0}$ and $(\bar{\varepsilon}_0/4, 3\bar{\varepsilon}_2/2, r_0)$ -sketch pivot non-degenerate localizing certificates at points t_i^0 such that $\|c_i^0\|_{\mathbb{C}^m} \leq C'_{\text{pivot}} C_{\text{switch}}$. Then the proof follows the same line as in Appendix A.5 substituting $\bar{\varepsilon}_0$ by $\bar{\varepsilon}_0/4$ and $\bar{\varepsilon}_2$ by $3\bar{\varepsilon}_2/2$.

A.7 Proof of Proposition 3.1

The proof is an application of Theorem 1 using the sinc-4 as pivot kernel. Let $\mu_0 \in \mathcal{M}(\mathcal{X})$ be a sparse measure of s_0 spikes $\{\mathbf{x}_1^0, \dots, \mathbf{x}_{s_0}^0\}$. First, recall Lemma 3.1 gives a control with probability $1 - \alpha$

$$\gamma_n := C_{\alpha} \left(\frac{1}{\tau} \right)^{\frac{d}{2}} \frac{1}{\sqrt{n}}, \quad (\text{A.15a})$$

where $C_{\alpha} := 2\sqrt{1 + C_1 \log(C_2/\alpha)}$ is a constant only depending on α , while C_1 and C_2 are universal constants.

By Theorem 3, the sinc-4 pivot kernel $K_{\text{pivot}}(\mathbf{s}, \mathbf{t}) = \Psi_{\tau}(\mathbf{s} - \mathbf{t})$ verifies the LPC assumption with radius $r_0 = 1/(4d)$, $\bar{\varepsilon}_0 \geq 1/(32d^3)$, $\bar{\varepsilon}_2 \geq 23/128$ and $\Delta_0 = 42.66 s_0^{1/4} d^{7/4}$, which are independent of the width τ . Moreover, its associated Fisher-Rao distance is a rescaled Euclidean $\mathfrak{d}_{\mathbf{g}_{\tau}}(\mathbf{s}, \mathbf{t}) = (2\sqrt{3}\tau)^{-1} \|\mathbf{s} - \mathbf{t}\|$. Thus, as discussed in Equation (4.10b), taking τ verifying Equation (3.3–H $_{\tau}$) implies that the target belongs to the model set $\mathcal{M}_{s_0, \Delta_0, \mathfrak{d}_{\tau}}$. Then, we may invoke Theorem 1 with regularisation

$$\kappa = c_{\kappa} \gamma / \sqrt{s_0} = \frac{1}{\sqrt{2} C_{\text{switch}}} C_{\alpha} \tau^{-\frac{d}{2}} \frac{1}{\sqrt{n}} \frac{1}{\sqrt{s_0}},$$

where we used the optimal proportionality constant $c_{\kappa} = 1/\sqrt{2} C_{\text{switch}}$ as discussed in Remark 4.2. Injecting this in Equations (4.7a) to (4.7c), one gets for any radius $r > 0$ such that $r < \min(r_0, \sqrt{\bar{\varepsilon}_0/\bar{\varepsilon}_2}) = \mathcal{O}(d^{-3/2}) =: c'_d$,

- Control of the far region:

$$|\mu|(\mathcal{F}^{\text{reg}}(r)) \leq 2\sqrt{2} C_{\text{switch}} \frac{128}{23} C_{\alpha} \tau^{-\frac{d}{2}} \left(\frac{1}{r^2 \sqrt{n}} \right) \sqrt{s_0}, \quad (\text{A.15b})$$

- Control of all the near regions: for all $k \in [s_0]$,

$$|\mu(\mathcal{N}_k^{\text{reg}}(r)) - a_k^0| \leq 2\sqrt{2} C_{\text{switch}} \frac{128}{23} C_{\alpha} \tau^{-\frac{d}{2}} \left(\frac{1}{r^2 \sqrt{n}} \right) \sqrt{s_0} + 4\sqrt{2} C_{\text{switch}} C_{\alpha} \tau^{-\frac{d}{2}} \frac{1}{\sqrt{n}}, \quad (\text{A.15c})$$

- Detection level: for all Borelian $A \subset \mathcal{X}$ such that $|\mu|(A) > 2\sqrt{2}C_{\text{switch}}\frac{128}{23}C_\alpha\tau^{-\frac{d}{2}}\left(\frac{1}{r^2\sqrt{n}}\right)\sqrt{s_0}$, there exists $\mathbf{x}_k^0$ such that

$$\min_{t \in A} \mathfrak{d}_{\mathfrak{g}}(t, \mathbf{x}_k^0) \leq r, \quad (\text{A.15d})$$

The last part of the proof consists in realizing that we may take the radius $r = r_n$ to depend on n . The only constraint is that $r_n^2\sqrt{n}$ goes to $+\infty$ as n grows so that the controls asymptotically make sense, *i.e.* the BLASSO estimator concentrates around the true support. Thus, we may consider a slowly diverging sequence δ_n such that $r_n^2\sqrt{n} = \delta_n^2$, or equivalently $r_n = n^{-1/4}\delta_n$. For comments over the choice of δ_n , we refer to Remark 3.4. Note that, for large n , $\delta_n^2 = r_n^2\sqrt{n} < \sqrt{n}$ so that the first term on the right-hand side of Equation (A.15c) is the dominant one. Hence, we obtain the final control in Equation (3.9b)

$$|\mu(\mathcal{N}_k^{\text{reg}}(r)) - a_k^0| \leq 6\sqrt{2}C_{\text{switch}}\frac{128}{23}C_\alpha\tau^{-\frac{d}{2}}\left(\frac{1}{r^2\sqrt{n}}\right)\sqrt{s_0}.$$

Finally, the condition $r_n < c'_d$ can be restated implicitly in n as $n \geq (c'_d)^{-4}\delta_n^4$, and the constant $c_d := (c'_d)^{-4} = \mathcal{O}(d^6)$ depends polynomially on the dimension. This concludes the proof.

A.8 Proof of Proposition 3.2

The proof is an application of Theorem 2 which gives the controls for sketched BLASSO problems. We first prove the control on the sketch noise level and then turn to proving the actual proposition.

A.8.1 Proof of Lemma 3.2

First, we recall and prove Lemma 3.2 giving the control over the sketched noise level with probability $1 - \alpha$ w.r.t. the joint draws of the sample $\mathbf{Z}$ and the sketch. Precisely, we prove that with probability $1 - \alpha$ it holds that

$$\|\Gamma_{\text{sketch}}\|_{\mathbb{C}^m} \leq C_{\alpha,m} \frac{1}{\sqrt{n}},$$

with

$$C_{\alpha,m} = 2\sqrt{\left[\left(\frac{1}{\tau}\right)^d + \frac{1}{2\sqrt{m}}\left\|\frac{U_\tau}{\Lambda}\right\|_\infty \log\left(\frac{2}{\alpha}\right)\right]\left[1 + C_1 \log\left(\frac{2C_2}{\alpha}\right)\right]},$$

and C_1, C_2 are universal constants.

We follow the proof of De Castro et al. [2021, Lemma 3] and adapt it with a union bound argument to take into account the extra-randomness from the sketch. First, introduce the i.i.d. random variables $(\mathbf{Y}_j)_{j=1}^n$

$$\mathbf{Y}_j := \mathcal{A}\delta_{\mathbf{Z}_j} - \mathbb{E}_{\mathbf{Z}} \mathcal{A}\delta_{\mathbf{Z}},$$

where $\mathcal{A} : \nu \in \mathcal{M}(\mathbb{R}^d) \mapsto \frac{1}{\sqrt{m}}(W(\omega_i) \mathfrak{F}[\nu](\omega_i))_{i=1}^m \in \mathbb{C}^m$ is the so-called *sketching* operator yielding the random Fourier features of a Radon measure ν . Then, we have

$$\left\|\Gamma_{\text{sketch}}\right\|_{\mathbb{C}^m}^2 = \left\|\frac{1}{n} \sum_{j=1}^n \mathbf{Y}_j\right\|_{\mathbb{C}^m}^2 = \frac{1}{n^2} \sum_{j=1}^n \|\mathbf{Y}_j\|_{\mathbb{C}^m}^2 + \frac{1}{n^2} \sum_{j \neq l}^n \langle \mathbf{Y}_j, \mathbf{Y}_l \rangle_{\mathbb{C}^m}, \quad (\text{A.16a})$$

Fix $\alpha \in (0, 1)$, and let us assume that there exists a constant M such that $\|\mathbf{Y}_j\|_{\mathbb{C}^m}^2 \leq M$. Then, using the same arguments as in the proof of De Castro et al. [2021, Lemma 3], the first term of Equation (A.16a) is bounded by M/n while the second term is a canonical U-process. Thus, Arcones and Giné [1993, Proposition 2.3] ensures that there exist two universal constants $C_1, C_2 > 0$ such that

$$\left\|\Gamma_{\text{sketch}}\right\|_{\mathbb{C}^m}^2 \leq M \left(1 + C_1 \log\left(\frac{2C_2}{\alpha}\right)\right) \frac{1}{n} \quad (\text{A.16b})$$

with probability at least $1 - \frac{\alpha}{2}$.

We then proceed to show that $\|\mathbf{Y}_j\|_{\mathbb{C}^m}^2$ is bounded with probability $1 - \alpha/2$ using Hoeffding's inequality. Indeed, for any $j \in [n]$ we may write

$$\|\mathbf{Y}_j\|_{\mathbb{C}^m}^2 = \frac{1}{m} \sum_{i=1}^m \left| W(\boldsymbol{\omega}_i) \left(e^{-i\boldsymbol{\omega}_i^\top \mathbf{Z}_j} - \mathfrak{F}[f^0](\boldsymbol{\omega}_i) \right) \right|^2 = \frac{1}{m} \sum_{i=1}^m X_i^j,$$

with $X_i^j = \left| W(\boldsymbol{\omega}_i) \left(e^{-i\boldsymbol{\omega}_i^\top \mathbf{Z}_j} - \mathfrak{F}[f^0](\boldsymbol{\omega}_i) \right) \right|^2$. Moreover, the expectation with respect to the draw of $\boldsymbol{\omega}_{1:m}$ is

$$\mathbb{E}_{\boldsymbol{\omega}_{1:m}} \|\mathbf{Y}_j\|_{\mathbb{C}^m}^2 = \|L_\tau \delta_{\mathbf{Z}_j} - L_\tau f^0\|_{\mathcal{F}_\tau}^2,$$

where we recall the operator $L_\tau \nu = \lambda_\tau \star \nu$ is the smoothing kernel operator, which can be seen as the kernel mean embedding of ν in the RKHS $\mathcal{F}_\tau$. The $(X_i^j)_{i=1}^m$ are independent and bounded random variables with

$$0 \leq X_i^j \leq W^2(\boldsymbol{\omega}_i) \left(|e^{-i\boldsymbol{\omega}_i^\top \mathbf{Z}_j}| + |\mathfrak{F}[f^0](\boldsymbol{\omega}_i)| \right)^2 \leq 4\|W^2\|_\infty,$$

using Jensen's inequality to get that

$$|\mathfrak{F}[f^0](\boldsymbol{\omega}_i)| = |\mathbb{E}_{\mathbf{Z}} e^{-i\boldsymbol{\omega}_i^\top \mathbf{Z}}| \leq \mathbb{E}_{\mathbf{Z}} |e^{-i\boldsymbol{\omega}_i^\top \mathbf{Z}}| = 1.$$

Applying Hoeffding's inequality yields that with probability greater than $1 - \alpha/2$

$$\|\mathbf{Y}_j\|_{\mathbb{C}^m}^2 \leq \|L_\tau \delta_{\mathbf{Z}_j} - L_\tau f^0\|_{\mathcal{F}_\tau}^2 + \frac{2\|W^2\|_\infty}{\sqrt{m}} \log\left(\frac{2}{\alpha}\right)$$

Now observe that

$$\begin{aligned} \|L_\tau \delta_{\mathbf{Z}_j}\|_{\mathcal{F}_\tau}^2 &= \lambda_\tau(\mathbf{Z}_j - \mathbf{Z}_j) = \lambda_\tau(\mathbf{0}) \\ \|L_\tau f^0\|_{\mathcal{F}_\tau}^2 &\leq \mathbb{E}_{\mathbf{Z}} \|L_\tau \delta_{\mathbf{Z}}\|_{\mathcal{F}_\tau}^2 = \lambda_\tau(\mathbf{0}) \end{aligned}$$

by the reproducing property of the RKHS $\mathcal{F}_\tau$ and Jensen's inequality. We deduce that

$$\|L_\tau \delta_{\mathbf{Z}_j} - L_\tau f^0\|_{\mathcal{F}_\tau}^2 \leq 4\lambda_\tau(\mathbf{0}),$$

so that we can conclude

$$\|\mathbf{Y}_j\|_{\mathbb{C}^m}^2 \leq M_{\alpha,m} := 4\lambda_\tau(\mathbf{0}) + \frac{2\|W^2\|_\infty}{\sqrt{m}} \log\left(\frac{2}{\alpha}\right) \quad (\text{A.16c})$$

with probability greater than $1 - \alpha/2$. A union bound yields that the joint probability of Equations (A.16b) and (A.16c) is greater than $1 - \alpha$. The proof ends by plugging $\lambda_\tau(\mathbf{0}) = (\frac{1}{\tau})^d$ and $W^2 = \frac{U_\tau}{\Lambda}$.

A.8.2 Proof of the main proposition

The proof of Proposition 3.2 follows the same line as the proof of Proposition 3.1 given above in Section A.7. First, remark that Theorem 4 ensures the sinc-4 kernel verifies Assumption 3 and is thus amenable to sketching with the following constants:

$$\begin{aligned} m_0 &= C s_0 \left(C_1 \log(s_0) \log\left(\frac{s_0}{\alpha}\right) + C_2 \log\left(\frac{(s_0 N)^d}{\alpha}\right) \right) \\ N &:= |\mathcal{X}| 32\sqrt{12}\sqrt{C_\Lambda} d^{7/2} + \frac{128}{23} (12\sqrt{12}C_\Lambda d^{1/2} + \sqrt{C_\Lambda} 12d) \\ C_1 &= \mathcal{O}\left(d^{15/2} C_\Lambda\right) \\ C_2 &= \mathcal{O}\left(d^6 C_\Lambda\right) \\ C_\Lambda &:= \sup_{\boldsymbol{\omega} \in [-\frac{1}{\tau}, \frac{1}{\tau}]^d} \frac{f_\tau^{(4)}(\boldsymbol{\omega})}{\Lambda} \end{aligned}$$

where the diameter $|\mathcal{X}| = \sup_{\mathbf{s}, \mathbf{t} \in \mathcal{X}} \mathfrak{d}_{\mathbf{g}_\tau}(\mathbf{s}, \mathbf{t})$, and C depends on Ψ_τ and polynomially on d .

With our hypothesis on the width τ Equation (3.3–H $_\tau$), the target μ^0 belongs to the model set $\mathcal{M}_{s_0, \Delta_0, \mathfrak{d}_\tau}$ where $\mathfrak{d}_\tau$ is the Fisher-Rao metric of the sinc-4 kernel Ψ_τ (see Equation (4.10b)). We can then invoke Theorem 2, which ensures a positive constant C'_{pivot} exists and yields the sketched BLASSO controls with high-probability. Precisely, take the sketch size $m \geq m_0$, and use the control on the sketched noise level γ_{sketch} to tune the regularisation

$$\kappa = \frac{c'_\kappa \gamma_{\text{sketch}}}{\sqrt{s_0}} = \frac{1}{C'_{\text{pivot}} C_{\text{switch}}} \frac{C_{\alpha, m}}{\sqrt{n}} \frac{1}{\sqrt{s_0}},$$

where we used the optimal $c'_\kappa = \frac{1}{C'_{\text{pivot}} C_{\text{switch}}}$. Injecting this in Equations (4.9a) to (4.9c), one gets for any radius $r > 0$ such that $r < \min(r_0, \sqrt{\varepsilon_0/6\varepsilon_2}) =: c'_d$,

- Control of the far region:

$$|\mu(\mathcal{F}^{\text{reg}}(r))| \leq 2C'_{\text{pivot}} C_{\text{switch}} \frac{256}{69} C_{\alpha, m} \left(\frac{1}{r^2 \sqrt{n}} \right) \sqrt{s_0}, \quad (\text{A.17a})$$

- Control of all the near regions: for all $k \in [s_0]$,

$$|\mu(\mathcal{N}_k^{\text{reg}}(r)) - a_k^0| \leq 2C'_{\text{pivot}} C_{\text{switch}} \frac{256}{69} C_{\alpha, m} \left(\frac{1}{r^2 \sqrt{n}} \right) \sqrt{s_0} + 4C'_{\text{pivot}} C_{\text{switch}} \frac{C_{\alpha, m}}{\sqrt{n}}, \quad (\text{A.17b})$$

- Detection level: for all Borelian $A \subset \mathcal{X}$ such that $|\mu|(A) > 2C'_{\text{pivot}} C_{\text{switch}} \frac{256}{69} C_{\alpha, m} \left(\frac{1}{r^2 \sqrt{n}} \right) \sqrt{s_0}$, there exists $\mathbf{x}_k^0$ such that

$$\min_{\mathbf{t} \in A} \mathfrak{d}_{\mathbf{g}}(\mathbf{t}, \mathbf{x}_k^0) \leq r, \quad (\text{A.17c})$$

Using a similar reasoning as in Section A.7, we finish the proof by taking the radius $r = r_n$ to depend on n . Let δ_n be a slowly diverging sequence such that $r_n^2 \sqrt{n} = \delta_n^2$, or equivalently $r_n = n^{-1/4} \delta_n$. Again, the first term on the right-hand side of Equation (A.17b) is the dominant one. Hence, we obtain the final control in Equation (3.9b)

$$|\mu(\mathcal{N}_k^{\text{reg}}(r)) - a_k^0| \leq 6C'_{\text{pivot}} C_{\text{switch}} \frac{256}{69} C_{\alpha, m} \left(\frac{1}{r^2 \sqrt{n}} \right) \sqrt{s_0}.$$

Again, the condition on the radius $r_n < c'_d$ is met for any n such that $n \geq (c'_d)^{-4} \delta_n^4$, with $c_d := (c'_d)^{-4} = \mathcal{O}(d^6)$ scaling polynomially in the dimension d . This concludes the proof.

B Technical results and remarks

B.1 Covariant derivatives: the translation invariant kernel case

The covariant derivative of the kernel $K(\mathbf{s}, \mathbf{t}) = \mathbb{E}_{\omega \sim \Lambda}[\psi_\omega(\mathbf{s}) \overline{\psi_\omega(\mathbf{t})}]$ is properly introduced by Poon et al. [2023, Section 4.1] as a “bi”-multilinear map $K^{(i,j)}(\mathbf{s}, \mathbf{t}) : (\mathbb{C}^d)^i \times (\mathbb{C}^d)^j \rightarrow \mathbb{C}$, and it involves the covariant derivatives of order i and j of the feature map $\psi_\omega : \mathbb{R}^d \rightarrow \mathbb{C}$. In the case of translation invariant kernels $K(\mathbf{s}, \mathbf{t}) = \rho(\mathbf{s} - \mathbf{t})$, the Riemannian covariant derivatives $K^{(i,j)}(\mathbf{s}, \mathbf{t})[U, V]$ and its Riemannian operator norm $\|K^{(i,j)}(\mathbf{s}, \mathbf{t})\|_{\mathbf{s}, \mathbf{t}}$ are given by the standard Euclidean derivatives $(-1)^j \nabla^{i+j} \rho(\mathbf{h})$ of ρ at point $\mathbf{h} = \mathbf{s} - \mathbf{t}$ and the Euclidean operator norm, namely

$$K^{(i,j)}(\mathbf{s}, \mathbf{t})[U, V] = (-1)^j \text{Trace}(\nabla^{i+j} \rho(\mathbf{h}) \times U_1 \otimes \cdots \otimes U_i \otimes V_1 \otimes \cdots \otimes V_j), \quad (\text{B.1a})$$

$$\|K^{(i,j)}(\mathbf{s}, \mathbf{t})\|_{\mathbf{s}, \mathbf{t}} = \left\| \nabla^{i+j} \rho(\mathbf{h}) \times_{l=1}^{i+j} \mathbf{g}^{-\frac{1}{2}} \right\|_2. \quad (\text{B.1b})$$

where $U \in (\mathbb{C}^d)^i$ (resp. $V \in (\mathbb{C}^d)^j$) are i (resp. j) tangent vectors at points $\mathbf{s}$ (resp. $\mathbf{t}$), $\otimes$ denotes the tensor product and

$$\forall T \in (\mathbb{C}^d)^{i+j}, \quad (\nabla^{i+j} \rho(h) \bigotimes_{l=1}^{i+j} \mathbf{g}^{-\frac{1}{2}})[T] := \nabla^{i+j} \rho(h) [\mathbf{g}^{-\frac{1}{2}} T_1, \dots, \mathbf{g}^{-\frac{1}{2}} T_{i+j}]. \quad (\text{B.1c})$$

In particular, one has

$$T_1^\top \nabla \rho(h) = K^{(1,0)}(\mathbf{s}, \mathbf{t})[T_1] = -K^{(0,1)}(\mathbf{s}, \mathbf{t})[T_1], \quad (\text{B.1d})$$

$$T_1^\top \nabla^2 \rho(h) T_2 = K^{(2,0)}(\mathbf{s}, \mathbf{t})[T_1, T_2] = K^{(0,2)}(\mathbf{s}, \mathbf{t})[T_1, T_2] = -K^{(1,1)}(\mathbf{s}, \mathbf{t})[T_1, T_2], \quad (\text{B.1e})$$

$$\left\| \mathbf{g}^{-\frac{1}{2}} \nabla \rho(h) \right\|_2 = \|K^{(1,0)}(\mathbf{s}, \mathbf{t})\|_{\mathbf{s}}, \quad (\text{B.1f})$$

$$\left\| \mathbf{g}^{-\frac{1}{2}} \nabla^2 \rho(h) \mathbf{g}^{-\frac{1}{2}} \right\|_2 = \|K^{(2,0)}(\mathbf{s}, \mathbf{t})\|_{\mathbf{s}, \mathbf{t}}. \quad (\text{B.1g})$$

B.2 Model RKHS and dual operator

Proposition B.1. Assume (2.3– $\mathbf{H}_{\text{mod}}$) holds true.

- The unique RKHS of K_{mod} is given by

$$\mathcal{H}_{\text{mod}} = \{\eta : \mathcal{X} \rightarrow \mathbb{R} \mid \exists c \in \mathcal{F}, \eta = \eta_c\} = \{\eta : \mathcal{X} \rightarrow \mathbb{R} \mid \exists! c \in \overline{\text{Im}(F)}, \eta = \eta_c\}, \quad (\text{B.2})$$

and for all c orthogonal to $\overline{\text{Im}(F)}$ in $\mathcal{F}$, $\eta_c = 0$.

- The isometry is given by the mapping $c \in \overline{\text{Im}(F)} \mapsto \eta_c \in \mathcal{H}_{\text{mod}}$.
- The norms satisfy

$$\forall c \in \overline{\text{Im}(F)}, \quad \|\eta_c\|_{\mathcal{H}_{\text{mod}}} = \|c\|_{\mathcal{F}}.$$

Proof. See [Steinwart and Christmann, 2008, Theorem 4.21]. $\square$

B.3 Proof of Proposition 4.1

By Assumption (2.3– $\mathbf{H}_{\text{mod}}$), it holds that $\mathcal{H}_{\text{mod}} \subset \mathcal{C}(\mathcal{X})$. Define $\text{Id}_{\text{mod}} : \mathcal{H}_{\text{mod}} \rightarrow \mathcal{C}(\mathcal{X})$ the identity mapping. One has

$$F^* c = \text{Id}_{\text{mod}}(\eta_c), \text{ for all } c \in \overline{\text{Im}(F)}, \quad (\text{B.3a})$$

$$\langle \text{Id}_{\text{mod}}(\eta_c), \nu \rangle_{\mathcal{C}(\mathcal{X}), \mathcal{M}(\mathcal{X})} = \langle c, F\nu \rangle_{\mathcal{F}}, \text{ for all } c \in \mathcal{F}, \nu \in \mathcal{M}(\mathcal{X}). \quad (\text{B.3b})$$

The result follows from (B.3) and Proposition B.1. For the sake of readability, we denote $\text{Id}_{\text{mod}}(\eta_c)$ by η_c .

C Functional framework

C.1 Radon measures

Recall that we consider sparse models built from measures in the form of Equation (2.4).

Definition 7 (Set $(\mathcal{M}(\mathbb{R}^d), \|\cdot\|_{\text{TV}})$). We work in the vector space $\mathcal{M}(\mathbb{R}^d)$ of finite signed Radon measures on $\mathbb{R}^d$, endowed with the total variation norm:

$$\|\mu\|_{\text{TV}} := \int_{\mathbb{R}^d} d|\mu|,$$

where $|\mu| = \mu^+ + \mu^-$ and $\mu = \mu^+ - \mu^-$ is the Jordan decomposition of $\mu \in \mathcal{M}(\mathbb{R}^d)$.

By the Riesz-Markov theorem, this space is the dual of real-valued continuous functions on $\mathbb{R}^d$ *vanishing at infinity*³ equipped with the supremum norm, which we denote as $(C_0(\mathbb{R}^d, \mathbb{R}), \|\cdot\|_\infty)$ or $(C_0(\mathbb{R}^d), \|\cdot\|_\infty)$. The total variation norm then admits a variational formulation as:

$$\|\mu\|_{\text{TV}} = \sup \left\{ \int \eta \, d\mu; \eta \in \mathcal{C}(\mathbb{R}^d), \|\eta\|_\infty \leq 1 \right\}, \quad (\text{C.1})$$

C.2 Fourier transform

Here, we detail the definition of the Fourier transform and its inverse used throughout the paper.

Definition 8 (Fourier transform). *For any $g \in L^1(\mathbb{R}^d)$, we define its Fourier transform $\mathfrak{F}[g]$ and its inverse as*

$$\forall \omega \in \mathbb{R}^d, \quad \mathfrak{F}[g](\omega) = \int_{\mathbb{R}^d} g(z) e^{-i\omega^\top z} \, dz. \quad (\text{C.2a})$$

Furthermore, for g continuous and both g and $\mathfrak{F}[g]$ in $L^1(\mathbb{R}^d)$, we define the inverse transform

$$\forall z \in \mathbb{R}^d, \quad \mathfrak{F}^{-1}[\mathfrak{F}[g]](z) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \mathfrak{F}[g](\omega) e^{i\omega^\top z} \, d\omega. \quad (\text{C.2b})$$

C.3 Reproducing kernel Hilbert space of translation invariant kernel

In this subsection, we remind particular results for translation invariant kernels $k(\mathbf{s}, \mathbf{t}) = \rho(\mathbf{t} - \mathbf{s})$ and their characterisation in the Fourier domain. We begin by stating Bochner's theorem, a well known result characterizing functions of the positive definite type as the characteristic function of a positive measure, coined the *spectral measure*.

Theorem 7 (Bochner). *Let $k(\mathbf{s}, \mathbf{t}) = \rho(\mathbf{t} - \mathbf{s})$. Then k is a reproducing kernel if and only if there exists a positive measure ν such that*

$$\rho(\mathbf{t} - \mathbf{s}) = \int e^{+i\omega^\top(\mathbf{t}-\mathbf{s})} \, d\nu(\omega).$$

Furthermore, if the kernel is normalized, i.e. $\rho(\mathbf{0}) = 1$, then ν is a probability distribution.

This allows us to define the RKHS associated to the translation invariant kernel ρ as functions with sufficiently fast decay rate in the Fourier domain.

Proposition C.1 (RKHS of translation invariant kernels). *The RKHS of the translation invariant kernel $k(\mathbf{s}, \mathbf{t}) = \rho(\mathbf{t} - \mathbf{s})$ is defined as*

$$\mathcal{H}_\rho := \left\{ g \in L^2(\mathcal{X}) : \|g\|_{\mathcal{H}_\rho}^2 = \frac{1}{(2\pi)^d} \int_{\mathcal{X}} \frac{|\mathfrak{F}[g](\omega)|^2}{\mathfrak{F}[\rho](\omega)} \, d\omega \right\}$$

with scalar product

$$\langle g_1, g_2 \rangle_{\mathcal{H}_\rho} = \frac{1}{(2\pi)^d} \int \frac{\mathfrak{F}[g_1](\omega) \overline{\mathfrak{F}[g_2](\omega)}}{\mathfrak{F}[\rho](\omega)} \, d\omega$$

Element of proof. A formal proof is given in Wendland [2005, Theorem 10.12, p. 139], here we focus on proving the reproducing property. Moreover, we refer to Bach [2024, Section 7.3.3, p.192] for readers interested in an intuitive construction of the RKHS norm.

³ f vanishes at infinity iff $\forall \varepsilon > 0$, there exists a compact $K_\varepsilon \subset \mathbb{R}^d$ such that $\forall x \in K_\varepsilon^c, |f(x)| < \varepsilon$

First, we prove that $(\mathcal{H}_\rho, \langle \cdot, \cdot \rangle_{\mathcal{H}_\rho})$ contains functions of the form $k(\mathbf{s}, \cdot)$. To see this, one needs to remember that the Fourier transform is covariant by translation, meaning $\mathfrak{F}[f(\cdot - \mathbf{s})](\boldsymbol{\omega}) = e^{-i\boldsymbol{\omega}^\top \mathbf{s}} \mathfrak{F}[f](\boldsymbol{\omega})$. Hence, we have that

$$\|k(\mathbf{s}, \cdot)\|_{\mathcal{H}_\rho} = \|\rho(\cdot - \mathbf{s})\|_{\mathcal{H}_\rho} = \frac{1}{(2\pi)^d} \int \frac{|\mathfrak{F}[\rho](\boldsymbol{\omega})|^2}{\mathfrak{F}[\rho](\boldsymbol{\omega})} |e^{-i\boldsymbol{\omega}^\top \mathbf{s}}|^2 d\boldsymbol{\omega} = \int \frac{\mathfrak{F}[\rho](\boldsymbol{\omega})}{(2\pi)^d} d\boldsymbol{\omega} = \rho(\mathbf{0}) < +\infty.$$

Thus, $k(\mathbf{s}, \cdot)$ belongs to the RKHS. Following the same line, we can prove that the reproducing property holds. Let $g \in \mathcal{H}_\rho$ and $\mathbf{s} \in \mathbb{R}^d$

$$\langle g, \rho(\cdot - \mathbf{s}) \rangle_{\mathcal{H}_\rho} = \frac{1}{(2\pi)^d} \int \frac{\mathfrak{F}[g](\boldsymbol{\omega}) \overline{\mathfrak{F}[\rho](\boldsymbol{\omega})}}{\mathfrak{F}[\rho](\boldsymbol{\omega})} e^{+i\boldsymbol{\omega}^\top \mathbf{s}} d\boldsymbol{\omega} = \mathfrak{F}^{-1}[\mathfrak{F}[g]](\mathbf{s}) = g(\mathbf{s}).$$

Completeness is treated in [Wendland \[2005, Theorem 10.12, p. 139\]](#). □

Remark C.1 (Link between the Fourier transform of ρ and its spectral measure ν). *With the choice of convention introduced in Section C.2 for the inverse Fourier transform, the following identity holds*

$$\rho = (2\pi)^d \mathfrak{F}^{-1}[\nu] \iff \nu = \frac{\mathfrak{F}[\rho]}{(2\pi)^d}.$$

As a consequence we have

$$\begin{aligned} \|\rho \star g\|_{\mathcal{H}_\rho}^2 &= \frac{1}{(2\pi)^d} \int_{\mathcal{X}} \frac{|\mathfrak{F}[\rho](\boldsymbol{\omega})|^2 |\mathfrak{F}[g](\boldsymbol{\omega})|^2}{\mathfrak{F}[\rho](\boldsymbol{\omega})} d\boldsymbol{\omega}, \\ &= \int_{\mathcal{X}} \frac{\mathfrak{F}[\rho](\boldsymbol{\omega})}{(2\pi)^d} |\mathfrak{F}[g](\boldsymbol{\omega})|^2 d\boldsymbol{\omega}, \\ &= \int_{\mathcal{X}} \nu(\boldsymbol{\omega}) |\mathfrak{F}[g](\boldsymbol{\omega})|^2 d\boldsymbol{\omega}, \end{aligned}$$

If the kernel is normalized, i.e. $\rho(0) = 1$, then ν is a probability distribution and the last line can be written as an expectation.

C.4 Fourier transform and spectral measure of the sinus cardinal smoothing kernel λ_τ

We recall our choice of smoothing kernel λ_τ which is the rescaled sinus cardinal

$$\lambda_\tau(\mathbf{x}) = \left(\frac{1}{\tau}\right)^d \text{sinc}\left(\frac{\mathbf{x}}{\tau}\right),$$

with its spectral measure from Theorem 7 denoted as U_τ . Denoting $\text{sinc}_\tau = \text{sinc}(\cdot/\tau)$, we have with our choice of Fourier conventions that $\Pi_1(\boldsymbol{\omega}) := \mathfrak{F}[\text{sinc}_1] = \pi^d \mathbb{1}_{[-1,1]^d}$. Thus, the Fourier transform of its scaled version is the gate function

$$\Pi_\tau(\boldsymbol{\omega}) := \mathfrak{F}[\text{sinc}_\tau](\boldsymbol{\omega}) = \mathfrak{F}[\text{sinc}_1(\cdot/\tau)](\boldsymbol{\omega}) = (\pi\tau)^d \mathbb{1}_{[-1/\tau, 1/\tau]^d}(\boldsymbol{\omega}),$$

with a frequency cut-off as $1/\tau$. Finally, by linearity, the Fourier transform of the smoothing kernel is

$$\mathfrak{F}[\lambda_\tau](\boldsymbol{\omega}) = \pi^d \mathbb{1}_{[-1/\tau, 1/\tau]^d}(\boldsymbol{\omega}), \tag{C.3}$$

and its spectral measure is

$$U_\tau(\boldsymbol{\omega}) = \frac{\mathfrak{F}[\lambda_\tau]}{(2\pi)^d}(\boldsymbol{\omega}) = \frac{1}{2^d} \mathbb{1}_{[-1/\tau, 1/\tau]^d}(\boldsymbol{\omega}). \tag{C.4}$$

C.5 Spectral measure of the sinc-4 kernel

The sinc-4 kernel $\Psi_\tau(\mathbf{x} - \mathbf{y}) = \text{sinc}(\frac{\mathbf{x} - \mathbf{y}}{4\tau})^4 = \text{sinc}_{4\tau}(\mathbf{x} - \mathbf{y})^4$ defined in Equation (2.9d) is a translation invariant kernel with associated RKHS given by

$$\mathcal{H}_{\text{sinc}} := \left\{ g : \mathbb{R}^d \rightarrow \mathbb{R} \text{ s.t. } \|g\|_{\mathcal{H}_{\text{sinc}}}^2 = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \frac{|\mathfrak{F}[g]|^2}{\mathfrak{F}[\Psi_\tau]} < +\infty \right\}. \quad (\text{C.5})$$

We denote its spectral measure as $f_\tau^{(4)} = \mathfrak{F}[\Psi_\tau]/(2\pi)^d$, which verifies by Bochner's theorem (Theorem 7)

$$\Psi_\tau(\mathbf{x}) = \int_{\mathbb{R}^d} e^{i\mathbf{x}^\top \boldsymbol{\omega}} f_\tau^{(4)}(\boldsymbol{\omega}) d\boldsymbol{\omega}.$$

Since $\Psi_\tau(\mathbf{0}) = 1$, then $f_\tau^{(4)}$ is a normalized probability density function. Moreover, it has an explicit analytical form. Indeed, remember that with our choice of Fourier conventions, the function $\text{sinc}_{4\tau} = \text{sinc}(\cdot/4\tau)$ is the inverse Fourier transform of the rescaled low-pass filter $\Pi_{4\tau}(\boldsymbol{\omega}) = (4\pi\tau)^d \mathbb{1}_{[-1/4\tau, 1/4\tau]^d}(\boldsymbol{\omega})$. Thus, $f_\tau^{(4)}$ is proportional to $\Pi_{4\tau}^{(\star 4)}$, which we define as the convolution of $\Pi_{4\tau}$ with itself taken four times. The proportionality constant is such that $f_\tau^{(4)}$ is a normalized probability density function. Precisely, we know that $f_\tau^{(4)}$ is even since Ψ_τ is even and real-valued, so that

$$f_\tau^{(4)} = \check{f}_\tau^{(4)} = \mathfrak{F}^{-1}[\Psi_\tau] = \mathfrak{F}^{-1}[\text{sinc}_{4\tau}^4] = \mathfrak{F}^{-1}[\mathfrak{F}^{-1}[\Pi_{4\tau}]^4] = (2\pi)^{-4d} \mathfrak{F}^{-1}[\mathfrak{F}[\Pi_{4\tau}](-\cdot)^4] = (2\pi)^{-4d} \Pi_{4\tau}^{(\star 4)}$$

where we used that $g^{(\star n)} = \mathfrak{F}^{-1}[\mathfrak{F}[g]^n]$ and $\mathfrak{F}[\Pi_{4\tau}]$ is even.

Note that $f_\tau^{(4)}$ may also be interpreted as the density of the sum of independent uniform random variables on $[-1/4\tau, 1/4\tau]^d$, also known as the (rescaled) Irwin-Hall distribution of order 4. It is even by symmetry around $\mathbf{0}$ and its expression is

$$\forall \boldsymbol{\omega} \in [-1/\tau, 1/\tau]^d, \quad f_\tau^{(4)}(\boldsymbol{\omega}) = (2\tau)^d \prod_{j=1}^d \frac{1}{6} \sum_{k=1}^{\lfloor 2\tau\omega_j + 2 \rfloor} (-1)^k \binom{4}{k} (2\tau\omega_j + 2 - k)^4 \quad (\text{C.6})$$

Moreover, its mode is in $\boldsymbol{\omega} = \mathbf{0}$ and its maximum value $(\frac{2}{3})^d (2\tau)^d$ so that

$$\sup_{\boldsymbol{\omega} \in [-1/\tau, 1/\tau]^d} \mathfrak{F}[\Psi_\tau](\boldsymbol{\omega}) = (2\pi)^d \sup_{\boldsymbol{\omega} \in [-1/\tau, 1/\tau]^d} f_\tau^{(4)}(\boldsymbol{\omega}) = (2\pi)^d \left(\frac{2}{3}\right)^d (2\tau)^d = \mathcal{O}(\tau^d)$$

C.6 Kernel switch between translation invariant RKHS

A direct consequence of Bochner's theorem and Proposition C.1 is that we can explicitly give the embedding constant C_{switch} between the RKHS of two translation invariant kernels. This is discussed *e.g.* in Zhang and Zhao [2013, Corollary 3.2] and the constant is precisely the essential supremum of the ratio of their Fourier transform as introduced in Equation (3.6– $\mathbf{H}_\phi$).

Consider two continuous and translation-invariant kernels:

$$K_{\text{pivot}}(\mathbf{s}, \mathbf{t}) = \rho_{\text{pivot}}(\mathbf{t} - \mathbf{s}) = \frac{1}{(2\pi)^d} \int e^{i\boldsymbol{\omega}^\top (\mathbf{t} - \mathbf{s})} \mathfrak{F}[\rho_{\text{pivot}}](\boldsymbol{\omega}) d\boldsymbol{\omega},$$

$$K_{\text{mod}}(\mathbf{s}, \mathbf{t}) = \rho_{\text{mod}}(\mathbf{t} - \mathbf{s}) = \frac{1}{(2\pi)^d} \int e^{i\boldsymbol{\omega}^\top (\mathbf{t} - \mathbf{s})} \mathfrak{F}[\rho_{\text{mod}}](\boldsymbol{\omega}) d\boldsymbol{\omega}$$

Then, assuming the support of $\mathfrak{F}[\rho_{\text{mod}}]$ contains that of $\mathfrak{F}[\rho_{\text{pivot}}]$, we have for any $\eta \in \mathcal{H}_{\text{pivot}}$

that

$$\begin{aligned}
\|\eta\|_{\mathcal{H}_{\text{mod}}}^2 &= \frac{1}{(2\pi)^d} \int \frac{|\mathfrak{F}[\eta]|^2}{\mathfrak{F}[\rho_{\text{mod}}]}(\omega) d\omega \\
&= \frac{1}{(2\pi)^d} \int \frac{\mathfrak{F}[\rho_{\text{pivot}}]}{\mathfrak{F}[\rho_{\text{mod}}]}(\omega) \frac{|\mathfrak{F}[\eta]|^2}{\mathfrak{F}[\rho_{\text{pivot}}]}(\omega) d\omega \\
&\leq \text{ess sup}_{\omega \in \text{Supp } \mathfrak{F}[\rho_{\text{pivot}}]} \frac{\mathfrak{F}[\rho_{\text{pivot}}]}{\mathfrak{F}[\rho_{\text{mod}}]}(\omega) \cdot \|\eta\|_{\mathcal{H}_{\text{pivot}}}^2.
\end{aligned}$$

Thus,

$$\frac{\|\eta\|_{\mathcal{H}_{\text{mod}}}}{\|\eta\|_{\mathcal{H}_{\text{pivot}}}} \leq \text{ess sup}_{\omega \in \text{Supp } \mathfrak{F}[\rho_{\text{pivot}}]} \sqrt{\frac{\mathfrak{F}[\rho_{\text{pivot}}]}{\mathfrak{F}[\rho_{\text{mod}}]}(\omega)} =: C_{\text{switch}}. \quad (\text{C.7})$$

C.7 Proof of Proposition 4.2

The proposition is a direct consequence of the preceding section above. Plugging $K_{\text{pivot}} = \Psi_\tau$ the sinc-4 kernel, and letting $K_{\text{mod}} = \rho$ be any translation-invariant kernel such that $\text{Supp } \mathfrak{F}[\Psi_\tau] = [-1/\tau, 1/\tau]^d \subset \text{Supp } \mathfrak{F}[\rho_{\text{mod}}]$. Then, Equation (C.7) becomes

$$C_{\text{switch}}(\Psi_\tau, \rho) = \text{ess sup}_{\omega \in [-1/\tau, 1/\tau]^d} \sqrt{\frac{\mathfrak{F}[\Psi_\tau]}{\mathfrak{F}[\rho_{\text{mod}}]}(\omega)}. \quad (\text{C.8})$$

In addition, we have the straightforward upper-bound:

$$C_{\text{switch}}(\Psi_\tau, \rho) \leq \frac{\text{ess sup}_{\omega \in [-1/\tau, 1/\tau]^d} \sqrt{\mathfrak{F}[\Psi_\tau]}(\omega)}{\text{ess inf}_{\omega \in [-1/\tau, 1/\tau]^d} \sqrt{\mathfrak{F}[\rho_{\text{mod}}]}(\omega)}.$$

We know that $\mathfrak{F}[\Psi_\tau]$ is upper-bounded with a mode $\max \mathfrak{F}[\Psi_\tau] = (\frac{8\pi}{3}\tau)^d$. Hence, the numerator scales as $\mathcal{O}(\tau^{d/2})$ and a sufficient condition for $C_{\text{switch}}(\Psi_\tau, \rho)$ to be finite is that $\mathfrak{F}[\rho_{\text{mod}}]$ is essentially bounded away from zero on $[-1/\tau, 1/\tau]^d$.

D Notation

<i>General notation</i>	
$[n]$	Set of integers $\{1, \dots, n\}$;
$\mathfrak{F}[\cdot], \mathfrak{F}^{-1}[\cdot]$	Fourier transform and its inverse;
Continuous sparse regression and BLASSO	
$\mathcal{X}$	Compact set of $\mathbb{R}^d$, parameter space;
$(\mathcal{C}(\mathcal{X}), \ \cdot\ _\infty)$	Continuous functions on $\mathcal{X}$;
$(\mathcal{M}(\mathcal{X}), \ \cdot\ _{\text{TV}})$	Signed Radon measures on $\mathcal{X}$;
$(\mathcal{F}, \langle \cdot, \cdot \rangle_{\mathcal{F}})$	Separable Hilbert space of observations;
μ^0	Target sparse measure $\sum_{k=1}^{s_0} a_k^0 \delta_{\mathbf{x}_k^0}$, Eq. (2.4);
$s_0 = s_0$	Sparsity (number of atoms of μ^0);
$(\mathbf{x}_k^0, a_k^0)_{k=1}^{s_0}$	Locations and weights of the atoms of μ^0 ;
Δ_0	Minimal separation between atoms of μ^0 , Eq. (2.4);
$\mathcal{M}_{s_0, \Delta_0, \mathfrak{d}_g}$	Class of sparse models, Eq. (2.4);
$\hat{\mu}$	Any near-optimal BLASSO solution, Eq. (2.1– J_κ);
κ	Regularisation parameter of (2.1– J_κ);
J_κ	BLASSO objective functional, Eq. (2.1– J_κ);

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Notation (continued)

Γ (resp. γ)	Noise (resp. noise level), Eq. (2.5);
$\mathcal{N}_k^{\text{reg}}(r)$ (resp. $\mathcal{F}^{\text{reg}}(r)$)	Near (resp. far) regions of radius r , Eq. (2.6);
η^0	Non-degenerate dual certificate of radius r ;
c^0	Pre-certificate such that $\eta^0 = F^\star c^0$;
$\mathcal{D}_{\eta^0}(\mu \ \mu^0)$	Bregman divergence of the TV norm at η^0 , Eq. (2.8);

Model kernel, pivot kernel and kernel switch

K_{mod}	Model kernel, Eq. (2.2);
$(\mathcal{H}_{\text{mod}}, \ \cdot\ _{\mathcal{H}_{\text{mod}}})$	RKHS associated to K_{mod} , Eq. (B.2);
K_{pivot}	Pivot kernel (used for the kernel switch), Eq. (4.5);
$(\mathcal{H}_{\text{pivot}}, \ \cdot\ _{\mathcal{H}_{\text{pivot}}})$	RKHS associated to K_{pivot} ;
C_{switch}	Kernel switch constant, Assumption 2;
$\tilde{K}_{\text{sketch}, \text{mod}}$	Sketched (Monte-Carlo) model kernel, Eq. (2.11a);
Θ_Ω	Radial basis function kernel with covariance Ω , Eq. (2.9b);
Ψ_τ	Sinc-4 kernel (sinus cardinal to the fourth power), Eq. (2.9d);
φ_ω (resp. ψ_ω)	Model (resp. pivot) sketching functions;
Λ	Sketching distribution, Eq. (2.10b);
m	Sketch size (number of random Fourier features);
F (resp. $F^\star$)	Forward measurement operator (resp. its adjoint), Eq. (2.10a);

Mixture models and smoothing

ϕ	Template distribution of the mixture, Eq. (3.1);
f^0	True mixture density $\sum_k a_k^0 \phi(\cdot - \mathbf{x}_k^0)$, Eq. (3.1);
n	Sample size;
$\mathbf{Z} = \{z_1, \dots, z_n\}$	i.i.d. sample drawn from $\phi \star \mu^0$;
$\hat{f}_n$	Empirical distribution of $\mathbf{Z}$;
τ	Width parameter of the smoothing kernel, Equation (3.3–H $_\tau$);
λ_τ	Smoothing kernel (sinus cardinal), Eq. (3.2);
L_τ	Smoothing operator $L_\tau f = \lambda_\tau \star f$;
$\mathcal{F}_\tau$	RKHS of the smoothing kernel λ_τ ;
$\tilde{\mu}$	Sketched BLASSO solution;
$r_n = \delta_n n^{-1/4}$	Effective near/far region radius, Remark 3.4;

Local positive curvature (LPC)

LPC	Local positive curvature assumption, Assumption 1;
B_{ij}	Uniform bounds on kernel covariant derivatives, Eq. (4.3a);
r_0	Radius of the near regions in LPC, Eq. (4.3b);
$\bar{\varepsilon}_0, \bar{\varepsilon}_2$	LPC positive curvature constants, Eq. (4.3b);

Differential geometry

$\nabla_1 K(\mathbf{s}, \mathbf{t})$	Gradient at point $\mathbf{s}$ of $\mathbf{u} \mapsto K(\mathbf{u}, \mathbf{t})$;
$\nabla_2 K(\mathbf{s}, \mathbf{t})$	Gradient at point $\mathbf{t}$ of $\mathbf{u} \mapsto K(\mathbf{s}, \mathbf{u})$;
$\mathbf{g}_\mathbf{x}$	Metric tensor $\nabla_1 \nabla_2 K(\mathbf{x}, \mathbf{x})$ at point $\mathbf{x}$;
$\mathfrak{d}_\mathbf{g}(\mathbf{s}, \mathbf{t})$	Fisher-Rao distance between $\mathbf{s}$ and $\mathbf{t}$ with respect to a kernel K with metric $\mathbf{g}$, Eq. (4.1);
$\mathfrak{d}_{\mathbf{g}, \tau}(\mathbf{s}, \mathbf{t})$	Fisher-Rao distance between $\mathbf{s}$ and $\mathbf{t}$ with respect to the sinc-4 kernel Ψ_τ ;